

Advanced Geovisualisation

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Welcome

This is the website for “Advanced Geovisualisation” (module **ENVS456**) at the University of Liverpool. This course is designed by Dr. Gabriele Filomena and Dr. Elisabetta Pietrostefani from the Geographic Data Science Lab at the University of Liverpool, United Kingdom.

The website is **free to use** and is licensed under the [Attribution-NonCommercial-NoDerivatives 4.0 International](#). A compilation of this web course is hosted as a GitHub repository that you can access:

- As an [html website](#).
- As a [GitHub repository](#).

Contact 2025-26

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Overview

Aims

This module aims to equip students with an advanced understanding of geovisualisation principles and techniques, combining theoretical knowledge of map design with practical computational skills. Students will apply these skills to address geographic research problems, utilizing the internet as both a source of geographical data and an analytical platform. The module also aims to help students create effective visual narratives on human and physical geography-related issues, by allowing them to employ advanced methods such as network analysis and basic simulation models.

Learning Outcomes

By the end of the module, students will:

- Acquire a deep understanding of theoretical principles in cartography.
- Develop advanced and grounded skills in the design of both static and dynamic maps in computational environments.
- Understand the infrastructure of spatial data formats and web-based applications.
- Master the acquisition and manipulation of data from the web, and the design of web-based visualisation and analysis.

Moreover, at the end of the module, they should be able to:

1. Visualise and represent geo-data through static and dynamic maps.
2. Collect Web-based data.
3. Generate interactive maps and dashboards.
4. Understand basic concepts of spatial network analysis.
5. Acquire introductory skills in satellite imagery, spatio-temporal, and mobility data visualisation.
6. Manipulate geo-data through scripting in Python.

Asking for Academic Help:

Verbal face-to-face feedback. Immediate face-to-face feedback will be provided during computer, discussion and clinic sessions in interaction with staff. This will take place in all live sessions during the semester.

Teams Forum. Asynchronous written feedback will be provided via Teams. Students are encouraged to contribute by asking and answering questions relating to the module content. Staff will monitor the forum Monday to Friday 9am-5pm, but it will be open to students to make contributions at all times. Response time will vary depending on the complexity of the question and staff availability.

Direct emails will not be replied to, unless they pertain personal questions.

Queries regarding deadlines, extenuating circumstances requests, etc. should be addressed to: envsci [at] liverpool.ac.uk

Assessment:

Two pieces of coursework ([Assignment 1](#) 50% and [Assignment 2](#) 50%). See guidelines on the assignment pages, submission and marking criteria [here](#).

Syllabus

Week 1 GF Central Teaching Lab: PC Teaching Centre BLUE ZONE (1 - 4pm)

- Lecture: Introduction to the module
- Lab: Powerful examples and Python Refresher

Week 2 GF Central Teaching Lab: PC Teaching Centre BLUE ZONE (1 - 4pm)

- Lecture: Geovisualisation: Design Principles and Statistical Visualisation
- Lab: Static Maps in Python.

Week 3 GF Central Teaching Lab: PC Teaching Centre BLUE ZONE (1 - 4pm)

- Lecture: Interactive Maps
- Lab: Interactive Maps with Folium.

Week 4 EP Central Teaching Lab: PC Teaching Centre BLUE ZONE (1 - 4pm)

- Lecture: The Web's Architecture
- Lab: Web Architectures and APIs in Python.

Week 5 EP Central Teaching Lab: PC Teaching Centre BLUE ZONE (2 - 4pm)

- Lab only: Review & Assignment I clinic.
- **Assignment I Due** 26th February 2.00pm

Week 6 GF Central Teaching Lab: PC Teaching Centre BLUE ZONE (1 - 4pm)

- Lecture: OpenStreetMap Data & Spatial Network.
- Lab: Retrieving Data From OpenStreetMap.

Week 7 EP Central Teaching Lab: PC Teaching Centre BLUE ZONE (1 - 4pm)

- Lecture: From Pixels to Patterns: Visualising Satellite Image Classification
- Lab: Satellite Image Classification in Python

Week 8

- NO TEACHING

Week 9 GF Central Teaching Lab: PC Teaching Centre BLUE ZONE (1 - 4pm)

- Lecture: Simulating agents.
- Lab: Agent-based modelling with Python.

Week 10 GF Central Teaching Lab: PC Teaching Centre BLUE ZONE (1 - 4pm)

- Lecture: Time and Flows.
- Lab: Visualising Flows in Python.

Week 11 GF Central Teaching Lab: PC Teaching Centre BLUE ZONE (2 - 4pm)

- Lab only: Review & Assignment II clinic.
- **Assignment II Due (Week 11)* 30th April 2.00pm

Assignment I

- **Title:** Exploring APIs and Interactive Maps in Python.
- **Type:** Coursework.
- **Due: Thursday February 26th (2.00 pm) - Week 5.**
- 50% of the final mark.
- Submission on Canvas, .html files only.

Context, Design, Data, and Assemblage

In this assessment, you will have the opportunity to explore different sources and combine them in a map that can be explored interactively through a web browser. This assignment requires you to identify a research problem from literature, source relevant data from the web in different formats, assemble them, and document the process. To be successful, you will need to demonstrate your understanding not only of the technical aspects involved in the process but also of the conceptual notions underpinning them. Below are the required components for your submission:

1. **Context and Problem:** Identify a research problem with a geographical connotation. Discuss concisely recent research around it in physical or human geography (around 7–8 references). Introduce how you will explore and visualise dimensions of the problem (e.g., gentrification, access to healthy food in cities, urban heat islands, etc.).
2. **Data and Backend:** Draft a list of spatial datasets relevant to your research problem and demonstrate your ability to develop your own API request function in Python. Include datasets containing spatial information or linkable to other spatial sources. Highlight the data/variables worth considering and their role in representing the problem. Demonstrate your understanding of core “backend” web mapping concepts. Include an explanation of how tilesets, client-server architecture, and APIs are implemented and contribute to your map’s functionality.
3. **Design:** Create Good Looking static maps to represent your datasets, focusing on spatial units (e.g., buildings, cities). Move onto interactive visualisation with folium, incorporating interactivity for categorical and numerical variables. Seek feedback to refine your ideas. Use inspiration from web map examples discussed in the course to ensure effective representation of data.
4. **Assemblage:** Enhance your map by incorporating widgets for dataset exploration and experimenting with tilesets, such as creating your own in Mapbox. Address design considerations, including the map’s extent, zoom levels, and variable visibility at different zoom levels. Ensure consistency and aesthetic appeal to complete this stage successfully.

Expected Content

Text and Code, alternated:

- ~250 w. introducing the research problem, the context, and existing research on the topic.
- Code: Introductory Static Maps (2), presenting the topic and the context.
- ~250 w. presenting and motivating the chosen data sources.
- Code: An API request written by your own - All the necessary steps for making your API work and for data cleaning/exploration.
- ~200 w. with your description of what your API is, how it works and how it will made your map possible.
- Code: An interactive final map. This should be fed with data obtained through the API request.
- ~200 w. description of how your interactive map works, its components and your design ideas.
- ~150 w. to summarise your research problem and how you tackled it by means of geovisualisation tools (Conclusion).

You CANNOT employ for your main maps the following libraries: Holoviews, Geoviews, and Plotly

Evaluation

The assignment will be evaluated based on 3 main pillars, on which you will have to be successful to achieve a good mark:

1. **Narrative.** The ability to identify and present a research problem, motivate and justify one's map, as well as the ability to bring each component of the assignment into a coherent whole that "fits together".
2. **Map design abilities.** The ability to demonstrate the understanding of geovisualisation and interactivity design principles.
3. **Technical skills.** The ability to master Python scripting and technologies that allow one to create a compelling map, but also to access interesting and sophisticated data sources.

How is this assignment useful?

This assessment includes several elements that will help you improve critical aspects of your web mapping skills:

- **Design:** this is not about making maps, this is about making good maps. And behind every good map there is a set of conscious choices that you will have to think through to be successful (what map? what data? how to present the data? etc.).
- **Technology:** at the end of the day, building good web maps requires solid understanding of current technology that goes beyond what the average person can be expected to know. In this assignment, you will need to demonstrate you are proficient in a series of tasks manipulating geospatial data in a web environment.
- **Presentation:** in many real-world contexts, your work is as good as it can come across to the audience it is intended to. This means that it is vital to be able to communicate not only what you are doing but why and on what building blocks it is based on.

Setting up the Working Environment

Follow the instructions for your Operating System and test your installation. If you experience any issues, write a message on the MS Teams channel of the module.

Setting up the Python environment is necessary for:

- Executing the [Jupyter Notebooks](#) of the Lab sessions of the course.
- Preparing your own Jupyter Notebooks for the assignments (one each).

We will use Miniconda to handle our working environment.

Miniconda is a free minimal installer for conda. It is a small bootstrap version of Anaconda that includes only conda, Python, the packages they both depend on, and a small number of other useful packages.

Set up Miniconda (and Python) on MS Windows

Installation

1. Install Miniconda:
 - *Option 1:* On a UoL Machine: Download and install Miniconda from [here](#). This will install Miniconda and Python in C:\. If this process is aborted because it requires administrator rights, press **Start**, select **Install University Applications**, type and choose **Miniconda**.
 - *Option 2, Recommended:* Install Miniconda on your personal laptop: Follow the instructions [here](#).
2. During the installation, leave the default settings. In particular, when asked whom to “Install Miniconda for”, choose “Just for me”.

Important: If you do choose to work on University Machines you will have to reinstall **Miniconda** every lab session unless you use a PC where Miniconda has been installed already.

Set up the Directories

1. Create a folder where you want to keep your work conducted throughout this course. For example, call it **geovis**. You can save it wherever you want. If you are working on a university machine, it could be worth creating it in **M:**, which should be your “virtual” hard-disk.

Example fictional path (Windows, personal laptop): **C:\Users\YourName\Documents\geovis**

Example path (Windows, UoL machine): **M:\geovis**

2. Download the [data](#) and the [images](#) for running and rendering the Jupyter notebooks.
3. Unzip the folders and move the nested folders into the folder **geovis**.
4. Create another folder called **labs**.

The folder structure should look like:

```
geovis/  
  data/  
  labs_img/  
  labs/
```

Set up the Python Environment

1. Download the **geovis.yml** from GitHub by clicking **Download raw file**, top right, [at this page](#).

2. Save it in the folder **geovis** created before.

Example path: **C:\Users\YourName\Documents\geovis\geovis.yml**; **UoL machine**

example: **M:\geovis\geovis.yml**

3. Type in the search bar and find the **Anaconda Prompt (miniconda 3)**. Launch it. The terminal should appear.
4. In the **Anaconda Prompt**, run: `conda env create -n geovis --file "C:\Users\YourName\Documents\geovis\geovis.yml"`
5. Then run: `conda env update -n geovis --file "C:\Users\YourName\Documents\geovis\geovis.yml" --prune`
6. If you are prompted any questions, press **y**. This process will install all the packages necessary to carry out the lab sessions.

7. In the **Anaconda Prompt** write and run: `conda activate geovis`. This activates your working environment.
 8. *Necessary on University machines, otherwise optional:* configuration of Jupyter Notebooks
 - In the **Anaconda Prompt**, run: `jupyter server --generate-config`. This should create a file at: `C:\Users\YourName\.jupyter\jupyter_server_config.py`
 - Open the file with a text editor (e.g. [Notepad++](#)). Do a `Ctrl+F` search for: `c.ServerApp.root_dir`
 - Uncomment it and set it to your course folder (the folder where you created `geovis`). Example: `c.ServerApp.root_dir = r"M:\geovis"` (use your own path if different).
 - Save the file and close it.
-

Start a Lab Session

1. Download the Jupyter Notebook of the session in your folder. Choose one Jupyter notebook and click **Download raw file** as shown below.
 2. Save the file in the `labs` folder within your `geovis` folder on your machine.
 3. Type in the search bar, find and open the **Anaconda Prompt** (`miniconda 3`).
 4. In the **Anaconda Prompt** run: `conda activate geovis`
 5. (*Recommended*) “Move” into your course folder so Jupyter opens in the right place: `cd "C:\Users\YourName\Documents\geovis\labs"` (or `cd "M:\geovis\labs"` on UoL machines).
 6. In the **Anaconda Prompt** run: `jupyter notebook`. This should open Jupyter Notebook in your default browser.
 7. Navigate to your course folder and double click on the notebook downloaded in step 1.
 8. You can now work on your copy of the notebook.
-

Follow these instructions and test your installation **prior to the first Lab Session**. If you experience any issues, write a message on the MS Teams channel of the module.

Set up Miniconda (and Python) on Mac

Installation

To install Miniconda on your personal laptop, follow the instructions [here](#). During the installation, leave the default settings. In particular, when asked whom to “Install Miniconda for”, choose “Just for me”.

Set up the Directories

1. Create a folder where you want to keep your work conducted throughout this course. For example, call it **geovis**. You can save it wherever you want.

Example fictional path (Mac): /Users/yourname/Documents/geovis/;

Example fictional path (Mac): /Users/yourname/Library/CloudStorage/Dropbox/geovis/

2. Download the [data](#) and the [images](#) for running and rendering the Jupyter notebooks.
3. Unzip the folders and move the nested folders into the folder **geovis**.
4. Create another folder called **labs**.

The folder structure should look like:

```
geovis/  
  data/  
  labs_img/  
  labs/
```

Set up the Python Environment

1. Download the `geovis.yml` from GitHub by clicking `Download raw file`, top right, [at this page](#).
 2. Save it in the folder `geovis` created before.
 3. Open the **Terminal**.
 4. In the **Anaconda Prompt**, run: `conda env create -n geovis --file "/Users/yourname/Documents/geovis/geovis.yml"`
 5. Then run: `conda env update -n geovis --file "/Users/yourname/Documents/geovis/geovis.yml" --prune`
 6. If you are prompted any questions, press `y`. This process will install all the packages necessary to carry out the lab sessions.
 7. Activate the environment: `conda activate geovis`
-

Start a Lab Session

1. Download the Jupyter Notebook of the session in your folder. Choose one Jupyter notebook from [here](#) and click `Download raw file` as shown below.
2. Save the file in the `labs` folder within your `geovis` folder on your machine.
3. Open the **Terminal**.
4. Activate the environment: `conda activate geovis`
5. (*Recommended*) “Move” into your `labs` folder so Jupyter opens in the right place: `cd "/Users/yourname/Documents/geovis/labs"`
6. Start Jupyter Notebook: `jupyter notebook`
7. This should open Jupyter Notebook in your default browser. You should see something like this:
8. Navigate to your folder. You can now work on your copy of the notebook.

1 Introduction & Python Refresher

The **Lecture slides** can be found [here](#).

This **lab's** notebook can be downloaded from [here](#).

1.1 Part I: Powerful (Web) Mapping Examples

This part of the lab has two main components: 1. The first one will require you to find a partner and work together with them 2. And the second one will involve group discussion.

1.1.1 Paired Activity

In pairs, find **three** examples where web maps are used to communicate an idea. Complete the following sheet for each example:

- **Substantive**
 - **Title:** Title of the map/project
 - **Author:** Who is behind the project?
 - **Big idea:** a “one-liner” on what the project tries to accomplish –
 - **Message:** what does the map try to get accross
- **Technical**
 - **URL:**
 - **Interactivity:** does the map let you interact with it in any way? Yes/No
 - **Zoomable:** can you explore the map at different scales? Yes/No
 - **Design:** what geovisualisation solutions have been adopted?
 - **Basemap:** Is there an underlying map providing geographical context? Yes/No. If so, who is it provided by?
 - **Technology:** can you guess what technology does this map rely on?

Post each sheet as a separate item on the Teams channel for Lab No.1

1.1.1.1 Example

The project “WHO Coronavirus (COVID-19) Dashboard”

- **Substantive**

- Title: WHO Coronavirus (COVID-19) Dashboard
- Author: World Health Organization
- Big idea: Shows confirmed COVID-19 cases and deaths by country to date
- Message: The project displays a map of the world where COVID-19 cases are shown by country. This element is used to show which countries have had more cases (large trends). A drop down button allows us to visualise the map by a) Total per 100,000 population b) % change in the last 7 days c) newly reported in the last 7 days d) newly reported in the last 24 hours.

- **Technical**

- URL: <https://covid19.who.int/>
- Interactivity: Yes
- Zoomable: Yes
- Tooltips: Yes
- Basemap: No
- Technology: Unknown

1.1.2 Class discussion

We will select a few examples posted and collectively discuss (some of) the following questions:

1. What makes them powerful, what “speaks” to us?
2. What could be improved, what is counter-intuitive?
3. What design elements do they rely on?
4. What technology do they use?

1.1.3 References

- For an excellent coverage of “visualisation literacy”, Chapter 11 of Andy Kirk’s “[Data Visualisation](#)” is a great start.
- A comprehensive overview of computational notebooks and how they relate to modern scientific work is available on [Ch.1 of the GDS book](#).
- A recent overview of notebooks in Geography is available in [Boeing & Arribas-Bel \(2021\)](#)

1.2 Part II: Python/Pandas (Refresher)

*Gabriele Filomena has prepared this notebook by readapting material shared on this [repository](#).
Copyright (c) 2013-2023 Geoff Boeing.*

1.2.1 Python

A quick overview of ubiquitous programming concepts including data types, for loops, if-then-else conditionals, and functions.

```
import numpy as np
import pandas as pd
```

```
# integers (int)
x = 100
type(x)
```

int

```
# floating-point numbers (float)
x = 100.5
type(x)
```

float

```
# sequence of characters (str)
x = 'Los Angeles, CA 90089'
len(x)
```

21

```
# list of items
x = [1, 2, 3, 'USC']
len(x)
```

4

```
# sets are unique
x = {2, 2, 3, 3, 1}
x
```

{1, 2, 3}

```
# tuples are immutable sequences
latlng = (34.019425, -118.283413)
type(latlng)
```

tuple

```
# you can unpack a tuple
lat, lng = latlng
type(lat)
```

float

```
# dictionary of key:value pairs
iceland = {'Country': 'Iceland', 'Population': 372520, 'Capital': 'Reykjavik', '% Foreign Pop
type(iceland)
```

dict

```
# you can convert types
x = '100'
print(type(x))
y = int(x)
print(type(y))
```

```
<class 'str'>
<class 'int'>
```

```
# you can loop through an iterable, such as a list or tuple
for coord in latlng:
    print('Current coordinate is:', coord)
```

```
Current coordinate is: 34.019425
Current coordinate is: -118.283413
```

```
# loop through a dictionary keys and values as tuples
for key, value in iceland.items():
    print(key, value)
```

```
Country Iceland
Population 372520
Capital Reykjavík
% Foreign Population 0.18
```

```
# booleans are trues/falsees
x = 101
x > 100
```

True

```
# use two == for equality and one = for assignment
x == 100
```

False

```
# if, elif, else for conditional branching execution
x = 101
if x > 100:
    print('Value is greater than 100.')
elif x < 100:
    print('Value is less than 100.')
else:
    print('Value is 100.')

```

Value is greater than 100.

```
# use functions to encapsulate and reuse bits of code
def convert_items(my_list, new_type=str):
    # convert each item in a list to a new type
    new_list = [new_type(item) for item in my_list]
    return new_list

l = [1, 2, 3, 4]
convert_items(l)
```

['1', '2', '3', '4']

1.2.2 pandas Series and DataFrames

`pandas` has two primary data structures we will work with: `Series` and `DataFrame`.

1.2.2.1 Pandas Series

```
# a pandas series is based on a numpy array: it's fast, compact, and has more functionality
# it has an index which allows you to work naturally with tabular data
my_list = [8, 5, 77, 2]
my_series = pd.Series(my_list)
my_series
```

```
0      8
1      5
2     77
3      2
dtype: int64
```

```
# look at a list-representation of the index
my_series.index.tolist()
```

```
[0, 1, 2, 3]
```

```
# look at the series' values themselves
my_series.values
```

```
array([ 8,  5, 77,  2])
```

```
# what's the data type of the series' values?
type(my_series.values)
```

```
numpy.ndarray
```

```
# what's the data type of the individual values themselves?
my_series.dtype
```

```
dtype('int64')
```

1.2.2.2 Pandas DataFrames

```
# a dict can contain multiple lists and label them
my_dict = {'hh_income' : [75125, 22075, 31950, 115400],
           'home_value' : [525000, 275000, 395000, 985000]}
my_dict
```

```
{'hh_income': [75125, 22075, 31950, 115400],
 'home_value': [525000, 275000, 395000, 985000]}
```

```
# a pandas dataframe can contain one or more columns
# each column is a pandas series
# each row is a pandas series
# you can create a dataframe by passing in a list, array, series, or dict
df = pd.DataFrame(my_dict)
df
```

	hh_income	home_value
0	75125	525000
1	22075	275000
2	31950	395000
3	115400	985000

```
# the row labels in the index are accessed by the .index attribute of the DataFrame object
df.index.tolist()
```

```
[0, 1, 2, 3]
```

```
# the column labels are accessed by the .columns attribute of the DataFrame object
df.columns
```

```
Index(['hh_income', 'home_value'], dtype='str')
```

```
# the data values are accessed by the .values attribute of the DataFrame object
# this is a numpy (two-dimensional) array
df.values
```

```
array([[ 75125, 525000],
       [ 22075, 275000],
       [ 31950, 395000],
       [115400, 985000]])
```

1.2.3 Loading data in Pandas

Usually, you'll work with data by loading a dataset file into pandas. CSV is the most common format. But pandas can also ingest tab-separated data, JSON, and proprietary file formats like Excel .xlsx files, Stata, SAS, and SPSS.

Below, notice what pandas's `read_csv` function does:

1. Recognize the header row and get its variable names.
2. Read all the rows and construct a pandas DataFrame (an assembly of pandas Series rows and columns).
3. Construct a unique index, beginning with zero.
4. Infer the data type of each variable (i.e., column).

```
# load a data file
# note the relative filepath! where is this file located?
# use dtype argument if you don't want pandas to guess your data types
df = pd.read_csv('../data/GTD_2022.csv', low_memory = False)
```

```
to_replace = [-9, -99, "-9", "-99"]
for value in to_replace:
    df = df.replace(value, np.nan)

df['eventid'] = df['eventid'].astype("Int64")
```

```
# dataframe shape as rows, columns
df.shape
```

```
(209706, 75)
```

```
# or use len to just see the number of rows
len(df)
```

```
209706
```

```
# view the dataframe's "head"
df.head()
```


	eventid	year	date	extended	country	country_txt	region	region_txt
0	201000000000	2005	28/05/2005	1	153	Pakistan	6	South Asia
1	201000000000	2005	29/05/2005	1	95	Iraq	10	Middle East & North Af
2	201000000000	2007	08/06/2007	1	95	Iraq	10	Middle East & North Af
3	201000000000	2010	15/06/2010	1	4	Afghanistan	6	South Asia
4	201000000000	2011	06/01/2011	1	92	India	6	South Asia

```
# view the dataframe's "tail"
df.tail()
```

	eventid	year	date	extended	country	country_txt	region	region_txt
209701	202000000000	2020	31/12/2020	0	228	Yemen	10	Middle East & No
209702	202000000000	2020	31/12/2020	0	228	Yemen	10	Middle East & No
209703	202000000000	2020	31/12/2020	0	75	Germany	8	Western Europe
209704	202000000000	2020	31/12/2020	0	4	Afghanistan	6	South Asia
209705	202000000000	2020	31/12/2020	1	33	Burkina Faso	11	Sub-Saharan Afric

```
# column data types
df.dtypes
```

```
eventid      Int64
year         int64
date          str
extended     int64
country      int64
...
INT_LOG      float64
INT_IDEO     float64
INT_MISC     float64
INT_ANY      float64
related      str
Length: 75, dtype: object
```

```
# or
for dt in df.columns[:10]:
    print(dt, type(dt))
```

```
eventid <class 'str'>
```

```

year <class 'str'>
date <class 'str'>
extended <class 'str'>
country <class 'str'>
country_txt <class 'str'>
region <class 'str'>
region_txt <class 'str'>
city <class 'str'>
latitude <class 'str'>

```

1.2.4 Selecting and slicing data from a DataFrame

# CHEAT SHEET OF COMMON TASKS		
# Operation	Syntax	Result
#-----		
# Select column by name	df[col]	Series
# Select columns by name	df[col_list]	DataFrame
# Select row by label	df.loc[label]	Series
# Select row by integer location	df.iloc[loc]	Series
# Slice rows by label	df.loc[a:c]	DataFrame
# Select rows by boolean vector	df[mask]	DataFrame

1.2.4.1 Select DataFrame's column(s) by name

```

# select a single column by column name
# this is a pandas series
df['country']

```

```

0      153
1       95
2       95
3        4
4       92
...
209701  228
209702  228
209703   75
209704    4

```

```
209705      33
Name: country, Length: 209706, dtype: int64
```

```
# select multiple columns by a list of column names
# this is a pandas dataframe that is a subset of the original
df[['country_txt', 'year']]
```

	country_txt	year
0	Pakistan	2005
1	Iraq	2005
2	Iraq	2007
3	Afghanistan	2010
4	India	2011
...	...	...
209701	Yemen	2020
209702	Yemen	2020
209703	Germany	2020
209704	Afghanistan	2020
209705	Burkina Faso	2020

```
# create a new column by assigning df['new_col'] to some values
# people killed every perpetrator
df['killed_per_attacker'] = df['nkill'] / df['nperps']

# inspect the results
df[['country', 'year', 'nkill', 'nperps', 'killed_per_attacker']].head(15)
```

	country	year	nkill	nperps	killed_per_attacker
0	153	2005	1.0	NaN	NaN
1	95	2005	1.0	NaN	NaN
2	95	2007	15.0	50.0	0.300000
3	4	2010	1.0	NaN	NaN
4	92	2011	1.0	NaN	NaN
5	160	2011	1.0	30.0	0.033333
6	95	2011	1.0	NaN	NaN
7	155	2002	7.0	3.0	2.333333
8	92	2010	5.0	100.0	0.050000
9	153	2011	4.0	NaN	NaN
10	4	2011	1.0	NaN	NaN

	country	year	nkill	nperps	killed_per_attacker
11	153	2011	2.0	NaN	NaN
12	186	2001	12.0	60.0	0.200000
13	4	2006	6.0	NaN	NaN
14	155	2007	0.0	NaN	NaN

1.2.4.2 Select row(s) by label

```
# use .loc to select by row label
# returns the row as a series whose index is the dataframe column names
df.loc[0]
```

```
eventid          201000000000
year              2005
date             28/05/2005
extended         1
country          153
...
INT_IDEO         NaN
INT_MISC         0.0
INT_ANY          NaN
related          NaN
killed_per_attacker NaN
Name: 0, Length: 76, dtype: object
```

```
# use .loc to select single value by row label, column name
df.loc[15, 'gname'] #group name
```

```
'Taliban'
```

```
# slice of rows from label 5 to label 7, inclusive
# this returns a pandas dataframe
df.loc[5:7]
```

	eventid	year	date	extended	country	country_txt	region	region_txt
5	201000000000	2011	08/01/2011	0	160	Philippines	5	Southeast
6	201000000000	2011	11/04/2011	1	95	Iraq	10	Middle Ea

	eventid	year	date	extended	country	country_txt	region	region_txt
7	2000000000000	2002	16/07/2002	0	155	West Bank and Gaza Strip	10	Middle Ea

```
# slice of rows from label 17 to label 27, inclusive
# slice of columns from country_txt to city, inclusive
df.loc[17:27, 'country_txt': 'city']
```

	country_txt	region	region_txt	city
17	India	6	South Asia	Mumbai
18	India	6	South Asia	Mumbai
19	Afghanistan	6	South Asia	Khost
20	India	6	South Asia	Unknown
21	Pakistan	6	South Asia	Lahore
22	Pakistan	6	South Asia	Lahore
23	Pakistan	6	South Asia	Mullazai
24	Colombia	3	South America	El Tambo
25	Nigeria	11	Sub-Saharan Africa	Unknown
26	Pakistan	6	South Asia	Quetta
27	Kenya	11	Sub-Saharan Africa	Nairobi

```
# subset of rows from with labels in list
# subset of columns with names in list
df.loc[[1, 350], ['country', 'gname']]
```

	country	gname
1	95	Ansar al-Sunna
350	147	Boko Haram

```
# you can use a column of identifiers as the index (indices do not *need* to be unique)
df_gname = df.set_index('gname')
df_gname.index.is_unique
```

False

```
df_gname.head(3)
```

	eventid	year	date	extended	country	country_txt	region	region_txt
gname								
Unknown	201000000000	2005	28/05/2005	1	153	Pakistan	6	South Asia
Ansar al-Sumna	201000000000	2005	29/05/2005	1	95	Iraq	10	Middle East
Al-Qaida in Iraq	201000000000	2007	08/06/2007	1	95	Iraq	10	Middle East

```
# .loc works by label, not by position in the dataframe
try:
    df_gname.loc[0]
except KeyError as e:
    print('label not found')
```

label not found

```
# the index now contains gname values, so you have to use .loc accordingly to select by row
df_gname.loc['Taliban'].head()
```

	eventid	year	date	extended	country	country_txt	region	region_txt	city
gname									
Taliban	201000000000	2010	15/06/2010	1	4	Afghanistan	6	South Asia	Sayd
Taliban	201000000000	2006	26/08/2006	0	4	Afghanistan	6	South Asia	Muq
Taliban	201000000000	2008	27/04/2008	0	4	Afghanistan	6	South Asia	Muh
Taliban	201000000000	2009	12/05/2009	0	4	Afghanistan	6	South Asia	Khos
Taliban	201000000000	2014	28/03/2014	0	4	Afghanistan	6	South Asia	Kabu

1.2.4.3 Select by (integer) position - Independent from actual Index

```
# get the row in the zero-th position in the dataframe
df.iloc[0]
```

```
eventid      201000000000
year          2005
date         28/05/2005
extended      1
country       153
...
INT_IDEO      NaN
```

```

INT_MISC          0.0
INT_ANY           NaN
related           NaN
killed_per_attacker  NaN
Name: 0, Length: 76, dtype: object

```

```

# you can slice as well
# note, while .loc is inclusive, .iloc is not
# get the rows from position 0 up to but not including position 3 (ie, rows 0, 1, and 2)
df.iloc[0:3]

```

	eventid	year	date	extended	country	country_txt	region	region_txt
0	2010000000000	2005	28/05/2005	1	153	Pakistan	6	South Asia
1	2010000000000	2005	29/05/2005	1	95	Iraq	10	Middle East & North Af
2	2010000000000	2007	08/06/2007	1	95	Iraq	10	Middle East & North Af

```

# get the value from the row in position 3 and the column in position 2 (zero-indexed)
df.iloc[3, 6] #country_txt

```

```
np.int64(6)
```

1.2.4.4 Select/filter by value

You can subset or filter a dataframe for based on the values in its rows/columns.

```

# filter the dataframe by urban areas with more than 25 million residents
df[df['nkill'] > 30].head()

```

	eventid	year	date	extended	country	country_txt	region	region_txt
17	2010000000000	2008	26/11/2008	1	92	India	6	South Asia
18	2010000000000	2008	26/11/2008	1	92	India	6	South Asia
27	2010000000000	2013	21/09/2013	1	104	Kenya	11	Sub-Saharan Africa
38	2010000000000	2014	17/07/2014	1	200	Syria	10	Middle East & North A
39	2010000000000	2014	24/07/2014	0	200	Syria	10	Middle East & North A

```
# you can chain multiple conditions together
# pandas logical operators are: | for or, & for and, ~ for not
# these must be grouped by using parentheses due to order of operations
df[['country','nkill', 'nwound']][(df['nkill'] > 200) & (df['nwound'] > 10)].head()
# columns on the left-hand side are here used to slice the resulting output
```

	country	nkill	nwound
427	229	339.0	28.0
1000	217	1385.0	10878.0
1001	217	1385.0	10878.0
2780	1004	287.0	400.0
3787	4	266.0	64.0

```
# ~ means not... it essentially flips trues to falses and vice-versa
df[['country','nkill', 'nwound']][~(df['nkill'] > 200) & (df['nwound'] > 10)]
```

	country	nkill	nwound
7	155	7.0	16.0
17	92	68.0	76.0
18	92	32.0	22.0
19	4	9.0	25.0
21	153	29.0	31.0
...	...	...	...
209528	4	10.0	15.0
209547	153	3.0	17.0
209636	4	1.0	13.0
209659	153	0.0	14.0
209681	228	30.0	60.0

1.2.5 Grouping and summarizing

```
# group by terroristic group name
groups = df.groupby('gname')
```

```
# what is the median number of people killed per event across the different groups?
groups['nkill'].median().sort_values(ascending=False)
```



```

gname
Samri militia                766.0
Gumuz militia                222.0
Dissident Military Mmbrs of Northern Tribal Group  150.0
Ansar al-Din Front           103.0
Issa and Gurgura Liberation Front  100.0
...
Western Somalia Liberation Front  NaN
Workers of the Company           NaN
World Punishment Organization    NaN
Zawiya Martyrs Brigade          NaN
Zimbabwe Guerrillas             NaN
Name: nkill, Length: 3725, dtype: float64

```

```

# look at several columns' medians by group
groups[['nkill', 'nwound', 'nperps']].median()

```

	nkill	nwound	nperps
gname			
01-May	0.0	0.0	1.5
09-Feb	0.0	0.0	NaN
14 K Triad	0.0	0.0	8.0
14 March Coalition	5.0	80.0	NaN
14th of December Command	0.0	0.0	NaN
...	...	...	...
Zulu Miners	1.0	10.0	NaN
Zuwar al-Imam Rida	0.0	0.0	4.0
Zviadists	0.0	0.0	NaN
Zwai Tribe	5.0	NaN	NaN
leftist guerrillas-Bolivarian militia	0.0	0.0	NaN

```

# you can create a new dataframe by directly passing columns between "[[ ]]", after the groupby
# to do so, you also need to pass a function that can deal with the values (e.g. sum..etc)
western_europe = df[df.region_txt == 'Western Europe']
western_europe.groupby('country_txt')[['nkill', 'nwound']].sum().sort_values('nkill', ascending=False)

```

	country_txt	nkill	nwound
0	United Kingdom	3426.0	6321.0
1	Spain	1289.0	4937.0

	country_txt	nkill	nwound
2	France	565.0	2565.0
3	Italy	420.0	1320.0
4	Greece	264.0	763.0
5	Ireland	108.0	26.0
6	Germany	103.0	735.0
7	West Germany (FRG)	97.0	862.0
8	Belgium	83.0	523.0
9	Norway	80.0	90.0
10	Switzerland	75.0	98.0
11	Cyprus	45.0	41.0
12	Netherlands	41.0	75.0
13	Austria	37.0	150.0
14	Portugal	32.0	95.0
15	Sweden	22.0	83.0
16	Finland	12.0	37.0
17	Malta	5.0	12.0
18	Denmark	5.0	31.0
19	Andorra	0.0	0.0
20	Iceland	0.0	0.0
21	Luxembourg	0.0	6.0
22	Vatican City	0.0	3.0

1.2.6 Indexes

Each `DataFrame` has an index. Indexes do not have to be unique (but that would be for the best)

```
# resetting index (when loading a .csv file pandas creates an index automatically, from 0 to
df.reset_index(drop = True).sort_index().head() # this does not assign the new index though,
```

	eventid	year	date	extended	country	country_txt	region	region_txt
0	201000000000	2005	28/05/2005	1	153	Pakistan	6	South Asia
1	201000000000	2005	29/05/2005	1	95	Iraq	10	Middle East & North Af
2	201000000000	2007	08/06/2007	1	95	Iraq	10	Middle East & North Af
3	201000000000	2010	15/06/2010	1	4	Afghanistan	6	South Asia
4	201000000000	2011	06/01/2011	1	92	India	6	South Asia

```
#this does assign the new index to your df
df = df.reset_index(drop = True).sort_index()
df.head()
```

	eventid	year	date	extended	country	country_txt	region	region_txt
0	201000000000	2005	28/05/2005	1	153	Pakistan	6	South Asia
1	201000000000	2005	29/05/2005	1	95	Iraq	10	Middle East & North Af
2	201000000000	2007	08/06/2007	1	95	Iraq	10	Middle East & North Af
3	201000000000	2010	15/06/2010	1	4	Afghanistan	6	South Asia
4	201000000000	2011	06/01/2011	1	92	India	6	South Asia

```
# index isn't unique
df.index.is_unique
```

True

```
# you can set a new index
# drop -> Delete columns to be used as the new index.
# append -> whether to append columns to existing index.
df = df.set_index('eventid', drop=True, append=False)
df.index.name = None # remove the index "name"
df.head()

# this index is not ideal, but it's the original source's id
```

	year	date	extended	country	country_txt	region	region_txt
201000000000	2005	28/05/2005	1	153	Pakistan	6	South Asia
201000000000	2005	29/05/2005	1	95	Iraq	10	Middle East & North Africa
201000000000	2007	08/06/2007	1	95	Iraq	10	Middle East & North Africa
201000000000	2010	15/06/2010	1	4	Afghanistan	6	South Asia
201000000000	2011	06/01/2011	1	92	India	6	South Asia

1.3 Part III: Geospatial Vector data in Python

*Gabriele Filomena has prepared this notebook by readapting material shared on this [repository](#).
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```
%matplotlib inline

import geopandas as gpd
```

1.3.1 Importing geospatial data

GeoPandas builds on Pandas types **Series** and **Dataframe**, by incorporating information about geographical space.

- **GeoSeries**: a Series object designed to store shapely geometry object
- **GeoDataFrame**: object is a pandas DataFrame that has a column with geometry (that contains a *Geoseries*)

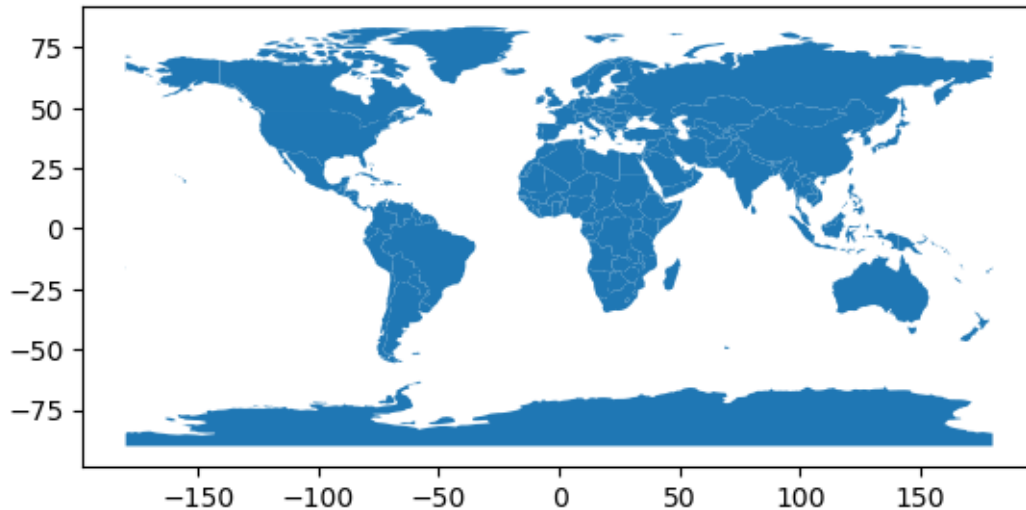
We can use the GeoPandas library to read many of GIS file formats (relying on the **fiona** library under the hood, which is an interface to GDAL/OGR), using the **gpd.read_file** function. For example, let's start by reading a shapefile with all the countries of the world (adapted from <http://www.naturalearthdata.com/downloads/110m-cultural-vectors/110m-admin-0-countries/>, zip file is available in the `/data` directory), and inspect the data:

```
countries = gpd.read_file("../data/ne_countries.zip")
# or if the archive is unpacked:
# countries = gpd.read_file("../data/ne_countries.shp")
```

```
countries.head()
```

	iso_a3	name	continent	pop_est	gdp_md_est	geometry
0	AFG	Afghanistan	Asia	34124811.0	64080.0	POLYGON (((61.21082 35.0...
1	AGO	Angola	Africa	29310273.0	189000.0	MULTIPOLYGON (((23.9...
2	ALB	Albania	Europe	3047987.0	33900.0	POLYGON ((21.02004 40.0...
3	ARE	United Arab Emirates	Asia	6072475.0	667200.0	POLYGON ((51.57952 24.0...
4	ARG	Argentina	South America	44293293.0	879400.0	MULTIPOLYGON (((-66.9...

```
countries.plot()
```



We observe that:

- Using `.head()` we can see the first rows of the dataset, just like we can do with Pandas.
- There is a **geometry** column and the different countries are represented as polygons
- We can use the `.plot()` (matplotlib) method to quickly get a *basic* visualization of the data

1.3.2 What's a GeoDataFrame?

We used the GeoPandas library to read in the geospatial data, and this returned us a GeoDataFrame:

```
type(countries)
```

```
geopandas.geodataframe.GeoDataFrame
```

A GeoDataFrame contains a tabular, geospatial dataset:

- It has a 'geometry' column that holds the geometry information (or features in GeoJSON).
- The other columns are the **attributes** (or properties in GeoJSON) that describe each of the geometries.

Such a GeoDataFrame is just like a pandas DataFrame, but with some additional functionality for working with geospatial data: * A **geometry** attribute that always returns the column with the geometry information (returning a GeoSeries). The column name itself does not necessarily need to be 'geometry', but it will always be accessible as the **geometry** attribute. *

It has some extra methods for working with spatial data (area, distance, buffer, intersection, ...) [see here, for example](#).

```
countries.geometry.head()
```

```
0    POLYGON ((61.21082 35.65007, 62.23065 35.27066...
1    MULTIPOLYGON (((23.90415 -11.72228, 24.07991 -...
2    POLYGON ((21.02004 40.84273, 20.99999 40.58, 2...
3    POLYGON ((51.57952 24.2455, 51.75744 24.29407,...
4    MULTIPOLYGON (((-66.95992 -54.89681, -67.56244...
Name: geometry, dtype: geometry
```

```
type(countries.geometry)
```

```
geopandas.geoseries.GeoSeries
```

```
countries.geometry.area
```

```
C:\Users\gfilo\AppData\Local\Temp\ipykernel_2500\3077649407.py:1: UserWarning: Geometry is in unprojected CRS. To avoid distortion in area calculations, please
project geometries to a projected CRS before this operation.
```

```
countries.geometry.area

0         63.593500
1        103.599439
2          3.185163
3          7.095047
4        278.923392
...
172         0.631326
173        38.475618
174       112.718524
175        62.789498
176        32.280371
Length: 177, dtype: float64
```

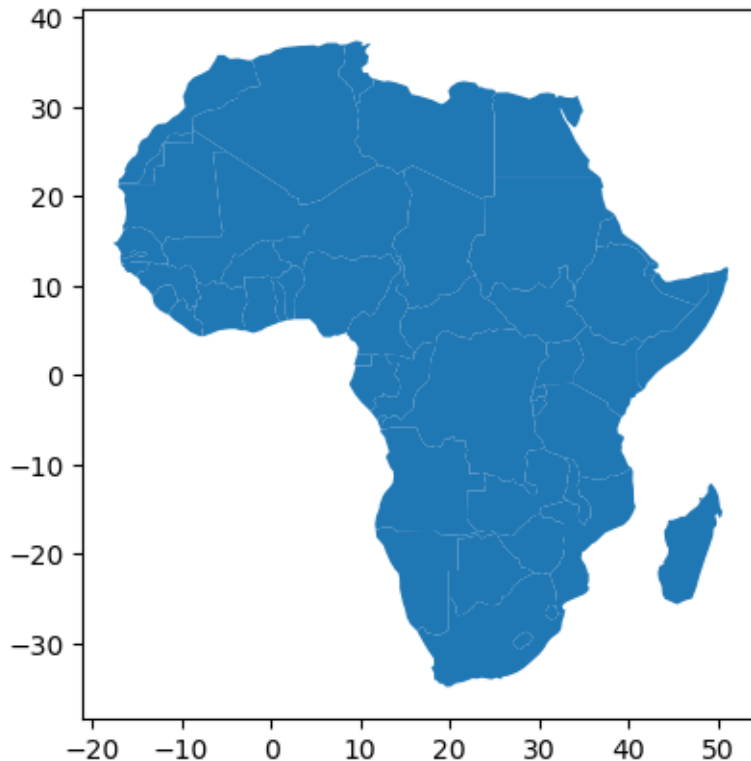
It's still a `DataFrame`, so we have all the `pandas` functionality available to use on the geospatial dataset, and to do data manipulations with the attributes and geometry information together. For example, we can calculate the average population over all countries (by accessing the 'pop_est' column, and calling the `mean` method on it):

```
countries['pop_est'].mean()
```

```
np.float64(41712369.84180791)
```

```
africa = countries[countries['continent'] == 'Africa']
```

```
africa.plot();
```



The rest of the tutorial is going to assume you already know some pandas basics, but we will try to give hints for that part for those that are not familiar.

Important:

- A `GeoDataFrame` allows to perform typical tabular data analysis together with spatial operations
- A `GeoDataFrame` (or *Feature Collection*) consists of:
 - **Geometries** or **features**: the spatial objects
 - **Attributes** or **properties**: columns with information about each spatial object

1.3.3 Geometries: Points, Linestrings and Polygons

Spatial **vector** data can consist of different types, and the 3 fundamental types are:

- **Point** data: represents a single point in space.
- **Line** data (“LineString”): represented as a sequence of points that form a line.
- **Polygon** data: represents a filled area.

And each of them can also be combined in multi-part geometries (See <https://shapely.readthedocs.io/en/stable/manual.html#geometric-objects> for extensive overview).

For the example we have seen up to now, the individual geometry objects are Polygons:

```
print(countries.geometry[2])
```

```
POLYGON ((21.0200403174764 40.84272695572588, 20.999989861747224 40.58000397395401, 20.67499
```

Let’s import some other datasets with different types of geometry objects.

A dataset about cities in the world (adapted from <http://www.naturalearthdata.com/downloads/110m-cultural-vectors/110m-populated-places/>, zip file is available in the `/data` directory), consisting of Point data:

```
cities = gpd.read_file("../data/ne_cities.zip")
```

```
print(cities.geometry[0])
```

```
POINT (12.453386544971766 41.903282179960115)
```

And a dataset of rivers in the world (from <http://www.naturalearthdata.com/downloads/50m-physical-vectors/50m-rivers-lake-centerlines/>, zip file is available in the `/data` directory) where each river is a (Multi-)LineString:

```
rivers = gpd.read_file("../data/ne_rivers.zip")
```

```
print(rivers.geometry[0])
```

```
LINESTRING (51.9371337598152 55.70106609892139, 51.880866467313695 55.68625891701544, 51.820
```


1.3.4 The shapely library

The individual geometry objects are provided by the [shapely](#) library

```
from shapely.geometry import Point, Polygon, LineString
```

```
type(countries.geometry[0])
```

```
shapely.geometry.polygon.Polygon
```

To construct one ourselves:

```
p = Point(0, 0)
```

```
print(p)
```

```
POINT (0 0)
```

```
polygon = Polygon([(1, 1), (2,2), (2, 1)])
```

```
polygon.area
```

```
0.5
```

```
polygon.distance(p)
```

```
1.4142135623730951
```

Important:

Single geometries are represented by **shapely** objects:

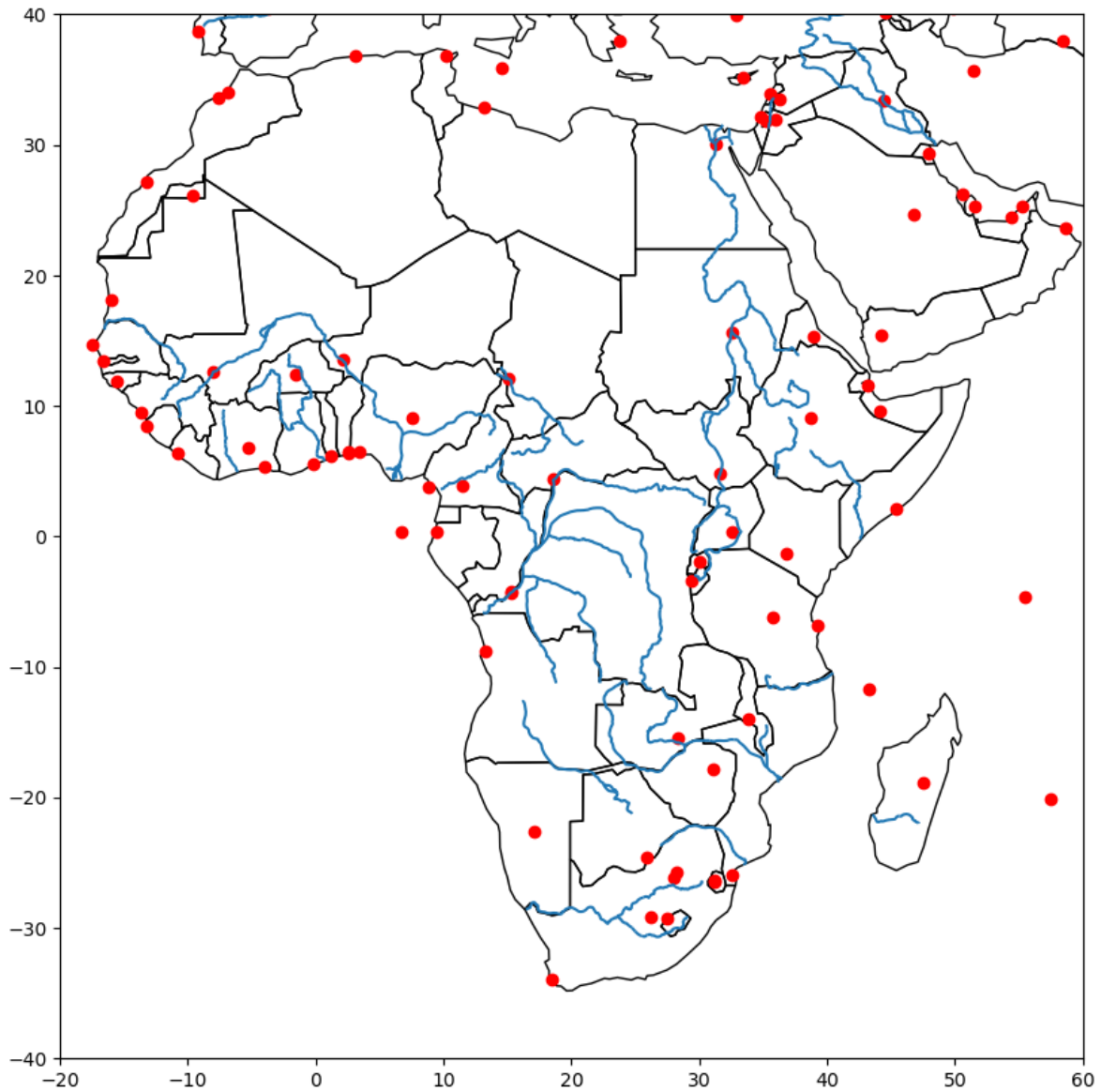
- If you access a single geometry of a GeoDataFrame, you get a shapely geometry object
- Those objects have similar functionality as geopandas objects (GeoDataFrame/GeoSeries).
For example:

- `single_shapely_object.distance(other_point)` -> distance between two points
- `geodataframe.distance(other_point)` -> distance for each point in the geodataframe to the other point

1.3.5 Plotting

```
%matplotlib inline
import matplotlib
import matplotlib.pyplot as plt

fig, ax = plt.subplots(1, 1, figsize=(15, 10))
countries.plot(ax = ax, edgecolor='k', facecolor='none')
rivers.plot(ax=ax)
cities.plot(ax=ax, color='red')
ax.set(xlim=(-20, 60), ylim=(-40, 40))
```



1.3.6 Creating GeoDataFrames (withouth specifying the CRS)

```
gpd.GeoDataFrame({  
    'geometry': [Point(1, 1), Point(2, 2)],  
    'attribute1': [1, 2],  
    'attribute2': [0.1, 0.2]})
```

	geometry	attribute1	attribute2
0	POINT (1 1)	1	0.1
1	POINT (2 2)	2	0.2

```
# Creating a GeoDataFrame from an existing dataframe
# For example, if you have lat/lon coordinates in two columns:
df = pd.DataFrame(
    {'City': ['Buenos Aires', 'Brasilia', 'Santiago', 'Bogota', 'Caracas'],
     'Country': ['Argentina', 'Brazil', 'Chile', 'Colombia', 'Venezuela'],
     'Latitude': [-34.58, -15.78, -33.45, 4.60, 10.48],
     'Longitude': [-58.66, -47.91, -70.66, -74.08, -66.86]})
```

```
gdf = gpd.GeoDataFrame(df, geometry=gpd.points_from_xy(df.Longitude, df.Latitude))
gdf
```

	City	Country	Latitude	Longitude	geometry
0	Buenos Aires	Argentina	-34.58	-58.66	POINT (-58.66 -34.58)
1	Brasilia	Brazil	-15.78	-47.91	POINT (-47.91 -15.78)
2	Santiago	Chile	-33.45	-70.66	POINT (-70.66 -33.45)
3	Bogota	Colombia	4.60	-74.08	POINT (-74.08 4.6)
4	Caracas	Venezuela	10.48	-66.86	POINT (-66.86 10.48)

2 Practice

Throughout the exercises in this course, we will work with several datasets about the city of Paris.

Here, we start with the following datasets:

- The administrative districts of Paris (https://opendata.paris.fr/explore/dataset/quartier__paris/): `paris_districts_utm.geojson`
- Real-time (at the moment I downloaded them ..) information about the public bicycle sharing system in Paris (vélib, <https://opendata.paris.fr/explore/dataset/stations-velib-disponibilites-en-temps-reel/information/>): `data/paris_bike_stations_mercator.gpkg`

Both datasets are provided as spatial datasets using a GIS file format.

Exercise 1:

We will start by exploring the bicycle station dataset (available as a GeoPackage file: `data/paris_bike_stations_mercator.gpkg`)

- Read the stations datasets into a GeoDataFrame called `stations`.
- Check the type of the returned object
- Check the first rows of the dataframes. What kind of geometries does this datasets contain?
- How many features are there in the dataset?

Hints

- Use `type(..)` to check any Python object type
- The `gpd.read_file()` function can read different geospatial file formats. You pass the file name as first argument.
- Use the `.shape` attribute to get the number of features

Exercise 2:

- Make a quick plot of the `stations` dataset.
- Make the plot a bit larger by setting the figure size to (12, 6) (hint: the `plot` method accepts a `figsize` keyword).

Exercise 3:

Next, we will explore the dataset on the administrative districts of Paris (available as a GeoJSON file: `../data/paris_districts_utm.geojson`)

- Read the dataset into a GeoDataFrame called `districts`.
- Check the first rows of the dataframe. What kind of geometries does this dataset contain?
- How many features are there in the dataset? (hint: use the `.shape` attribute)
- Make a quick plot of the `districts` dataset (set the figure size to (12, 6)).

Exercise 4:

What are the largest districts (biggest area)?

- Calculate the area of each district.
- Add this area as a new column to the `districts` dataframe.
- Sort the dataframe by the area column from largest to smallest values (descending).

Hints

- Adding a column can be done by assigning values to a column using the same square brackets syntax: `df['new_col'] = values`
- To sort the rows of a DataFrame, use the `sort_values()` method, specifying the column to sort on with the `by='col_name'` keyword. Check the help of this method to see how to sort ascending or descending.

2.1 Part IV: Coordinate reference systems & Projections

Gabriele Filomena has prepared this notebook by readapting material shared on this [repository](#). Copyright (c) 2018, Joris Van den Bossche.

```
countries = gpd.read_file("../data/ne_countries.zip")
cities = gpd.read_file("../data/ne_cities.zip")
rivers = gpd.read_file("../data/ne_rivers.zip")
```

2.1.1 Coordinate reference systems

Up to now, we have used the geometry data with certain coordinates without further wondering what those coordinates mean or how they are expressed.

The **Coordinate Reference System (CRS)** relates the coordinates to a specific location on earth.

For an in-depth explanation, see https://docs.qgis.org/2.8/en/docs/gentle_gis_introduction/coordinate_reference_systems.html

2.1.1.1 Geographic coordinates

Degrees of latitude and longitude.

E.g. 48°51 N, 2°17 E

The most known type of coordinates are geographic coordinates: we define a position on the globe in degrees of latitude and longitude, relative to the equator and the prime meridian. With this system, we can easily specify any location on earth. It is used widely, for example in GPS. If you inspect the coordinates of a location in Google Maps, you will also see latitude and longitude.

Attention!

in Python we use (lon, lat) and not (lat, lon)

- Longitude: $[-180, 180]\{\{1\}\}$
- Latitude: $[-90, 90]\{\{1\}\}$

2.1.2 Projected coordinates

(x , y) coordinates are usually in meters or feet

Although the earth is a globe, in practice we usually represent it on a flat surface: think about a physical map, or the figures we have made with Python on our computer screen. Going from the globe to a flat map is what we call a *projection*.

We project the surface of the earth onto a 2D plane so we can express locations in cartesian x and y coordinates, on a flat surface. In this plane, we then typically work with a length unit such as meters instead of degrees, which makes the analysis more convenient and effective.

However, there is an important remark: the 3 dimensional earth can never be represented perfectly on a 2 dimensional map, so projections inevitably introduce distortions. To minimize such errors, there are different approaches to project, each with specific advantages and disadvantages.

Some projection systems will try to preserve the area size of geometries, such as the Albers Equal Area projection. Other projection systems try to preserve angles, such as the Mercator projection, but will see big distortions in the area. Every projection system will always have some distortion of area, angle or distance.

Projected size vs actual size (Mercator projection):

2.1.3 Coordinate Reference Systems in Python / GeoPandas

A GeoDataFrame or GeoSeries has a `.crs` attribute which holds (optionally) a description of the coordinate reference system of the geometries:

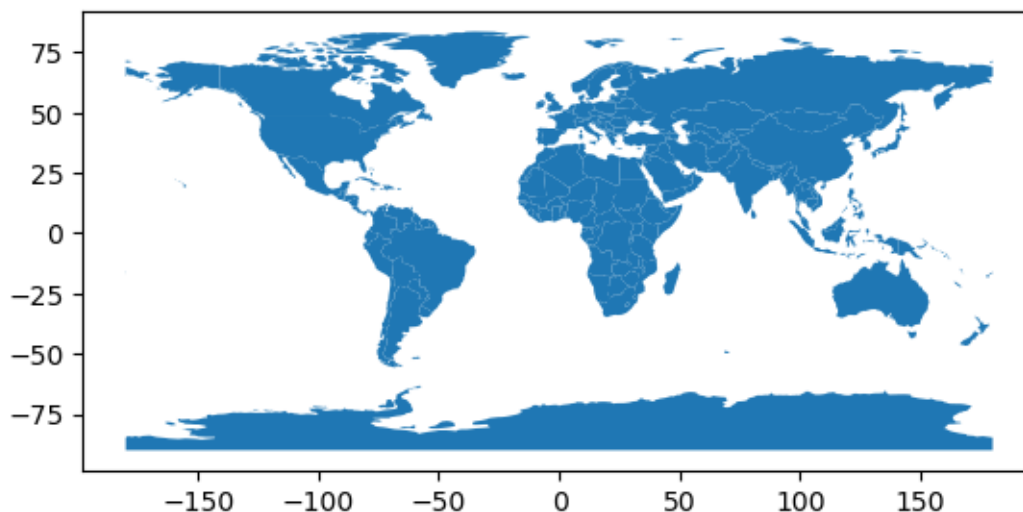
```
countries.crs
```

```
<Geographic 2D CRS: EPSG:4326>  
Name: WGS 84  
Axis Info [ellipsoidal]:  
- Lat[north]: Geodetic latitude (degree)  
- Lon[east]: Geodetic longitude (degree)  
Area of Use:  
- name: World.  
- bounds: (-180.0, -90.0, 180.0, 90.0)  
Datum: World Geodetic System 1984 ensemble  
- Ellipsoid: WGS 84  
- Prime Meridian: Greenwich
```

For the `countries` dataframe, it indicates that it uses the EPSG 4326 / WGS84 lon/lat reference system, which is one of the most used for geographic coordinates.

It uses coordinates as latitude and longitude in degrees, as can you be seen from the x/y labels on the plot:

```
countries.plot()
```



The `.crs` attribute returns a `pyproj.CRS` object. To specify a CRS, we typically use some string representation:

- **EPSG code** Example: `EPSG:4326` = WGS84 geographic CRS (longitude, latitude)

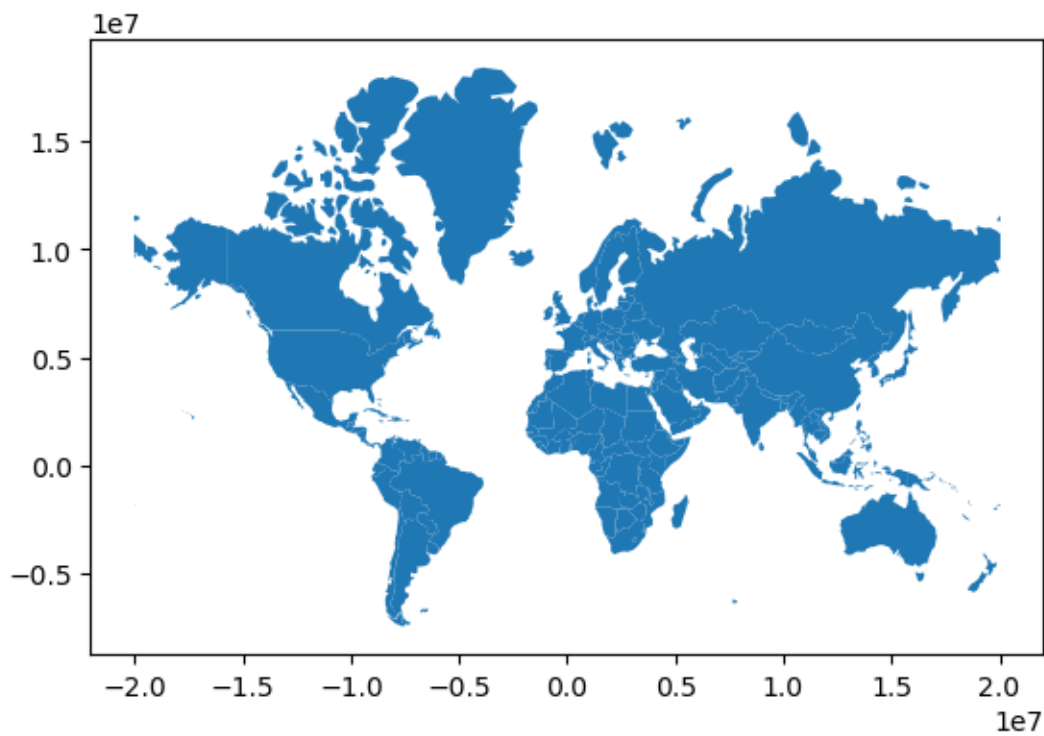
For more information, see also <http://geopandas.readthedocs.io/en/latest/projections.html>.

2.1.3.1 Transforming to another CRS

We can convert a `GeoDataFrame` to another reference system using the `to_crs` function.

For example, let's convert the countries to the World Mercator projection (<http://epsg.io/3395>):

```
# remove Antarctica, as the Mercator projection cannot deal with the poles
countries = countries[(countries['name'] != "Antarctica")]
countries_mercator = countries.to_crs(epsg=3395) # or .to_crs("EPSG:3395")
countries_mercator.plot()
```



Note the different scale of x and y.

2.1.3.2 Why using a different CRS?

There are sometimes good reasons you want to change the coordinate references system of your dataset, for example:

- Different sources with different CRS -> need to convert to the same crs.
- Different countries/geographical areas with different CRS.
- Mapping (distortion of shape and distances).
- Distance / area based calculations -> ensure you use an appropriate projected coordinate system expressed in a meaningful unit such as meters or feet (**not degrees!**).

Important:

All the calculations (e.g. distance, spatial operations, etc.) that take place in **GeoPandas** and **Shapely** assume that your data is represented in a 2D cartesian plane, and thus the result of those calculations will only be correct if your data is properly projected.

2.2 Practice

Again, we will go back to the Paris datasets. Up to now, we provided the datasets in an appropriate projected CRS for the exercises. But the original data were actually using geographic coordinates. In the following exercises, we will start from there.

Going back to the Paris districts dataset, this is now provided as a GeoJSON file ("`../data/paris_districts.geojson`") in geographic coordinates.

For converting the layer to projected coordinates, we will use the standard projected CRS for France is the RGF93 / Lambert-93 reference system, referenced by the **EPSG:2154** number.

Exercise: Projecting a GeoDataFrame

- Read the districts datasets ("`../data/paris_districts.geojson`") into a GeoDataFrame called **districts**.
- Look at the CRS attribute of the GeoDataFrame. Do you recognize the EPSG number?
- Make a plot of the **districts** dataset.
- Calculate the area of all districts.
- Convert the **districts** to a projected CRS (using the **EPSG:2154** for France). Call the new dataset **districts_RGF93**.
- Make a similar plot of **districts_RGF93**.
- Calculate the area of all districts again with **districts_RGF93** (the result will now be expressed in m²).

Hints

- The CRS information is stored in the `.crs` attribute of a `GeoDataFrame`.
- Making a simple plot of a `GeoDataFrame` can be done with the `.plot()` method.
- Converting to a different CRS can be done with the `.to_crs()` method, and the CRS can be specified as an EPSG number using the `epsg` keyword.

3 Static Maps in Python

The **Lecture slides** can be found [here](#).

This **lab's** notebook can be downloaded from [here](#).

3.1 Part I: Basic Maps

In this session, we will use the libraries `matplotlib` and `contextily` to plot the information represented into different `GeoDataFrames`. We will look into plotting `Point`, `LineString` and `Polygon GeoDataFrames`. Most of the plots here are rather ugly but, at this point, the goal is to get familiar with the parameters of the plot function and what can be done with them.

```
%matplotlib inline
import matplotlib
import matplotlib.pyplot as plt
import geopandas as gpd
import pandas as pd
import osmnx as ox
import contextily as ctx
import seaborn as sns
```

3.1.1 Plotting Points

Load the data of terrorist attacks 1970-2020 and choose a country. Germany is used as a case study here but feel free to change the country. If you do so, also change the `crs` (see <https://epsg.io>).

```
attacks = pd.read_csv("../data/GTD_2022.csv", low_memory = False)
```

Creating the `GeoDataFrame` from the `DataFrame`

```

germany = ['West Germany (FRG)', 'Germany', 'East Germany (GDR)'] # Germany was split till 19
df = attacks[attacks.country_txt.isin(germany)].copy()

# Uncomment the lines below for other countries that haven't changed their denominations/bou
# country = 'France'
# df = attacks[attacks.country_txt == country].copy()#
wgs = 'EPSG:4326'
germany_crs = 'EPSG:4839'
gdf = gpd.GeoDataFrame(df, geometry=gpd.points_from_xy(df.longitude, df.latitude), crs = wgs)
gdf = gdf[~gdf.geometry.is_empty] # remove empty geometries
gdf.to_file("../data/germany.gpkg", driver="GPKG", layer="germany")
gdf = gdf.to_crs(germany_crs)

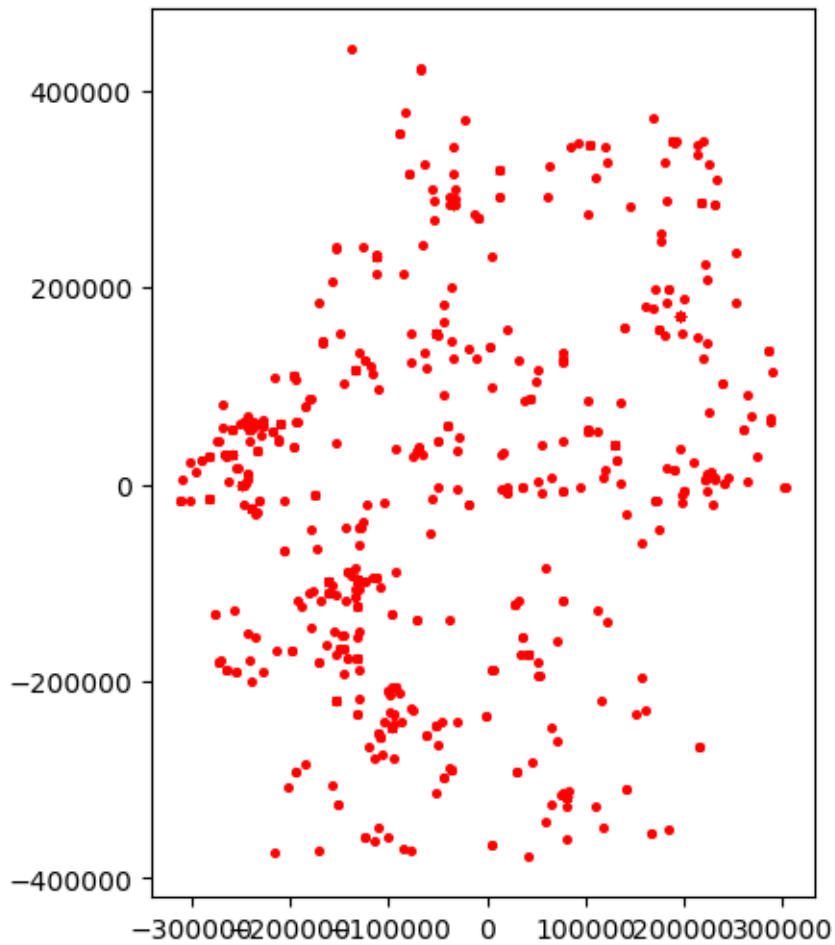
```

Basic plotting

```

# prepare the axis and coordinate
nr_rows = 1
nr_cols = 1
fig, ax = plt.subplots(nr_cols, nr_rows, figsize=(8, 6))
gdf.plot(ax=ax, color='red', markersize=7)

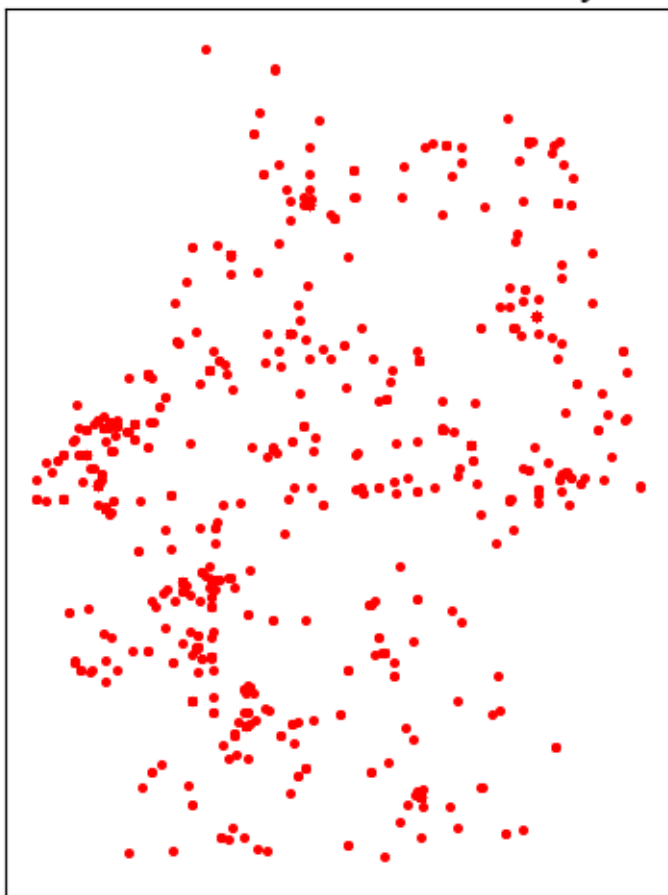
```



Slightly improving the plot:

```
# removing ticks
ax.xaxis.set_ticklabels([])
ax.yaxis.set_ticklabels([])
ax.tick_params(axis= 'both', which= 'both', length=0)
title_parameters = {'fontsize':'16', 'fontname':'Times New Roman'}
ax.set_title("Terroristic Attacks in Germany", **title_parameters)
fig
```

Terroristic Attacks in Germany

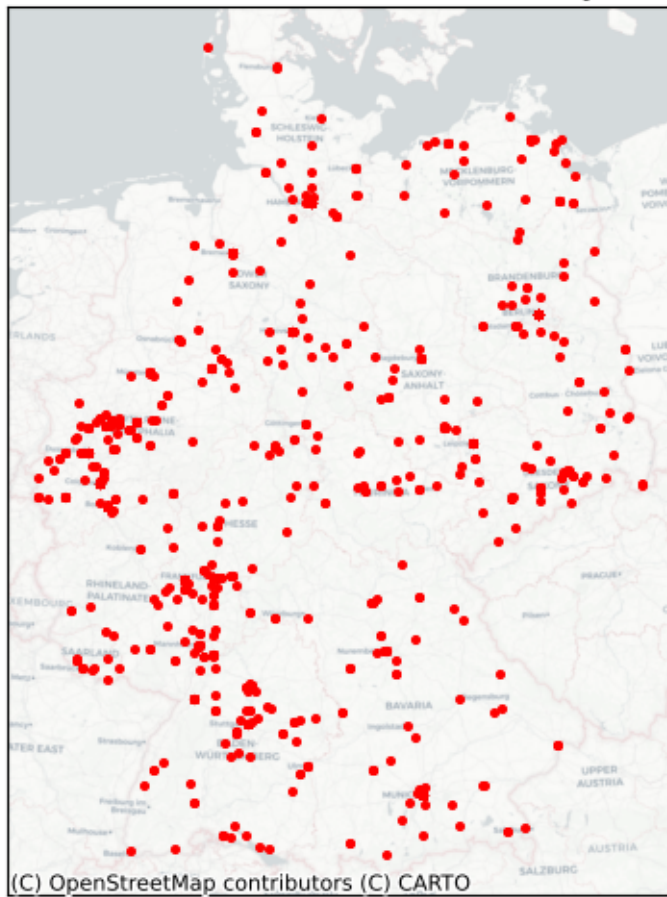


3.1.1.1 Adding some context: Base Maps with Contextily

see providers and options here <https://xyzservices.readthedocs.io/en/stable/introduction.html>

```
source = ctx.providers.CartoDB.Positron
ctx.add_basemap(ax, crs= gdf.crs.to_string(), source= source)
# replot
fig
```

Terroristic Attacks in Germany



<Figure size 640x480 with 0 Axes>

3.1.1.2 Parameters specific to Point in the plot method

- **markersize:** numerical value (for now)
- **marker:** see https://matplotlib.org/stable/api/markers_api.html

3.1.1.2.1 Other properties, shape independent:

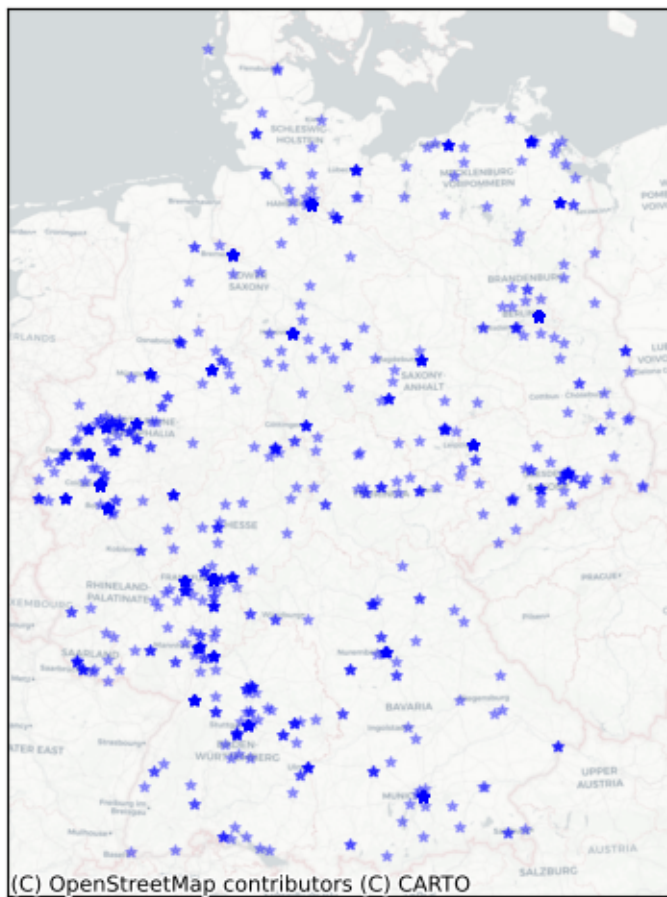
- **color:** https://matplotlib.org/3.1.0/gallery/color/named_colors.html
- **alpha:** regulates transparency of the shape: 0 to 1


```
# first, let's make a function
```

```
def ax_ticks_off(ax):  
    ax.xaxis.set_ticklabels([])  
    ax.yaxis.set_ticklabels([])  
    ax.tick_params(axis= 'both', which= 'both', length=0)
```

```
# prepare the axis and coordinate
```

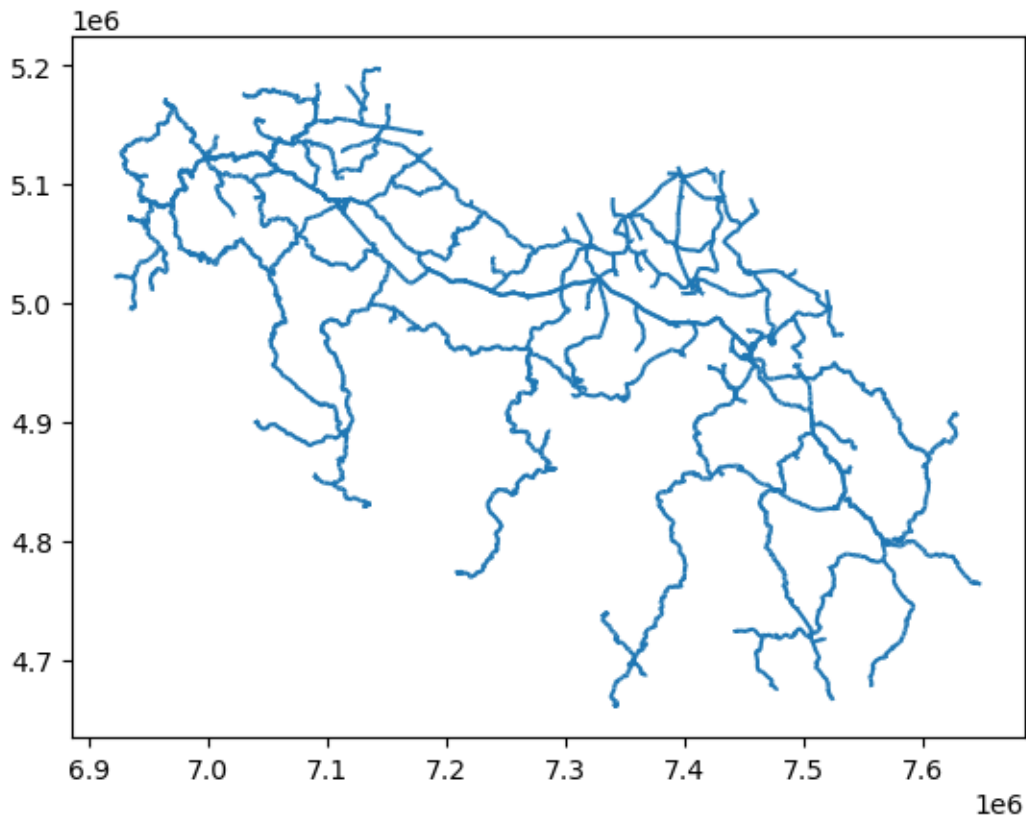
```
fig, ax = plt.subplots(1, 1, figsize=(8, 6))  
gdf.plot(ax=ax, markersize = 15, color = 'blue', marker = '*', alpha = 0.3)  
ctx.add_basemap(ax, crs= gdf.crs.to_string(), source= source)  
ax_ticks_off(ax)
```



3.1.2 Plotting LineStrings

Let's import railway tracks in the Western Balkans (Slovenia, Croatia, Bosnia & Herzegovina, Montenegro, Serbia, Kosovo)

```
wb_crs = 'EPSG:31277'
lines_gdf = gpd.read_file("../data/wb_railways.gpkg")
lines_gdf.plot()
```



```
# prepare the plot
fig, ax = plt.subplots(1, 1, figsize=(8, 6))
lines_gdf.plot(ax=ax, linewidth = 0.8, color = 'blue', alpha = 1)
ctx.add_basemap(ax, crs= lines_gdf.crs.to_string(), source = ctx.providers.Esri.WorldGrayCan
ax_ticks_off(ax)
ax.set_title("Railway infrastructure in the West Balkans", **title_parameters) #parameters a
```

```
Text(0.5, 1.0, 'Railway infrastructure in the West Balkans')
```

Railway infrastructure in the West Balkans



One can also filter prior to plotting, based on the columns in the GeoDataFrame. First we download Serbia's Boundary with OSMNX, more on that later on. Then we filter `lines_gdf` with a `within` operation.

```
serbia = ox.geocode_to_gdf('Serbia')
serbia = serbia.to_crs(wb_crs)
serbia_lines = lines_gdf[lines_gdf.geometry.within(serbia.iloc[0].geometry)].copy() #there's

# prepare the plot
fig, ax = plt.subplots(1, 1, figsize=(8, 6))
serbia_lines.plot(ax=ax, linewidth = 0.8, color = 'blue', alpha = 1)
ctx.add_basemap(ax, crs= lines_gdf.crs.to_string(), source = ctx.providers.Esri.WorldGrayCanvas)
ax_ticks_off(ax)
ax.set_title("Railway infrastructure in Serbia and Kosovo", **title_parameters) #parameters a
```

```
Text(0.5, 1.0, 'Railway infrastructure in Serbia and Kosovo')
```

Railway infrastructure in Serbia and Kosovo



3.1.2.1 Parameters specific to LineString:

- `linewidth`: numerical value (for now).
- `capstyle`: controls how Matplotlib draws the corners where two different line segments meet. See https://matplotlib.org/stable/gallery/lines_bars_and_markers/capstyle.html
- `joinstyle`: controls how Matplotlib draws the corners where two different line segments meet. https://matplotlib.org/stable/gallery/lines_bars_and_markers/joinstyle.html

```
# prepare the plot
fig, ax = plt.subplots(1, 1, figsize=(10, 10))
serbia_lines.plot(ax=ax, linewidth = 0.9, color = 'black', alpha = 1, capstyle = 'round', joinstyle = 'miter')
ax.set_axis_off() # we don't need the ticks function
ax.set_title("Railway infrastructure in Serbia", **title_parameters) #parameters as above
```

Text(0.5, 1.0, 'Railway infrastructure in Serbia')

Railway infrastructure in Serbia



3.1.3 Plotting Polygons

We are again using OSMNX to download data from OpenStreetMap automatically. In this case, we will get building footprints from the city of Algiers in Alageria.

3.1.3.1 Parameter specific to Polygon:

- `edgecolor`: the outline of the polygon, by default = `None` (often better).
- `linewidth`: the width of the outline of the polygon.

```
algeria_crs = 'EPSG:30729'
tags = {"building": True} #OSM tags
buildings = ox.features_from_address("Algiers, Algeria", tags = tags, dist = 2000)
buildings = buildings.reset_index()
# sometimes building footprints are represented by Points, let's disregard them
buildings = buildings[(buildings.geometry.geom_type == 'Polygon') | (buildings.geometry.geom_type == 'MultiPolygon')]
buildings = buildings.to_crs(algeria_crs)
```

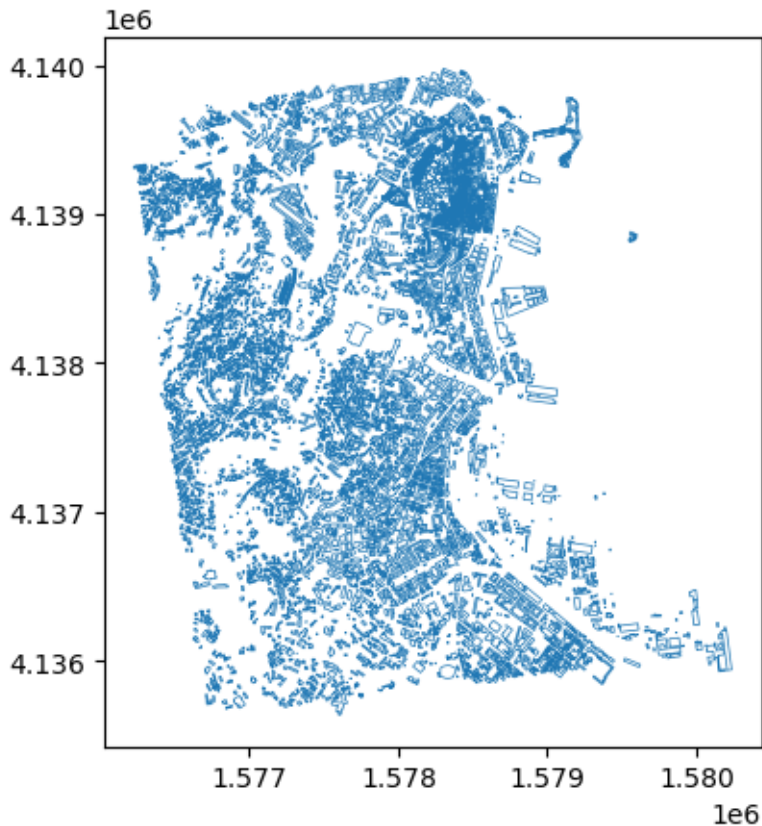
```
fig, ax = plt.subplots(1, 1, figsize=(15, 10))
ax.set_title("Buildings in Algiers", **title_parameters)
ax.set_axis_off() # we don't need the ticks function
buildings.plot(ax=ax, color = 'orange', edgecolor = 'black', lw = 0.2)
source = ctx.providers.CartoDB.PositronNoLabels
ctx.add_basemap(ax, crs= buildings.crs.to_string(), source= source)
```

Buildings in Algiers



For polygons, you can also plot just the boundaries of the geometries by:

```
buildings.boundary.plot(lw = 0.5)
```

3.1.4 Plotting more than one layer together

Let's also download roads for Algiers

```
tags = {"highway": True} #OSM tags
roads = ox.features_from_address("Algiers, Algeria", tags = tags, dist = 2000)
roads = roads.reset_index()
roads = roads.to_crs(algeria_crs)
# sometimes building footprints are represented by Points, let's disregard them
roads = roads[roads.geometry.geom_type == 'LineString']
```

And plot everything together. It's important to keep in mind that the last layer is always rendered on top of the others. In other words, they may cover the previous ones.

However, you can prevent this by passing arguments to the parameter `zorder` in the `plot` method. The layer with the higher `zorder` value will be plotted on top.

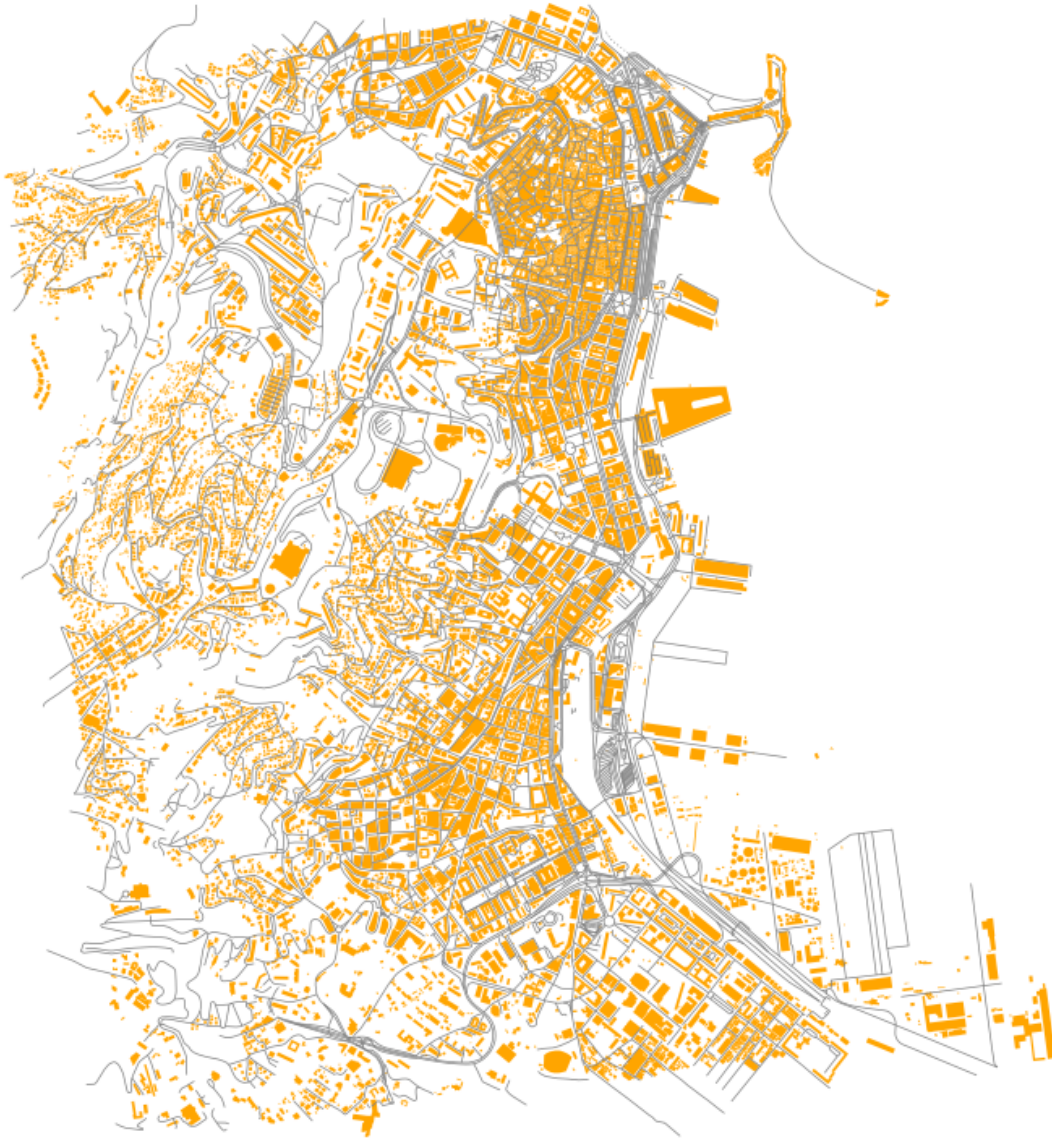
```

fig, ax = plt.subplots(1, 1, figsize=(15, 10))
ax.set_title("Buildings and Roads in Algiers", **title_parameters)
ax.set_axis_off() # we don't need the ticks function
# only roads within the extent of the buildings layer
roads[roads.geometry.within(buildings.unary_union.envelope)].plot(ax=ax, color = 'grey', lw=2)
buildings.plot(ax=ax, color = 'orange')

```

C:\Users\gfilo\AppData\Local\Temp\ipykernel_27180\1248606569.py:5: DeprecationWarning: The 'l' parameter of 'plot' is deprecated. Use 'linewidth' instead.
roads[roads.geometry.within(buildings.unary_union.envelope)].plot(ax=ax, color = 'grey', lw=2)

Buildings and Roads in Algiers



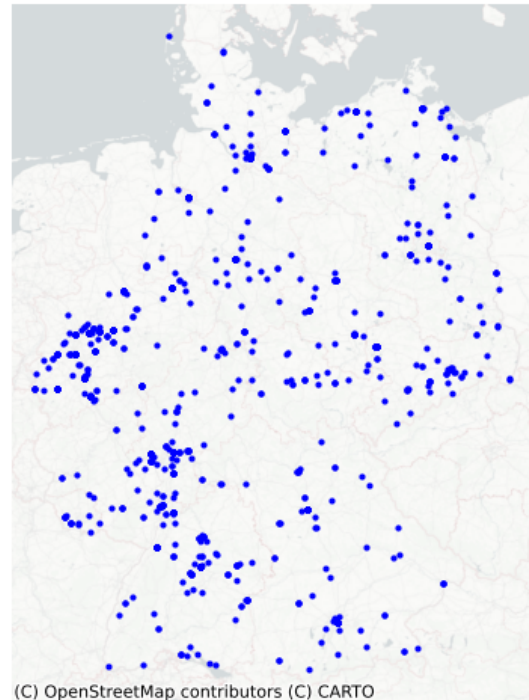
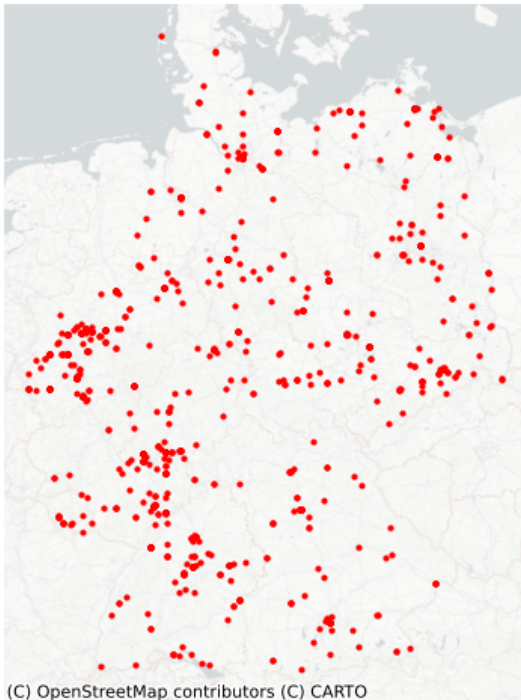
3.1.5 Sub-plots

To obtain multiple sub-plots, we manipulate the `nrows`, `ncols` parameters. We can use this approach to:

- * Plot the same layer with different properties.

```
fig, axes = plt.subplots(1, 2, figsize=(10, 6))
colors = ['red', 'blue']

for n, ax in enumerate(axes):
    gdf.plot(ax=ax, markersize = 4, color = colors[n])
    ax.set_axis_off()
    ctx.add_basemap(ax, crs= gdf.crs.to_string(), source= source)
```



- Plot different layers.

```
fig, axes = plt.subplots(1, 2, figsize=(10, 6))
gdfs = [buildings, roads]
colors = ['orange', 'grey']

buildings.plot(ax=axes[0], color = 'orange', edgecolor = 'none')
roads.plot(ax=axes[1], color = 'grey', lw = 0.5)

for ax in axes:
    ax.set_axis_off()
    ctx.add_basemap(ax, crs= buildings.crs.to_string(), source = source)
```

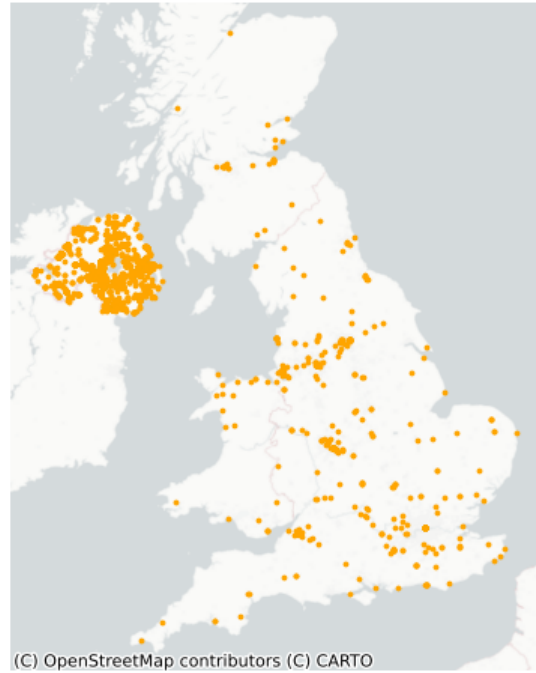
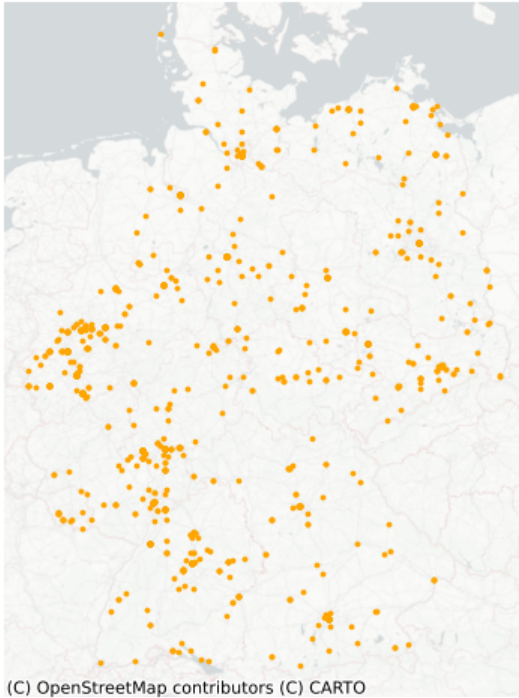


- Analyse phenomena across different geographical areas. For example, terrorism in Germany and in the UK.

```
# let's prepare the gdf for the UK
df_uk = attacks[attacks.country_txt == 'United Kingdom'].copy()
uk_crs = 'EPSG:27700'
gdf_uk = gpd.GeoDataFrame(df_uk, geometry=gpd.points_from_xy(df_uk.longitude, df_uk.latitude))
gdf_uk = gdf_uk.to_crs(uk_crs)
```

```
fig, axes = plt.subplots(1, 2, figsize=(10, 6))
gdfs = [gdf, gdf_uk]

for n, ax in enumerate(axes):
    gdf_tmp = gdfs[n]
    gdf_tmp.plot(ax=ax, color = 'orange', markersize = 3)
    ax.set_axis_off()
    ctx.add_basemap(ax, crs= gdf_tmp.crs.to_string(), source= source)
```

Exercise:

- Think about the plots above and how they could be improved.
- Copy and paste the code and execute the functions playing with the different parameters.
- Produce a neat map using the `GeoDataFrames` available in this notebook or the ones employed in the previous sessions, making use of the elements/parameters discussed here.
- Try out different tiles for the basemap to familiarise yourself with what's available.

3.2 Part II: Choropleth Mapping

```
import geoplot.crs as gcrs
import geoplot as gplt
```

Data

For this second part of the tutorial, we will use some data at the municipality level for Serbia. The data contains information regarding poverty level, average income, population and tourism. The data is taken from <https://data.stat.gov.rs/?caller=SDDB&languageCode=en-US> and can be associated to the polygons representing the administrative boundaries of the municipalities.

These boundaries can be found here https://data.humdata.org/dataset/geoboundaries-admin-boundaries-for-serbia?force_layout=desktop. While most of the data refers to 2023, the admin boundaries file traces back to 2017. Thus, it may contain obsolete information (few changes may occur).

Later on, we will go back to the terrorism dataset.

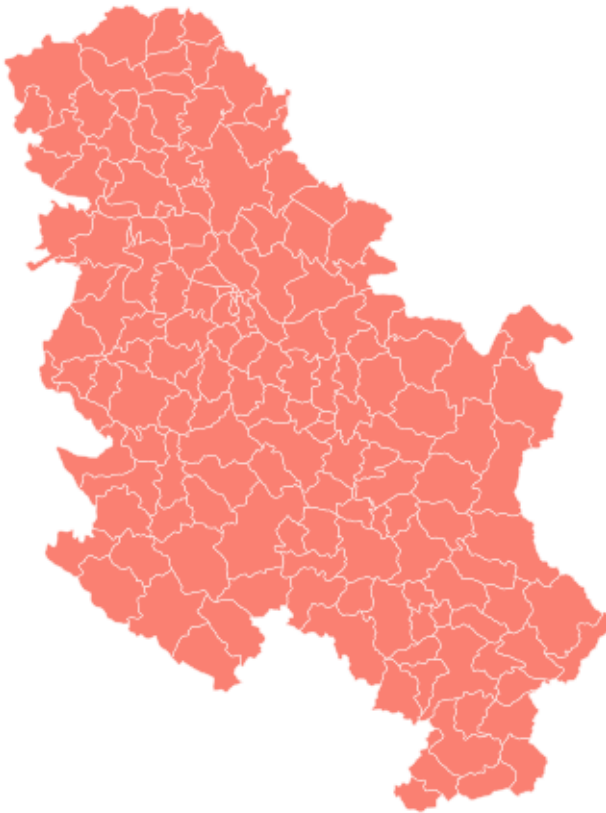
```
# This will be different on your computer and will depend on where
# you have downloaded the files
serbia_crs = 'EPSG:31277'
wgs = 'EPSG:4326'
serbia_admin = gpd.read_file('../data/serbia_admin.gpkg')
serbia_admin.set_index('townID', inplace = True, drop = True)
serbia_admin = serbia_admin.to_crs(serbia_crs)
```

Let's plot the GeoDataFrame following the last session's steps.

```
fig, ax = plt.subplots(1, 1, figsize=(8, 6))
serbia_admin.plot(ax = ax, color = 'salmon', linewidth = 0.3, edgecolor = 'white')
ax.set_axis_off()
title_parameters = {'fontsize':'16', 'fontname':'Times New Roman'}
ax.set_title("Serbian Municipalities", **title_parameters) #parameters as above
```

```
Text(0.5, 1.0, 'Serbian Municipalities')
```

Serbian Municipalities



Then we load the data and merge it into the **GeoDataFrame**, before getting rid of municipalities that do not have a corresponding shape/record in the GeoDataFrame (probably the result of changes in the national subdivisions).

```
data = pd.read_csv("../data/serbia_data.csv") #some slavic characters
data.drop('name_en', axis = 1, inplace = True)
serbia_admin = pd.merge(serbia_admin, data, left_on = "townID", right_on = "id")
serbia_admin = serbia_admin[serbia_admin.id.notna()]
serbia_admin['id'] = serbia_admin['id'].astype('int64')
serbia_admin.head()

#let's save the so-obtained gdf for later (encoding for dealing with slavic characters).
serbia_admin.to_file("../data/serbia_data.gpkg", encoding='utf-8', driver="GPKG", layer="serbia_admin")
```

C:\Users\gfilo\AppData\Local\miniconda3\envs\geovis\Lib\site-packages\pyogrio\raw.py:733: RuntimeWarning: ...

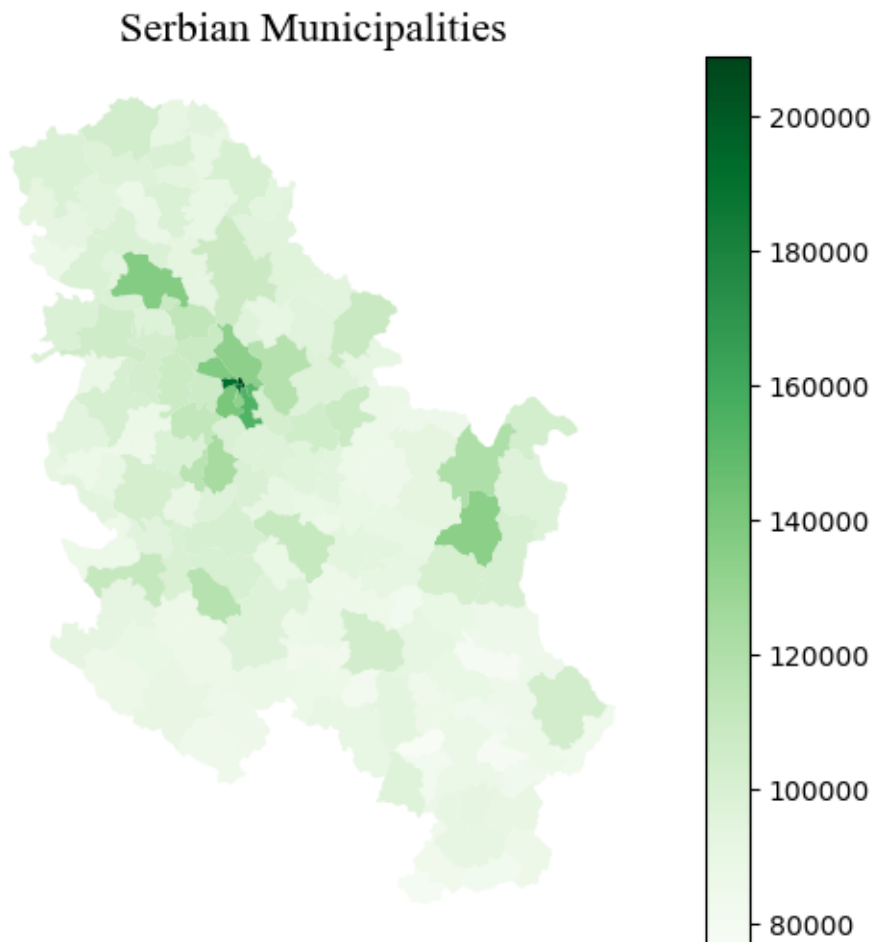
EPSG entry into the database.

```
ogr_write(
```

Creating a choropleth map is rather straightforward and can be done by using few other parameters. Reflect on what you see and whether the map below is informative.

```
fig, ax = plt.subplots(1, 1, figsize=(8, 6))
serbia_admin.plot(ax = ax, column = 'gross', linewidth = 0.3, cmap = 'Greens', legend = True)
ax.set_axis_off()
title_parameters = {'fontsize':'16', 'fontname':'Times New Roman'}
ax.set_title("Serbian Municipalities", **title_parameters) #parameters as above
```

```
Text(0.5, 1.0, 'Serbian Municipalities')
```



3.2.1 Choropleth Maps for Numerical Variables

We are essentially using the same approach employed for creating basic maps, the method `plot`, but we now need to pass arguments to some new parameters to specify which column is to be represented and how. As an optional argument, one can set `legend` to `True` and the resulting figure will include a colour bar.

- `column`: the name of the column representing the variable that we want to use to colour-code our shapes.
- `scheme`: the scheme used to colour the shapes based on the variable values.
- `cmap`: the colormap used to show variation.

3.2.1.1 Colormaps

Built-in colour maps can be found here https://matplotlib.org/stable/gallery/color/colormap_reference.html. However one can create new ones as follows from a list of colours:

```
from seaborn import palplot
from matplotlib.colors import LinearSegmentedColormap

colors = [(0.00, 0.00, 0.00,1), (0.248, 0.0271, 0.569, 1), (0.0311, 0.258, 0.646,1),
          (0.019, 0.415, 0.415,1), (0.025, 0.538, 0.269,1), (0.0315, 0.658, 0.103,1),
          (0.331, 0.761, 0.036,1),(0.768, 0.809, 0.039,1), (0.989, 0.862, 0.772,1),
          (1.0, 1.0, 1.0)]
palplot(colors)
```



```
kindlmann = LinearSegmentedColormap.from_list('kindlmann', colors)
```

or from colour names:

```
colors = ["white", "yellow", "red"]
palplot(colors)
```

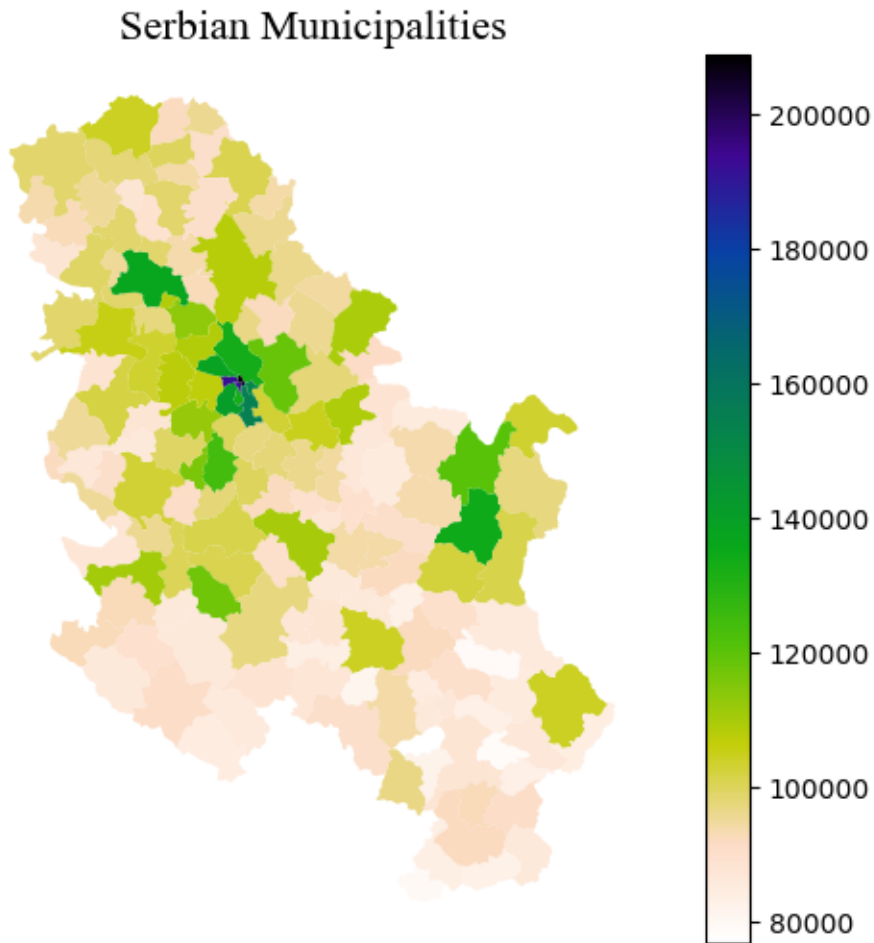


Let's try a new colormap and let's also set a number of classes to divide the data in, through the parameter `k`.

```
white_to_red = LinearSegmentedColormap.from_list("name", ["yellow","red"])
```

```
fig, ax = plt.subplots(1, 1, figsize=(8, 6))
serbia_admin.plot(ax = ax, column = 'gross', linewidth = 0.3, cmap = kindlmann.reversed(), 1
ax.set_axis_off()
title_parameters = {'fontsize':'16', 'fontname':'Times New Roman'}
ax.set_title("Serbian Municipalities", **title_parameters) #parameters as above
```

```
Text(0.5, 1.0, 'Serbian Municipalities')
```



With **GeoPandas**, when you use the `plot` method with `legend=True` the type of legend that appears depends on the data being visualized:

- **Continuous Data:** For columns with continuous data (like population estimates, temperatures, etc.), a colour bar is generated as the legend. This color bar represents a range of values with a gradient, indicating how data values correspond to colours on the map.
- **Categorical Data:** For columns with categorical data (like country names, types of land use, etc.), if you specify `legend=True`, **GeoPandas** will try to create a legend that categorizes these distinct values with different colours. However, creating legends for categorical data is not as straightforward as with continuous data and might require additional handling for a clear and informative legend (see below).

3.2.1.2 Scheme

It is important to keep in mind that choropleth maps strongly depend on the scheme that it is passed (or the default one) to classify the data in groups. The plot above only shows one municipality coloured in dark blue.

Look at the following plots and how three different classifiers produce different results for the same data.

Refer to <https://geopandas.org/en/stable/gallery/choropleths.html> and https://geographic-data.science/book/notebooks/05_choropleth.html for further details

```
# Function for plotting the map and the distribution of the value in bins

from mapclassify import Quantiles, EqualInterval, FisherJenks

def plot_scheme(gdf, column, scheme, figsize=(10, 6)):
    """
    Arguments
    -----
    gdf: GeoDataFrame
        The GeoDataFrame to plot
    column: str
        Variable name
    scheme: str
        Name of the classification scheme to use
    figsize: Tuple
        [Optional. Default = (10, 6)] Size of the figure to be created.

    """
    schemes = {'equal_interval': EqualInterval, 'quantiles': Quantiles, 'fisher_jenks': FisherJenks}
    classification = schemes[scheme](gdf[column], k=7)
    fig, (ax1, ax2) = plt.subplots(1, 2, figsize=figsize)
    # KDE
    sns.kdeplot(gdf[column], fill=True, color='purple', ax=ax1)
    sns.rugplot(gdf[column], alpha=0.5, color='purple', ax=ax1)
    for cut in classification.bins:
        ax1.axvline(cut, color='blue', linewidth=0.75)
    ax1.set_title('Value distribution')
    # Map
    p = gdf.plot(column=column, scheme=scheme, alpha=0.75, k=7, cmap='RdPu', ax=ax2, linewidth=0.75)
    ax2.axis('equal')
    ax2.set_axis_off()
```

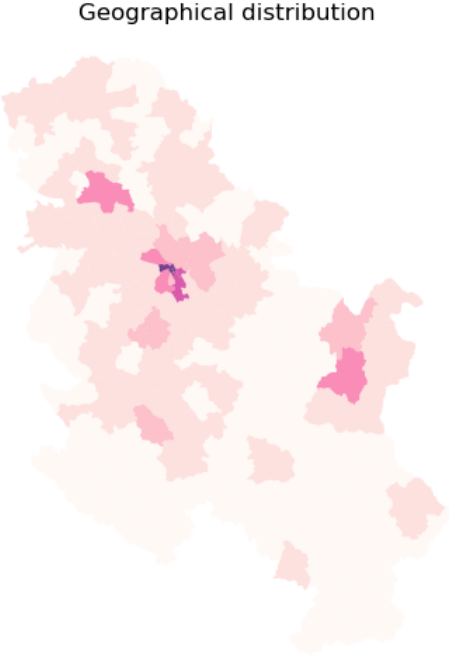
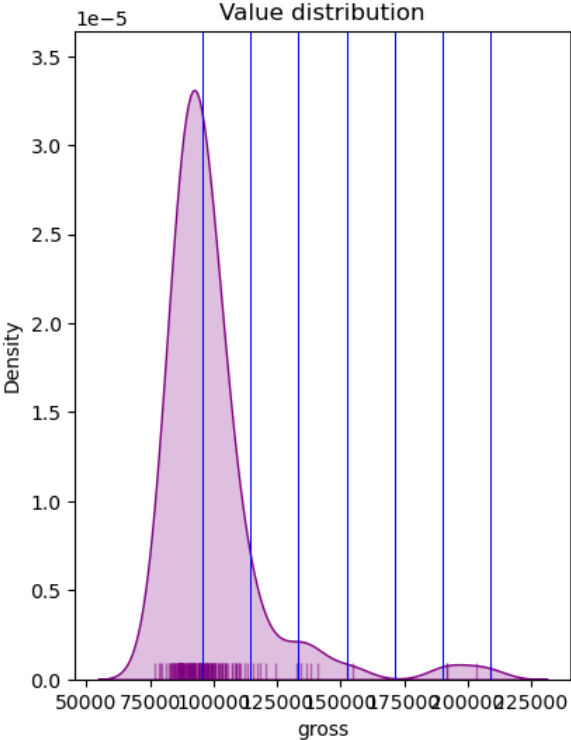
```
ax2.set_title('Geographical distribution')
fig.suptitle(scheme, size=25)
plt.show()
```

- The *Equal intervals* method splits the range of the distribution, the difference between the minimum and maximum value, into equally large segments and to assign a different colour to each of them according to a palette that reflects the fact that values are ordered.
- To obtain a more balanced classification, one can use the *Quantiles* scheme. This assigns the same amount of values to each bin: the entire series is laid out in order and break points are assigned in a way that leaves exactly the same amount of observations between each of them. This “observation-based” approach contrasts with the “value-based” method of equal intervals and, although it can obscure the magnitude of extreme values, it can be more informative in cases with skewed distributions.
- Amongst many other, the *Fisher Jenks* dynamically minimises the sum of the absolute deviations around class medians. The Fisher-Jenks algorithm is guaranteed to produce an optimal classification for a prespecified number of classes.

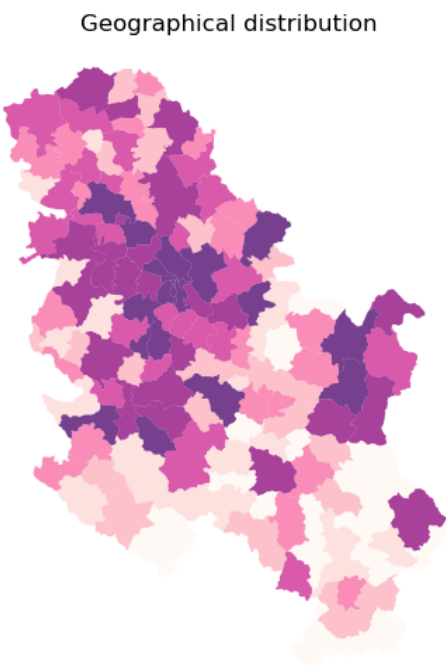
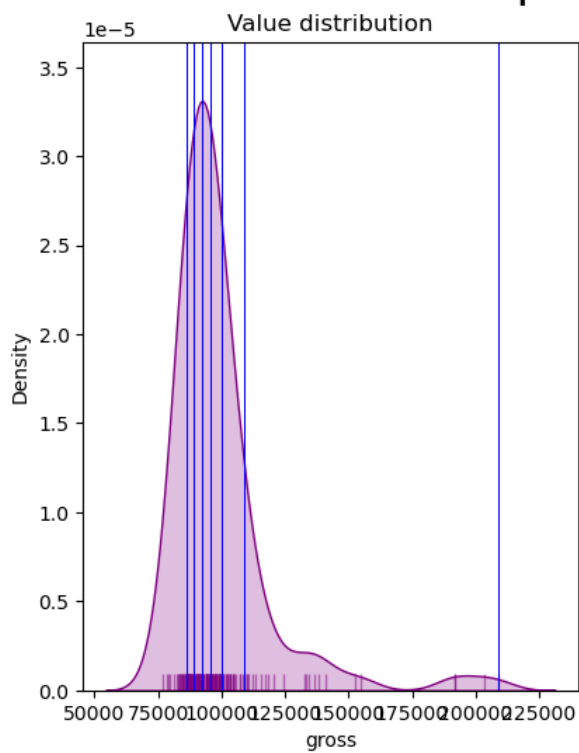
The only additional arguments to pass for producing a choropleth, therefore, are the actual variable we would like to classify and the number of segments we want to create, *k*. This is, in other words, the number of colours that will be plotted on the map so, although having several can give more detail, at some point the marginal value of an additional one is fairly limited, given the ability of the human brain to tell any differences.

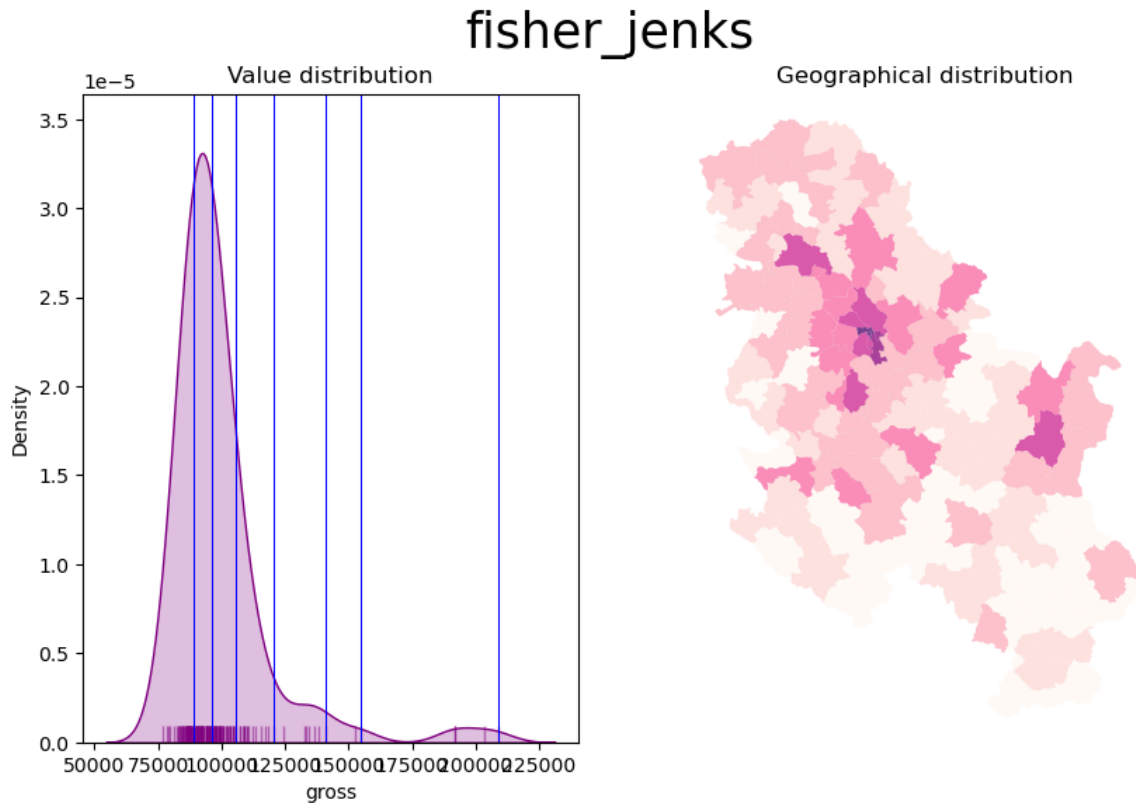
```
schemes = ['equal_interval', 'quantiles', 'fisher_jenks']
for scheme in schemes:
    plot_scheme(serbia_admin, 'gross', scheme)
```

equal_interval



quantiles





Also consider the [Modifiable Areal Unit Problem](#) and how the geographies of the administrative boundaries, in this case, may impact the visualisation.

For example, the most populated area is a municipality in the north that corresponds to the city of Novi Sad. Let's have a look at the data

```
serbia_admin[['name', 'pop', 'Province']].sort_values(by = 'pop', ascending = False).iloc[:10]
```

	name	pop	Province
48	Novi Sad	341625.0	Južno-Bački
24	Novi Beograd	186667.0	Grad Beograd
19	Čukarica	154854.0	Grad Beograd
3	Kragujevac	154290.0	Šumadijski
26	Palilula	148292.0	Grad Beograd
34	Zemun	143173.0	Grad Beograd
32	Voždovac	137315.0	Grad Beograd
35	Zvezdara	130225.0	Grad Beograd
39	Leskovac	123201.0	Jablanički

	name	pop	Province
120	Subotica	121250.0	Severno-Bački

In our dataset, the city of Novi Sad is categorised as a municipality by itself, because the administrative boundaries file is not updated. In reality, “since 2002, when the new statute of the city of Novi Sad came into effect, Novi Sad is divided into two city municipalities, Petrovaradin and Novi Sad. From 1989 until 2002, the name Municipality of Novi Sad meant the whole territory of the present-day city of Novi Sad.” (see: [wikipedia](#)).

On the contrary, Grad Beograd, that is Belgrade, is correctly split into different municipalities and its population, when visualised, is spread out across the different geometries of its municipalities. In other words, our map depends on the geometries of the areas and on how the data was collected. While it could be that these areas were indeed identified by population size in the first place, the point is that the fact that Novi Sad is not split into more areas, as Belgrade is, makes it stand out more clearly from the map (and to some extent a bit unfairly)

This may happen with different types of data, particularly with administrative boundaries and it is crucial to reflect on how Choropleth maps may be impacted. One can look for more granular data or consider to weight the continuous value with the extent of the area (i.e. obtaining density values).

3.2.1.3 An alternative to scheme: ColorMap Normalisation

The `mpl.colors.Normalize` function in `matplotlib` creates a normalization object, which adjusts data values into a range that is ideal for colour mapping in a colormap. This function is particularly beneficial in scenarios where precise control over the mapping of data values to colour representations is needed.

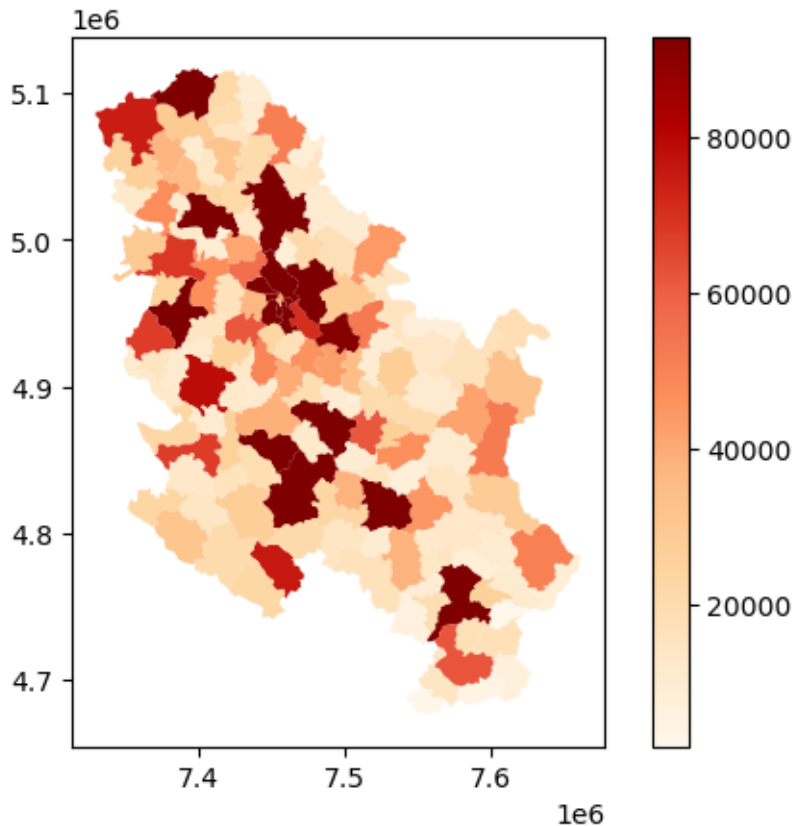
When employed in a plotting function, this normalization object ensures that the data values are scaled to fit a pre-defined range (for instance, `norm = mpl.colors.Normalize(vmin=0, vmax=40)`). Any values falling below 0 are mapped to the lowest colour on the colormap scale, while values exceeding 40 are mapped to the highest colour. This approach is especially useful when aiming to highlight differences within a specific data range; it can significantly enhance the visualization of data, by, for example, emphasizing temperature variations between 0°C and 40°C. This becomes crucial in instances where a few data points with high values (e.g., 50°C) might otherwise lead to a less informative visualization if not ‘normalized’ and treated as if they corresponded to 40° C values.

For our dataset, we can use as `vmax` the value corresponding to the 90th percentile.

```
serbia_admin['pop'].quantile(0.90)
```

```
np.float64(93014.0)
```

```
import matplotlib as mpl
fig, ax = plt.subplots(1, 1)
vmin = serbia_admin['pop'].min()
vmax = serbia_admin['pop'].quantile(0.90) #
norm = mpl.colors.Normalize(vmin=vmin, vmax=vmax)
serbia_admin.plot(ax = ax, column='pop', cmap='OrRd', legend=True, norm = norm)
```



Important:

When passing `norm` in the `plot` method, do not pass the arguments to the `scheme` parameter. For continuous variables, `norm` maps each value directly to a color, making discrete categorization redundant. In other words, it allows for a direct mapping of data values to the color map, eliminating the need for intermediary classification schemes. `norm` ensures a smooth gradient in the color map without artificially segmenting the data.

3.2.1.4 Customising the colorbar

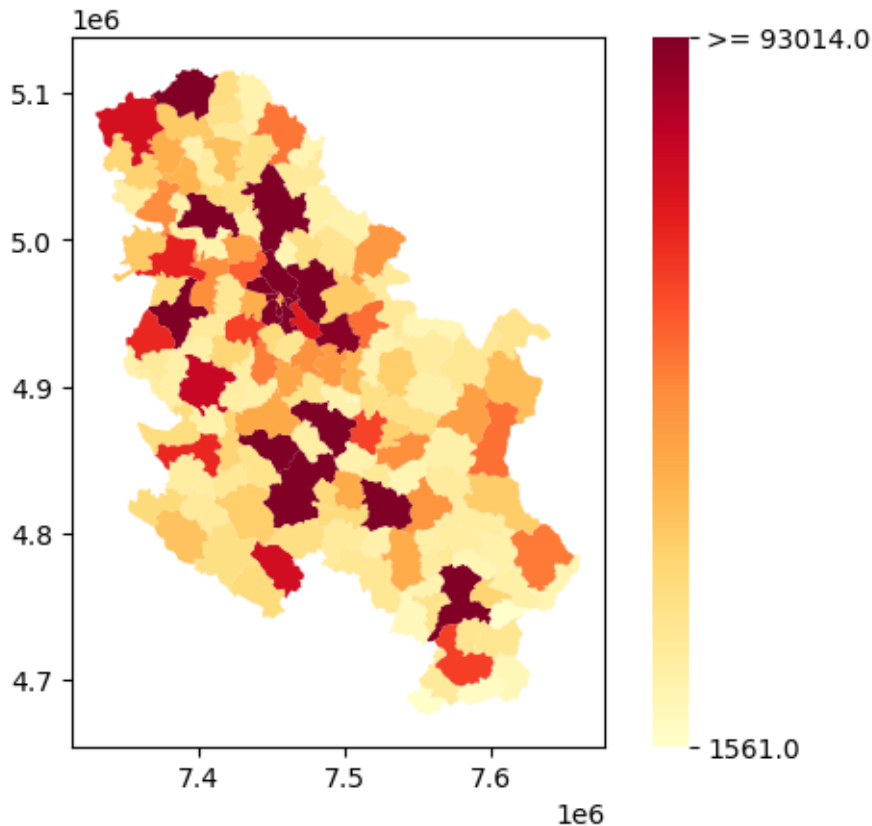
```
import matplotlib.cm as cm

fig, ax = plt.subplots(1, 1)
cmap = 'YlOrRd'
# we leave the legend out
serbia_admin.plot(column='pop', cmap=cmap, norm = norm, ax=ax)

# we add the colorbar separately passing the norm and cmap
cbar = fig.colorbar(cm.ScalarMappable(norm=norm, cmap=cmap), ax = ax)
cbar.outline.set_visible(False)

# updating ticks VALUES
ticks = [norm.vmin, norm.vmax]
cbar.set_ticks(ticks = ticks)

# updating ticks LABELS
cbar.ax.set_yticklabels([round(t,1) for t in ticks])
cbar.ax.set_yticklabels([round(t,1) if t < norm.vmax else ">= "+str(round(t,1)) for t in cba
```



Above, we removed the outline of the color bar. Then we set the tick values to the min and the max population values, based on our norm object. Then, for the vmax value's label we added a “>=” to remind us that other, higher values are displayed with the darkest color.

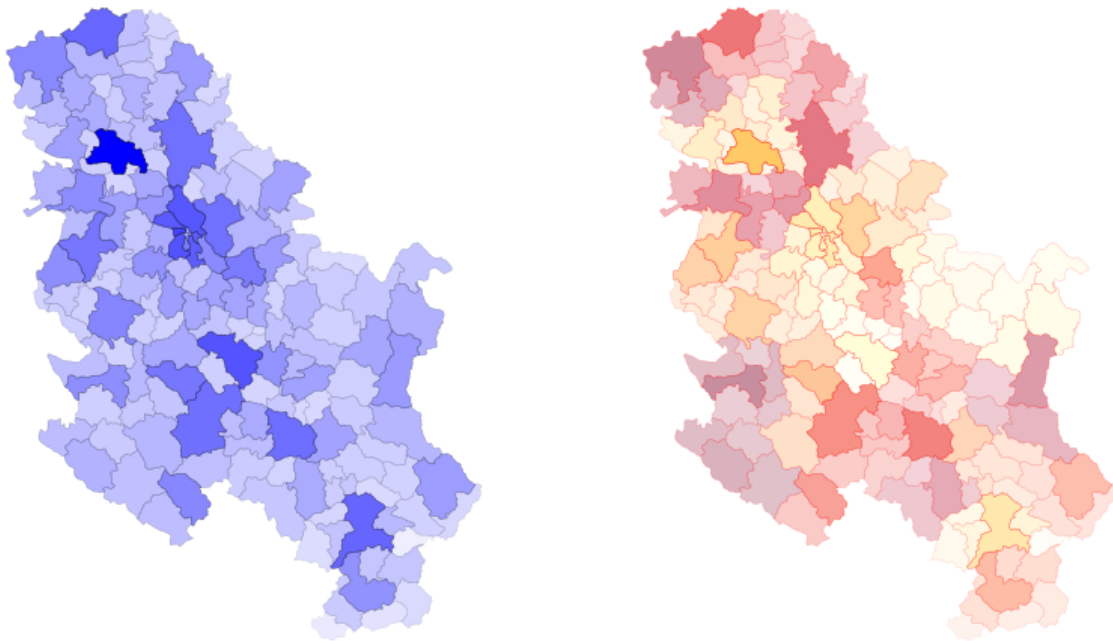
3.2.1.5 Varying alpha transparency based on an array

Finally, we can also convey variation in a continuous scale through transparency. `alpha` doesn't expect column names, so we cannot just pass the name of the column containing the variable. Instead, we have to create an array from 0.0 to 1.0 values. To so we can a) use normalisation methods, or b) rescale the original values within 0 to 1 based on the original min and max values.

For example, with square root normalization:

```
# 1. Create an alpha array based on a normalized value (e.g., population)
import numpy as np
pop_max = serbia_admin['pop'].max()
alpha = np.sqrt(serbia_admin['pop'] / pop_max)
```

```
# Plot with varying alpha values
fig, axes = plt.subplots(1, 2, figsize=(10, 6))
serbia_admin.plot(color = 'blue', ax=axes[0], alpha=alpha, edgecolor='black', linewidth = 0.3)
serbia_admin.plot(cmap = 'YlOrRd', ax=axes[1], alpha=alpha, edgecolor='red', linewidth = 0.3)
for ax in axes:
    ax.set_axis_off()
```



Important:

`matplotlib` would not be able to plot a color bar from variations in the alpha value since no column is passed directly. We would need, in this case, to build a color bar manually as demonstrated above.

3.2.2 Choropleth Maps for Categorical Variables

A choropleth for categorical variables assigns a different color to every potential value in the series based on certain colormaps (`cmap`). We don't need to specify a scheme in this case, but just to the categorical `column`. Using last's week `GeoDataFrame`, we can plot terrorist attacks in Germany, for example, by group.

```

gdf = gpd.read_file("../data/germany.gpkg").to_crs(germany_crs)
fig, ax = plt.subplots(1, 1, figsize=(8, 6))
gdf.plot(ax = ax, column = 'gname', legend = True)
ax.set_axis_off()
title_parameters = {'fontsize':'16', 'fontname':'Times New Roman'}
ax.set_title("Terrorist Attacks in Germany, by Group", **title_parameters) #parameters as ab

```

```

Text(0.5, 1.0, 'Terrorist Attacks in Germany, by Group')

```



The map above is what you would get from datasets that are not cleaned/manipulated directly or when there are too many categories in the selected column. First, let's get a slimmer slice of the gdf that only contains attacks that cause a number of fatalities and wounded higher than 10.

```
condition = (gdf.nkill + gdf.nwound) > 10
gdf_filtered = gdf[condition].copy()
```

Then, let's build a function that creates a random color map based on the number of categories. This creates random HUE-based colors:

```
# Generate random colormap
def rand_cmap(nlabels):
    """
    It generates a categorical random color map, given the number of classes

    Parameters
    -----
    nlabels: int
        The number of categories to be coloured.
    type_color: str {"soft", "bright"}
        It defines whether using bright or soft pastel colors, by limiting the RGB spectrum.

    Returns
    -----
    cmap: matplotlib.colors.LinearSegmentedColormap
        The color map.
    """
    # Generate color map for bright colors, based on hsv
    randHSVcolors = [(np.random.uniform(low=0.20, high=0.80),
                      np.random.uniform(low=0.20, high=0.80),
                      np.random.uniform(low=0.20, high=0.80)) for i in range(nlabels)]

    random_colormap = LinearSegmentedColormap.from_list('new_map', randHSVcolors, N=nlabels)

    return random_colormap
```

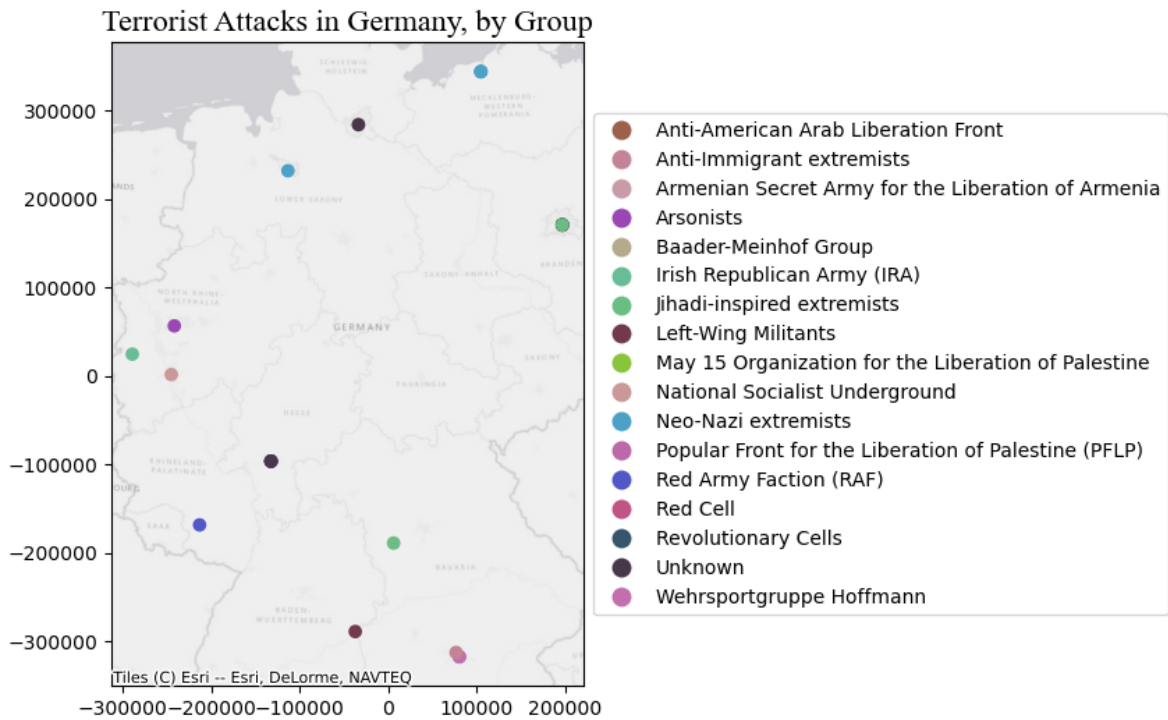
```
cmap = rand_cmap(len(gdf_filtered.gname.unique()))
cmap
```



We also place the legend on the centre left. This is done automatically, but the legend and its items can be manipulated directly. Legends in `matplotlib` are extremely complex to personalise. However, do have a look at https://matplotlib.org/stable/api/_as_gen/matplotlib.pyplot.legend.html#matplotlib.pyplot.legend for both automatic and explicit manipulation.

```
legend_kwds={"loc": "center left", "bbox_to_anchor": (1, 0.5)}
```

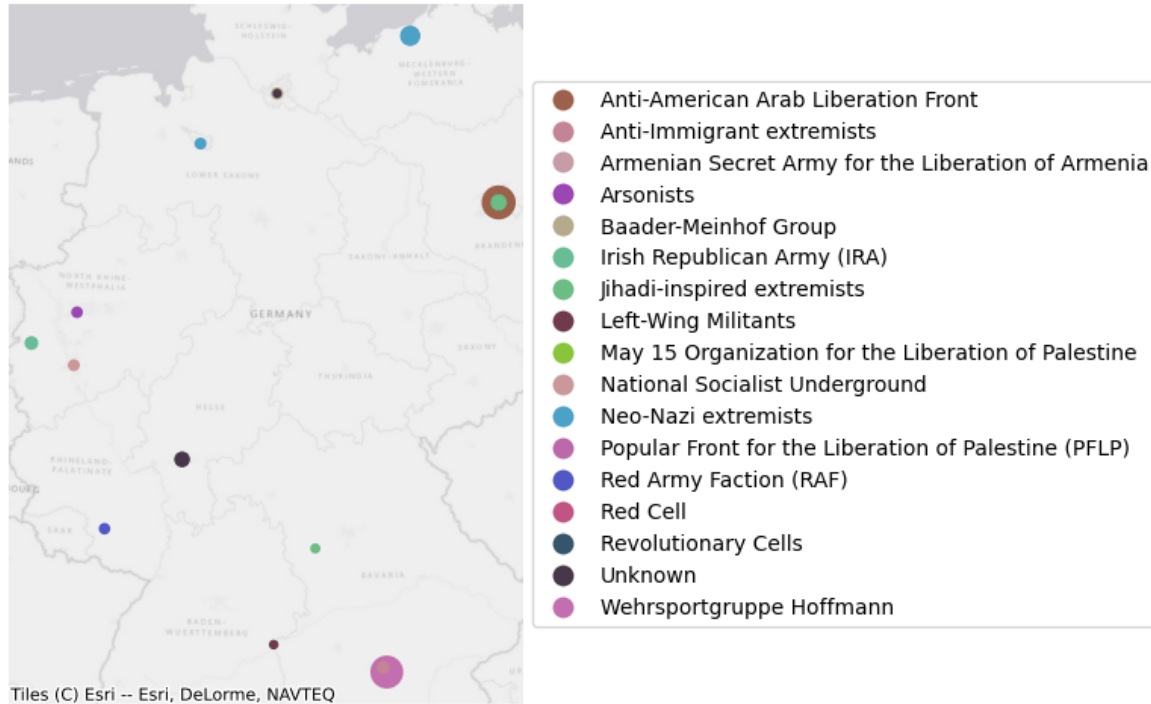
```
fig, ax = plt.subplots(1, 1, figsize=(8, 6))
gdf_filtered.plot(ax = ax, column = 'gname', legend = True, cmap = cmap, legend_kwds = legend_kwds)
title_parameters = {'fontsize': '16', 'fontname': 'Times New Roman'}
ax.set_title("Terrorist Attacks in Germany, by Group", **title_parameters) #parameters as above
ctx.add_basemap(ax, crs= gdf_filtered.crs.to_string(), source = ctx.providers.Esri.WorldGrayShadedReliefMap)
```



We can also convey the impact of the events through the `markersize`. This introduces the concept of *cartogram* (see below).

```
fig, ax = plt.subplots(1, 1, figsize=(8, 6))
gdf_filtered.plot(ax = ax, column = 'gname', markersize = 'nwound', legend = True, cmap = cm
ax.set_title("Terrorist Attacks in Germany, by Group", **title_parameters) #parameters as abo
ctx.add_basemap(ax, crs= gdf_filtered.crs.to_string(), source = ctx.providers.Esri.WorldGray
ax.set_axis_off()
```

Terrorist Attacks in Germany, by Group



3.3 Part III: Cartograms - Manipulating the Geometry size for showing the magnitude of a value

Cartograms are maps that represent the spatial distribution of a variable not by encoding it in a color palette but rather by modifying geographical objects. There are many algorithms to distort the shapes of geographical entities according to values, some of them are rather complex.

3.3.1 Polygons

You can obtain cartograms for Polygon with **geoplot**: see <https://residentmario.github.io/geoplot/>

geoplot functions pretty much work as plot

```
# this library needs the GeoDataFrame to be reverted to WGS
ax = gplt.cartogram(serbia_admin.to_crs(wgs), scale='pop', projection=gcrs.Mercator(), color
# see for projections that work with gplt https://scitools.org.uk/cartopy/docs/v0.15/crs/pro
gplt.polyplot(serbia_admin.to_crs(wgs), facecolor='lightgray', edgecolor='white', ax=ax, lw =
```



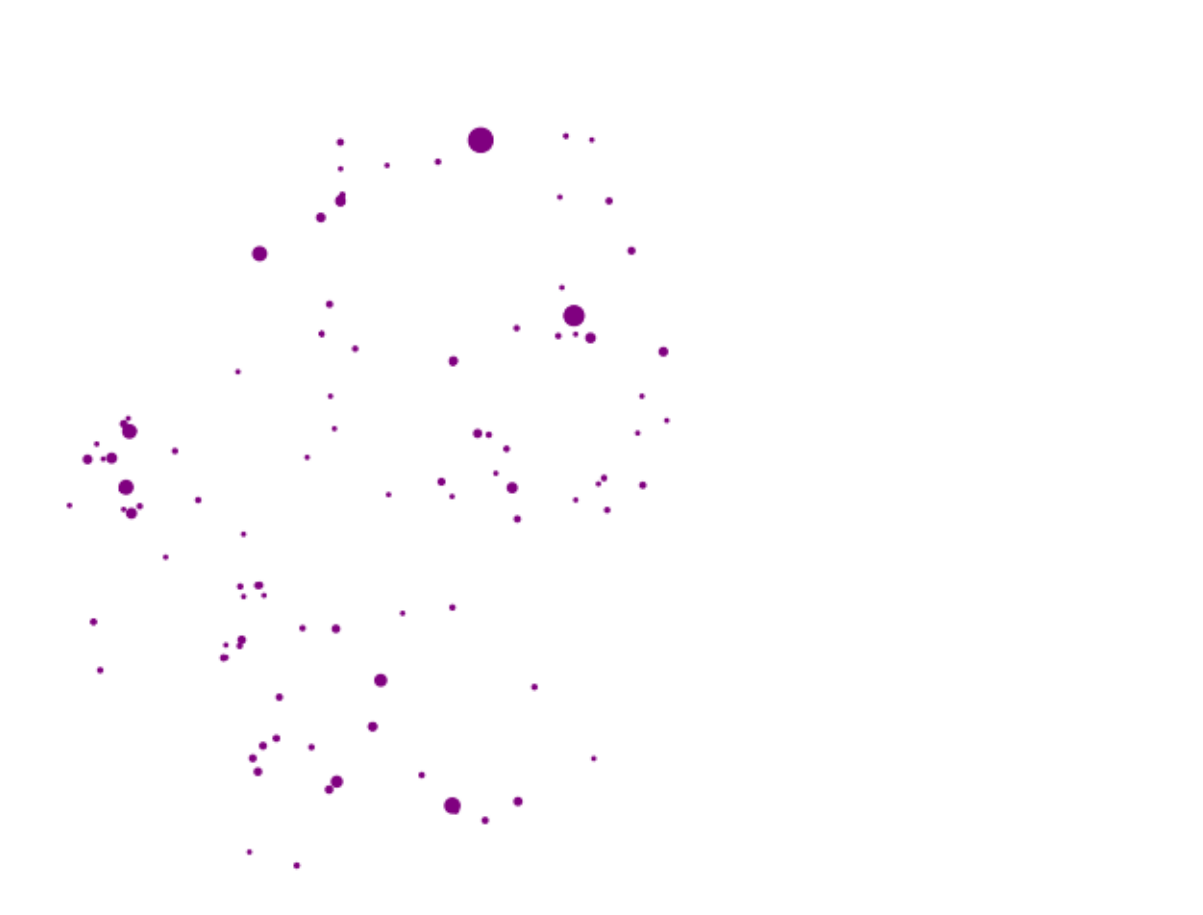
3.3.2 Points

For Point GeoDataFrames we can just go back to plot and pass a column name to markersize.

```
attacks = pd.read_csv("../data/GTD_2022.csv", low_memory = False)
country = 'Germany'
```

```
df = attacks[attacks.country_txt == country].copy()
wgs = 'EPSG:4326'
germany_crs = 'EPSG:4839'
gdf = gpd.GeoDataFrame(df, geometry=gpd.points_from_xy(df.longitude, df.latitude), crs = wgs)
gdf = gdf.to_crs(germany_crs)
```

```
fig, ax = plt.subplots(1, 1, figsize=(8, 6))
gdf.plot(ax = ax, markersize = 'nwound', color = 'purple', legend = True)
ax.set_axis_off()
```



One can also convert polygons into points by using their centroids, and then define the size of the dot proportionally to the value of the variable we want to display.

3.3.3 LineString

For `LineString` we pass the column name to `linewidth`.

Let's load a shapefile of lines. These lines represent frequency of train connections from/to train stations in the region of Liguria (Italy) to other stations within or outside the region. Each line refers to a connection between two specific stations, through a certain type of service and contains information about the frequency of that type of service. For example, the cities of Savona and Finale Ligure might be connected by 5 InterCity trains and 50 regional services. These services correspond to 2 different records.

```
trains_freq = gpd.read_file("../data/trains_fromLiguria.gpkg" )
trains_freq.crs
```

```
<Projected CRS: EPSG:3003>
Name: Monte Mario / Italy zone 1
Axis Info [cartesian]:
- X[east]: Easting (metre)
- Y[north]: Northing (metre)
Area of Use:
- name: Italy - onshore and offshore - west of 12°E.
- bounds: (5.93, 36.53, 12.0, 47.04)
Coordinate Operation:
- name: Italy zone 1
- method: Transverse Mercator
Datum: Monte Mario
- Ellipsoid: International 1924
- Prime Meridian: Greenwich
```

Let's check the type of services contained here.

```
trains_freq['train_type'].unique()
```

```
<StringArray>
['REG', 'IC', 'FB', 'ICN', 'U', 'EC/EN', 'AV']
Length: 7, dtype: str
```

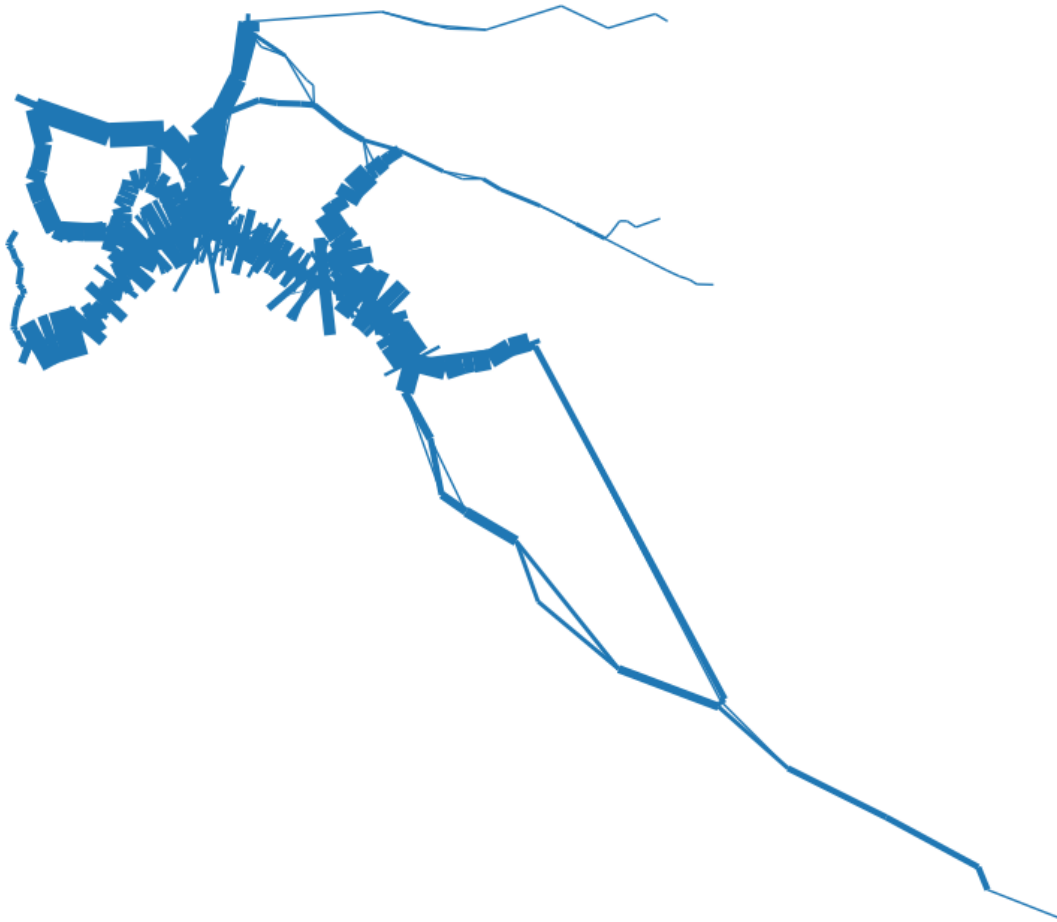
We have: - 'REG': regional trains. - 'IC': intercity trains. - 'FB': similar to IC, but slightly faster. - 'ICN': sleeper trains. - 'U': urban trains (Genoa). - 'EC/EN': international trains. - 'AV': High-speed trains.

Let's keep just regional, intercity, and high-speed trains.

```
to_keep = ['REG', 'IC', 'AV']  
trains_freq = trains_freq[trains_freq['train_type'].isin(to_keep)]
```

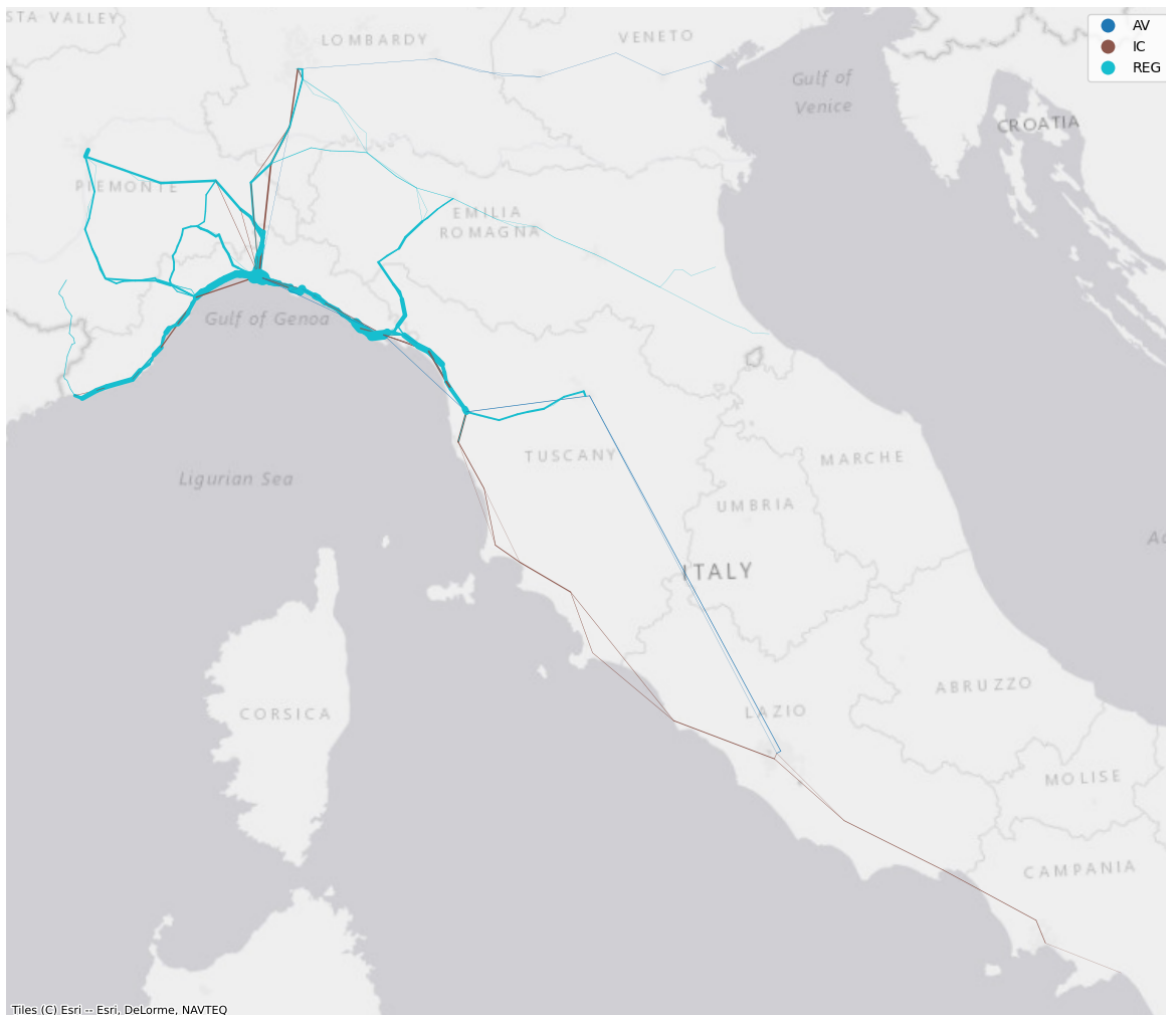
The usage of `linewidth` is a bit different from `markersize` for some reason. We have to pass an `array` of `N` values, where `N` is equal to the `GeoDataFrame` size. In other words, we have to pass the column we want to use to regulate the line width directly as a list/array. Specifying the column name is not enough.

```
fig, ax = plt.subplots(1, 1, figsize=(10, 15))  
trains_freq.plot(ax = ax, linewidth = trains_freq['freq'])  
ax.set_axis_off()
```



As you can see, the default arguments and simply passing the column values do not produce pretty results. The first thing to look at is the values that are passed to `linewidth`. In some cases, the min and max values, as well as their distribution, are not ideal for visually conveying the magnitude of the variable attached to the geometry. One option is to use a multiplier factor (see below), or to rescale the values from 0 to 1, for example, and then, again, if necessary use a multiplier.

```
fig, ax = plt.subplots(1, 1, figsize=(15, 20))
lw = trains_freq['freq'] * 0.15
trains_freq.plot(ax = ax, linewidth = lw, capstyle = 'round', joinstyle = 'round', column =
ctx.add_basemap(ax, crs= trains_freq.crs.to_string(), source = ctx.providers.Esri.WorldGrayC
ax.set_axis_off()
```



While this looks a bit better, this visualisation is not ideal because the frequencies are not snapped to the actual railway network. The lines represent, instead, connection between train stops and therefore their coordinates only include the ones corresponding to the stations where the different services call at. One can devise approaches to:

- Assigning the frequencies, or any other value, to the corresponding infrastructure's section. For example, the railway section between two stations could be associated with a value representing the total number of regional/local services travelling along it.
- Smoothing the lines representing the services by adding further coordinates along the line.

Both these processes go beyond the scopes of this lab and require several considerations depending on the data, the scale, and what information one wants to displays.

Exercise:

Today we've seen how to exploit `matplotlib` to plot `GeoDataFrame` layers. Go through the notebook again if you feel that there's something you need to review. You are not expected to remember each step/method/parameter. Rather, this notebook should be used as a reference for producing maps in Python. Do keep in mind that most of the maps above have been produced with just a bunch of rows, so each of them can be improved and embellished with some more effort.

Now, if you are not overwhelmed, have a look at the very last map and produce some nice visualisation using the same data. You can further improve its clarity, add a legend that refers to the line width, visualise only a certain type of services, or add information/context, for example. In the folder `\data` you can also find a `.gpkg` file containing all the train stations in Italy, should you need that.

3.3.3.1 Saving figures (check [here for details](#))

```
fig.savefig("fig1.pdf", dpi='figure', format="pdf", bbox_inches = 'tight')
```

4 Interactive Maps

Gabriele Filomena has readapted parts of this [notebook](#) to prepare this notebook. Copyright (c) Henrikki Tenkanen, Vuokko Heikinheimo, Håvard Wallin Aagesen, Christoph Fink, Kamyar Hasanzadeh.

The **Lecture slides** can be found [here](#).

This **lab's** notebook can be downloaded from [here](#).

[Online maps](#) have been interactive for a long time: virtually all online maps allow to zoom in and out, to pan the map extent, and to select map features, or otherwise query information about them. Interactive content in web pages, such as online maps, are typically implemented using [JavaScript/ECMAScript](#), a scripting language originally targeted at web pages, primarily, but used for many other applications.

In the open source realm, there exists a number of different *JavaScript* libraries for interactive web cartography, including [Leaflet](#) and [OpenLayers](#). We will not write a single line of *JavaScript*; instead, we will take advantage of the [Folium](#) Python package: it helps create interactive *Leaflet* maps from data stored in `geopandas.GeoDataFrame`. Folium is a library that bridges the capabilities of data manipulation in Python with the interactive mapping strengths of Leaflet, a leading open-source JavaScript library for creating dynamic, interactive maps in the browser. Let's explore both Folium and Leaflet in more detail to understand their roles and relationship.

Leaflet is at the heart of Folium's mapping capabilities. It's an open-source JavaScript library used extensively in web mapping. Known for its simplicity, performance, and ease of use, Leaflet has become a popular choice among developers for embedding maps in web applications. It allows users to create maps with various interactive features like zooming, panning, markers, popups, and different layers.

Folium builds on Leaflet's foundation, bringing its interactive mapping capabilities into the Python environment. This integration is significant for several reasons:

- With Folium, Python users can use libraries like Pandas and directly create interactive Leaflet maps from within Python. This seamless integration means data can be manipulated, analyzed, and visualized using Python, without the need for extensive JavaScript knowledge.
- Folium abstracts the complexity of Leaflet, allowing users to create sophisticated maps using simple Python commands.

- Folium inherits Leaflet's interactivity, enabling the creation of feature-rich maps with various interactive elements such as markers, layers, and choropleths. These maps can be customized and embedded in web applications or Jupyter notebooks, making them highly versatile for data visualization tasks.
- Folium can render maps with different tilesets (like OpenStreetMap, Stamen Terrain), overlay data in various formats, and even incorporate advanced features like heatmaps or GeoJSON layers.

Folium acts as a bridge between Python and Leaflet, enabling Python users to incorporate and develop web maps within their environment. It leverages Leaflet's core functionalities and extends them into the Python ecosystem, making geospatial data visualization both powerful and accessible within Python workflows.

Find more information about the capabilities of the *Folium* package on its official web pages: - [Documentation](#) - [Example gallery](#) - [Quickstart tutorial](#)

```
import folium
import pandas as pd
import geopandas as gpd
```

4.1 Simple interactive web maps with Folium

We will start by creating a simple interactive web map that contains nothing but a base map. This is so that we can get accustomed to how Folium's syntax works, and which steps we have to take. We create a `folium.Map` object, and specify centred around which `location` and at which initial zoom level (~0-20) a map shall be displayed. By setting `control_scale` to `True`, we make Folium display a scale bar.

```
location = (39.6508, 66.9654) # Samarkand, Uzbekistan
map = folium.Map(location=location, zoom_start=13, control_scale=True)
map
```

```
<folium.folium.Map at 0x22aa2e43e10>
```

4.1.0.1 Save the resulting map

To save this map to an HTML file that can be opened in any web browser, use `folium.Map.save()`:

```
map.save("base-map.html")
```

4.1.1 Basemap API

Folium, Leaflet.js and `contextily`, the library that we used for adding basemaps to our static maps in the 2nd session, rely on built-in tilesets borrowed from the `xyzservices` package. The XYZ protocol exposes maps as images for portions of the Earth, which we refer to as ‘tiles.’ The term ‘XYZ’ originates from the coordinates used to locate a specific tile. Think of the entire planet divided into squares, each identified by a unique combination of X and Y numbers. Now, add a third component (Z) for the zoom level: lower values use fewer tiles to cover the world, while higher zoom levels (higher Z) provide progressively smaller areas with more detail. Most XYZ APIs serve tiles directly over HTTP, enabling us to access them from the browser. In a nutshell, XYZ Tiles is a method of expressing map tile mosaics, indexed by (x,y) offsets and zoom levels (z).

```
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xyz
```

```
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```

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```

```

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  'attribution': 'Map data: (C) dl-de/by-2-0',
  'variant': 'web',
  'name': 'TopPlusOpen.Color'},
  'Grey': {'url': 'http://sgx.geodatenzentrum.de/wmts_topplus_open/tile/1.0.0/{variant}/default',
  'max_zoom': 18,
  'html_attribution': 'Map data: &copy; <a href="http://www.govdata.de/dl-de/by-2-0">dl-de/by-2-0</a>',
  'attribution': 'Map data: (C) dl-de/by-2-0',
  'variant': 'web_grau',
  'name': 'TopPlusOpen.Grey'}},
'Gaode': {'Normal': {'url': 'http://webrd01.is.autonavi.com/appmaptile?lang=zh_cn&size=1&scale={s}&type={t}&y={y}&x={x}&z={z}',
  'max_zoom': 19,
  'attribution': '&copy; <a href="http://www.gaode.com/">Gaode.com</a>',
  'html_attribution': '&copy; <a href="http://www.gaode.com/">Gaode.com</a>',
  'name': 'Gaode.Normal'},
  'Satellite': {'url': 'http://webst01.is.autonavi.com/appmaptile?style=6&x={x}&y={y}&z={z}',
  'max_zoom': 19,
  'attribution': '&copy; <a href="http://www.gaode.com/">Gaode.com</a>',
  'html_attribution': '&copy; <a href="http://www.gaode.com/">Gaode.com</a>',
  'name': 'Gaode.Satellite'}},
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  'max_zoom': 15,
  'attribution': 'Map tiles by <a href="https://labs.strava.com/heatmap">Strava 2021</a>',
  'html_attribution': 'Map tiles by <a href="https://labs.strava.com/heatmap">Strava 2021</a>',
  'name': 'Strava.All'},
  'Ride': {'url': 'https://heatmap-external-a.strava.com/tiles/ride/hot/{z}/{x}/{y}.png',
  'max_zoom': 15,
  'attribution': 'Map tiles by <a href="https://labs.strava.com/heatmap">Strava 2021</a>',
  'html_attribution': 'Map tiles by <a href="https://labs.strava.com/heatmap">Strava 2021</a>',
  'name': 'Strava.Ride'},
  'Run': {'url': 'https://heatmap-external-a.strava.com/tiles/run/bluered/{z}/{x}/{y}.png',
  'max_zoom': 15,
  'attribution': 'Map tiles by <a href="https://labs.strava.com/heatmap">Strava 2021</a>',
  'html_attribution': 'Map tiles by <a href="https://labs.strava.com/heatmap">Strava 2021</a>',
  'name': 'Strava.Run'}},

```

```

'max_zoom': 15,
'attribution': 'Map tiles by <a href="https://labs.strava.com/heatmap">Strava 2021</a>',
'html_attribution': 'Map tiles by <a href="https://labs.strava.com/heatmap">Strava 2021</a>',
'name': 'Strava.Run'},
'Water': {'url': 'https://heatmap-external-a.strava.com/tiles/water/blue/{z}/{x}/{y}.png',
'max_zoom': 15,
'attribution': 'Map tiles by <a href="https://labs.strava.com/heatmap">Strava 2021</a>',
'html_attribution': 'Map tiles by <a href="https://labs.strava.com/heatmap">Strava 2021</a>',
'name': 'Strava.Water'},
'Winter': {'url': 'https://heatmap-external-a.strava.com/tiles/winter/hot/{z}/{x}/{y}.png',
'max_zoom': 15,
'attribution': 'Map tiles by <a href="https://labs.strava.com/heatmap">Strava 2021</a>',
'html_attribution': 'Map tiles by <a href="https://labs.strava.com/heatmap">Strava 2021</a>',
'name': 'Strava.Winter'}},
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'html_attribution': 'Contains OS data &copy Crown copyright and database right 2025',
'attribution': 'Contains OS data (C) Crown copyright and database right 2025',
'key': '<insert your valid OS MapsAPI Key. Get a free key here - https://osdatahub.os.uk/>',
'min_zoom': 7,
'max_zoom': 16,
'max_zoom_premium': 20,
'bounds': [[49.766807, -9.496386], [61.465189, 3.634745]],
'name': 'OrdnanceSurvey.Road'},
'Road_27700': {'url': 'https://api.os.uk/maps/raster/v1/zxy/Road_27700/{z}/{x}/{y}.png?key=',
'html_attribution': 'Contains OS data &copy Crown copyright and database right 2025',
'attribution': 'Contains OS data (C) Crown copyright and database right 2025',
'key': '<insert your valid OS MapsAPI Key. Get a free key here - https://osdatahub.os.uk/>',
'crs': 'EPSG:27700',
'min_zoom': 0,
'max_zoom': 9,
'max_zoom_premium': 13,
'bounds': [[0, 0], [700000, 1300000]],
'name': 'OrdnanceSurvey.Road_27700'},
'Outdoor': {'url': 'https://api.os.uk/maps/raster/v1/zxy/Outdoor_3857/{z}/{x}/{y}.png?key=',
'html_attribution': 'Contains OS data &copy Crown copyright and database right 2025',
'attribution': 'Contains OS data (C) Crown copyright and database right 2025',
'key': '<insert your valid OS MapsAPI Key. Get a free key here - https://osdatahub.os.uk/>',
'min_zoom': 7,
'max_zoom': 16,
'max_zoom_premium': 20,
'bounds': [[49.766807, -9.496386], [61.465189, 3.634745]],
'name': 'OrdnanceSurvey.Outdoor'},
'Outdoor_27700': {'url': 'https://api.os.uk/maps/raster/v1/zxy/Outdoor_27700/{z}/{x}/{y}.p

```

```

'html_attribution': 'Contains OS data &copy Crown copyright and database right 2025',
'attribution': 'Contains OS data (C) Crown copyright and database right 2025',
'key': '<insert your valid OS MapsAPI Key. Get a free key here - https://osdatahub.os.uk/',
'crs': 'EPSG:27700',
'min_zoom': 0,
'max_zoom': 9,
'max_zoom_premium': 13,
'bounds': [[0, 0], [700000, 1300000]],
'name': 'OrdnanceSurvey.Outdoor_27700'},
'Light': {'url': 'https://api.os.uk/maps/raster/v1/zxy/Light_3857/{z}/{x}/{y}.png?key={key}',
'html_attribution': 'Contains OS data &copy Crown copyright and database right 2025',
'attribution': 'Contains OS data (C) Crown copyright and database right 2025',
'key': '<insert your valid OS MapsAPI Key. Get a free key here - https://osdatahub.os.uk/',
'min_zoom': 7,
'max_zoom': 16,
'max_zoom_premium': 20,
'bounds': [[49.766807, -9.496386], [61.465189, 3.634745]],
'name': 'OrdnanceSurvey.Light'},
'Light_27700': {'url': 'https://api.os.uk/maps/raster/v1/zxy/Light_27700/{z}/{x}/{y}.png?key={key}',
'html_attribution': 'Contains OS data &copy Crown copyright and database right 2025',
'attribution': 'Contains OS data (C) Crown copyright and database right 2025',
'key': '<insert your valid OS MapsAPI Key. Get a free key here - https://osdatahub.os.uk/',
'crs': 'EPSG:27700',
'min_zoom': 0,
'max_zoom': 9,
'max_zoom_premium': 13,
'bounds': [[0, 0], [700000, 1300000]],
'name': 'OrdnanceSurvey.Light_27700'},
'Leisure_27700': {'url': 'https://api.os.uk/maps/raster/v1/zxy/Leisure_27700/{z}/{x}/{y}.png?key={key}',
'html_attribution': 'Contains OS data &copy Crown copyright and database right 2025',
'attribution': 'Contains OS data (C) Crown copyright and database right 2025',
'key': '<insert your valid OS MapsAPI Key. Get a free key here - https://osdatahub.os.uk/',
'crs': 'EPSG:27700',
'min_zoom': 0,
'max_zoom': 5,
'max_zoom_premium': 9,
'bounds': [[0, 0], [700000, 1300000]],
'name': 'OrdnanceSurvey.Leisure_27700'}},
'UN': {'ClearMap': {'url': 'https://geoservices.un.org/arcgis/rest/services/ClearMap_WebTopoMap/MapServer',
'name': 'UN.ClearMap',
'html_attribution': '&copy; <a href="https://www.un.org/geospatial/mapsgeo/webservices">United Nations',
'attribution': 'United Nations'}}}

```

In Folium, you can pass one of the custom `xyzservices.TileProvider` listed above or a Leaflet-style URL to the `tiles` parameter: `http://{s}.yourtiles.com/{z}/{x}-{y}.png`. You can find a list of free tile providers here: <http://leaflet-extras.github.io/leaflet-providers/preview/>. Be sure to check their terms and conditions and to provide attribution (see below)

```
map = folium.Map(location=location, zoom_start=13, tiles="CartoDB.DarkMatterNoLabels") # folium.Map
map
```

```
<folium.folium.Map at 0x22aa3359950>
```

Or we can point to a custom *tileset URL*:

```
map = folium.Map(location=location, zoom_start=12, tiles="https://mt1.google.com/vt/lyrs=r&x=0&y=0")
map
```

```
<folium.folium.Map at 0x22aa335bad0>
```

Also, the *Mapbox Static Tiles API* serves raster tiles generated from Mapbox Studio styles. Raster tiles can be used in traditional web mapping libraries like Mapbox.js, Leaflet, OpenLayers, and others to create interactive slippy maps. The Static Tiles API is well-suited for maps with limited interactivity or use on devices that do not support WebGL.

Usage of the Mapbox APIs is governed by the Mapbox Terms of Service. Please visit <https://www.mapbox.com/legal/tos/> for more information.

```
# Your Mapbox access token
mapbox_access_token = 'YOUR_MAPBOX_ACCESS_TOKEN_HERE'

# Mapbox tile URL
tile_url = f'https://api.mapbox.com/styles/v1/mapbox/light-v9/tiles/{z}/{x}/{y}?access_token={mapbox_access_token}'

# Create a folium map
map = folium.Map(location=[53.406872, -2.973286], zoom_start=13, tiles = tile_url, attr='Mapbox')
map
```

```
<folium.folium.Map at 0x22aa2f94210>
```

4.1.2 Add a point marker

To add a single marker to a *Folium* map, create a `folium.Marker`. Supply a `folium.Icon` as the argument of the parameter `icon` to influence how the marker is styled, and set `tooltip` to display a text when the mouse pointer hovers over it.

```
center = folium.Marker(location=location, tooltip="City Centroid", icon=folium.Icon(color="green"))
center.add_to(map)
map
```

<folium.folium.Map at 0x22aa2f94210>

```
# Similarly..
popup = ["Tom", "Kendall", "Sean", "Zachary", "Karla"]

# Longitude and Latitude coordinates
lng = [-3.2031323, -0.2416811, -3.4924087, -4.3725404, -2.6607571]
lat = [53.4118332, 51.5285582, 55.940874, 55.8553807, 51.4684681]

# Creating a folium map
map = folium.Map(location=[53.4118332, -3.2031323], zoom_start=5, tiles = "NASAGIBS.ViirsEarth-NetGlobal")

# Adding markers with popups
for i in range(len(popup)):
    folium.Marker([lat[i], lng[i]], popup=popup[i]).add_to(map)

map
```

<folium.folium.Map at 0x22aa356ba50>

4.1.3 Add a layer of points

Folium also supports to add entire layers, for instance, as `geopandas.GeoDataFrames`. Folium implements *Leaflet's geoJSON layers* in its `folium.features.GeoJson` class. We can initialise such a class (and layer) with a `GeoDataFrame`, and add it to a map. In the example below, we work again with the `GeoDataFrame` of terrorist attacks in Germany.

```
wgs = 'EPSG:4326'
gdf = gpd.read_file("../data/germany.gpkg")
gdf = gdf.set_crs(wgs)
```

Let's explore the data around Berlin

```
berlin = (52.5200, 13.4050)
map = folium.Map(location= berlin, zoom_start=10)

attacks_layer = folium.features.GeoJson(gdf, name="Terrorist attacks")
attacks_layer.add_to(map)
map
```

```
<folium.folium.Map at 0x22aa35de0d0>
```

We can also add a pop-up window to our map which would show some details when clicking on the location of the attacks:

```
popup = folium.GeoJsonPopup(fields=["year", "gname", "nwound"], aliases =["Year", "Terrorist attacks", "Number of wounds"],
                             style="background-color: yellow;")
map = folium.Map(location= berlin, zoom_start=10)
attacks_layer = folium.features.GeoJson(gdf[gdf.nwound.notna()], name="Terrorist attacks", popup=popup)
attacks_layer.add_to(map)

map
```

```
<folium.folium.Map at 0x22aa2f34c10>
```

4.1.4 Add a polygon layer

We can do the same with polygons. Let's load municipalities in Serbia.

```
serbia = gpd.read_file("../data/serbia_data.gpkg")
serbia.head()
```

	name	id	poverty	name_sr	arr_dom	arr_int	pop	male	female	net	g
0	Arandelovac	1	23.3	Arandelovac	4042.0	958.0	39456.0	19166.0	20290.0	72348.0	1
1	Batočina	2	36.1	Batočina	11.0	0.0	10085.0	4995.0	5090.0	63243.0	8
2	Knić	3	40.1	Knić	753.0	55.0	12533.0	6312.0	6221.0	65463.0	9
3	Kragujevac	4	23.8	Kragujevac	2244.0	2417.0	154290.0	74515.0	79775.0	79415.0	1
4	Lapovo	5	23.9	Lapovo	39.0	36.0	6785.0	3294.0	3491.0	65350.0	9

Important:

We need to project the GeoDataFrame back to the WGS crs; web-maps, because of their interactivity and because users can pan to different areas, work with the WGS projection and cannot handle planar crs.

```
wgs = 'EPSG:4326'  
serbia.to_crs(wgs, inplace = True)
```

We will use the `folium.Choropleth` to display arrivals of tourists in July 2023, per municipality. Choropleth maps are more than simply geometries in Folium, which could be displayed as a `folium.features.GeoJson` layer. Rather, this class takes care of categorising data, adding a legend, and a few more small tasks to quickly create beautiful thematic maps.

The class expects an input data set that has an explicit, `str`-type, index column, as it treats the geospatial input and the thematic input as separate data sets that need to be joined (see also, below, how we specify both `geo_data` and `data`). A good approach to create such a column is to copy the data frame's index into a new column, for instance `id_str`.

```
serbia["id_str"] = serbia.index.astype(str)
```

Now we can create the polygon choropleth layer, and add it to a map object. Due to the slightly complex architecture of *Folium*, we have to supply a number of parameters: - `geo_data` and `data`, the geospatial and thematic input data sets, respectively. Can be the same `GeoDataFrame`. - `columns`: a tuple of the names of relevant columns in `data`: a unique index column, and the column containing thematic data - `key_on`: which column in `geo_data` to use for joining `data` (this is basically identical to `columns`, except it's only the first value)

```
# getting the location  
centroid = serbia.unary_union.centroid  
location = (centroid.coords[0][1], centroid.coords[0][0])
```

```
C:\Users\gfilo\AppData\Local\Temp\ipykernel_16604\1078060503.py:2: DeprecationWarning: The '  
centroid = serbia.unary_union.centroid
```

Let's look at the number of arrivals of tourists across the municipalities. - `arr_dom` indicates the number of domestic tourists (from other areas in Serbia) in July 2023. - `arr_int` indicates the number of international tourists in July 2023.

```
serbia[['arr_dom', 'arr_int', 'Province', 'name']].head()
```

	arr_dom	arr_int	Province	name
0	4042.0	958.0	Šumadijski	Arandelovac
1	11.0	0.0	Šumadijski	Batočina
2	753.0	55.0	Šumadijski	Knić
3	2244.0	2417.0	Šumadijski	Kragujevac
4	39.0	36.0	Šumadijski	Lapovo

```
map = folium.Map(location=location, zoom_start=7)
arrivals_choro = folium.Choropleth(geo_data=serbia, data=serbia, columns=("id_str", "arr_int",
                                "arr_dom", "Province", "name"),
                                fill_color="Oranges", line_weight=0.2, line_color='black',
                                legend_name="Nr of International Tourists, July 2023",
                                highlight = True) ## look at what this does!
arrivals_choro.add_to(map)
map
```

```
<folium.folium.Map at 0x22aa7411110>
```

In such an interactive map, it would be nice to display some property of each polygon when hovering with the mouse over it. `folium.Choropleth` does not support this. Yet, we can add a “transparent” layer using `folium.features.GeoJson`, and configure it to display `GeoJsonTooltip`. We can keep the `map` we created above, and simply add the layer to it.

```
# folium GeoJson layers expect a styling function, that receives each of the map's feature and
# It can, however, also return a static style:
def style_function(feature):
    return {
        "color": "transparent",
        "fillColor": "transparent",
    }

tooltip = folium.features.GeoJsonTooltip(fields=("name",), aliases=("Municipality:",))
tooltip_layer = folium.features.GeoJson(serbia, style_function=style_function, tooltip=tooltip)
tooltip_layer.add_to(map)
map
```

```
<folium.folium.Map at 0x22aa7411110>
```

The problem now is that we cannot highlight the polygons and fully figure out the boundaries of the areas we are hovering on. However, we can also tackle that. First we create again our `map`.


```
# base map
map = folium.Map(location=[serbia.geometry.centroid.y.mean(), serbia.geometry.centroid.x.mean()])
```

C:\Users\gfilo\AppData\Local\Temp\ipykernel_16604\3770209863.py:2: UserWarning: Geometry is not in a projected CRS. To avoid distortion, please project geometries to a projected CRS before this operation.

```
map = folium.Map(location=[serbia.geometry.centroid.y.mean(), serbia.geometry.centroid.x.mean()])
```

Then, the choropleth object as before. This time we also set `use_jenks` as `True`. The number of desired classes needs to be passed by using the parameter `bins`.

```
# Add the choropleth layer
arrivals_choro = folium.Choropleth(geo_data=serbia, data = serbia, columns=['id_str', 'arr_int'],
                                   use_jenks = True, bins = 7, fill_color='Oranges', line_weight=1,
                                   fill_opacity=0.7, legend_name='Arrivals Intensity').add_to(map)
## no highlight here
```

Then, below, we define two functions that always take a feature as a parameter (each geographic element on the map, the municipality polygons).

1. `style_function`:

- Purpose: Defines the default style for each polygon on the map.
- Sets the fill color, border color, border weight, and fill opacity for polygons. The provided colors and values are applied to all polygons by default.

2. `highlight_function`:

- Purpose: Changes the style of a polygon when it is hovered over.
- Alters the fill color and opacity when the mouse hovers over a polygon, making it stand out.

```
# Define style function
def style_function(feature):
    return {
        'fillColor': arrivals_choro.color_scale(feature['properties']['arr_int']), # fill color
        'color': 'black', # Border color
        'weight': 0.5, # Border weight
        'fillOpacity': 0.7 # Fill color opacity
    }

# Define highlight function
```

```
def highlight_function(feature):
    return {
        'fillColor': '#ff0000', # Fill color when highlighted
        'color': 'black', # Border color when highlighted
        'weight': 0.5, # Border weight when highlighted
        'fillOpacity': 1.0 # Fill color opacity
    }
```

Essentially, here, we use the `choropleth`, `arrivals_choro` just to get the colours assigned to the feature, on the basis of the `bins` that we created with the scheme. Then we pass these colors to the same features in the `folium.GeoJson` class below. In this case the layer is not transparent. Finally, as before, we add a `tooltip` and a `popup`.

The popup appears when the polygon is clicked but this aspect can be personalised too.

```
# Add popups and pass styling options
folium.GeoJson(serbia, style_function = style_function,
               highlight_function=highlight_function,
               tooltip= folium.features.GeoJsonTooltip(fields=['name']),
               popup=folium.features.GeoJsonPopup(fields=['arr_dom', 'arr_int'], aliases = [
               ]).add_to(map)

map
```

```
<folium.folium.Map at 0x22aa0556c10>
```

4.2 Interactive Maps and API

Let's go back to the Bike Points example we starting looking at in the Web's architecture session.

```
import requests
import pandas as pd
import geopandas as gpd
from shapely.geometry import Point

# Request to Transport for London API for BikePoint data
response = requests.get("https://api.tfl.gov.uk/BikePoint/")
bike_stations = pd.DataFrame(response.json())
```

We extract the numeric part from the `id` column

```
bike_stations['stationID'] = [id.split('_')[1] if '_' in id else None for id in bike_stations['id']]
bike_stations['stationID'] = bike_stations['stationID'].astype('int64')
```

and we create a `GeoDataFrame`

```
bike_stations_gdf = gpd.GeoDataFrame(bike_stations, geometry=[Point(xy) for xy in zip(bike_stations['x'], bike_stations['y'])])
bike_stations_gdf.set_crs('EPSG:4326', inplace = True) # Set CRS
bike_stations_gdf.head()
```

	\$type	id	url	commonName
0	Tfl.Api.Presentation.Entities.Place, Tfl.Api.P...	BikePoints_1	/Place/BikePoints_1	River Street , Cle
1	Tfl.Api.Presentation.Entities.Place, Tfl.Api.P...	BikePoints_2	/Place/BikePoints_2	Phillimore Garden
2	Tfl.Api.Presentation.Entities.Place, Tfl.Api.P...	BikePoints_3	/Place/BikePoints_3	Christopher Stree
3	Tfl.Api.Presentation.Entities.Place, Tfl.Api.P...	BikePoints_4	/Place/BikePoints_4	St. Chad's Street
4	Tfl.Api.Presentation.Entities.Place, Tfl.Api.P...	BikePoints_5	/Place/BikePoints_5	Sedding Street, SI

Now let's add some data about trips made by hire bikes. We need to use the station IDs for the beginning and end of the trips. Transport for London publishes online all trips made by hire bikes along many other datasets related to [bike usage in London](#). The files are published weekly. They have information on starting and ending stations, exact time of the trips.

We can download the files for August 2018 and do some cleaning to map the most used routes in London. We first need to filter for completed trips and select trips with different origins/destinations. The next step is to aggregate the trips by pairs of origin and destination stations. The results should be how many trips have originated and ended from a specific pair in August 2018.

```
import os

# URLs of hire bikes data in August 2018
urls = [
    "https://cycling.data.tfl.gov.uk/usage-stats/121JourneyDataExtract01Aug2018-07Aug2018.csv",
    "https://cycling.data.tfl.gov.uk/usage-stats/122JourneyDataExtract08Aug2018-14Aug2018.csv",
    "https://cycling.data.tfl.gov.uk/usage-stats/123JourneyDataExtract15Aug2018-21Aug2018.csv",
    "https://cycling.data.tfl.gov.uk/usage-stats/124JourneyDataExtract22Aug2018-28Aug2018.csv"
]

# Destination folder
folder = "../data/tfl"
```

```

# Create the destination folder if it doesn't exist
os.makedirs(folder, exist_ok=True)

# Function to download and save a file
def download_file(url, folder):
    response = requests.get(url)
    filename = url.split('/')[-1]
    filepath = os.path.join(folder, filename)
    with open(filepath, 'wb') as file:
        file.write(response.content)

# Download each file
for url in urls:
    download_file(url, folder)

```

The code above performs the following actions:

- Define URLs: A list of URLs is created, each pointing to a CSV file containing data on hire bikes for different weeks in August 2018.
- Destination Folder: It sets a destination folder where the downloaded files will be stored and creates the directory if it doesn't exist
- Define Download Function: `download_file` is a function defined to download a file from a given URL and save it to the specified destination folder. - It uses `requests.get` to fetch the file and writes it to the destination folder.
- Download: A loop goes through each URL in the `urls` list and calls `download_file` to download and save each file in the defined destination folder.

Now let's load them with `glob`, a library that loads all the files from a directory, and then create the dataframe of journeys.

```

import glob

# Reading all CSV files in the directory
files = glob.glob(f'{folder}/*JourneyDataExtract*.csv')

# Concatenating the CSV files into a single DataFrame
journeys = pd.concat((pd.read_csv(file) for file in files))
journeys.rename(columns = {"Duration" : "duration", "StartStation Id": "fromStation_id", "EndStation Id": "toStation_id"})
journeys = journeys[journeys['fromStation_id'] != journeys['toStation_id']] # same station journeys
journeys.head()

```

	Rental Id	duration	Bike Id	End Date	toStation_id	EndStation Name	
0	78826223	780	13775	02/08/2018 19:13	291	Claverton Street, Pimlico	0
1	78986704	660	3498	06/08/2018 17:45	742	Blenheim Crescent, Ladbroke Grove	0
2	78990055	1620	922	06/08/2018 18:36	609	Sugden Road, Clapham	0
3	78940313	5340	12074	05/08/2018 17:27	441	Sail Street, Vauxhall	0
4	79004458	1440	13220	07/08/2018 07:32	309	Embankment (Savoy), Strand	0

Finally, let's get a summary dataframe thanks to `groupby` and other `pandas` functions.

```
# Filter and aggregate the data
journeys_agg = (
    journeys.dropna(subset=['toStation_id', 'fromStation_id'])
    .query("`fromStation_id` in @bike_stations['stationID'] and `toStation_id` in @bike_stat")
    .query("duration > 0 and duration <= 180*60")
    .groupby(['fromStation_id', 'toStation_id'])
    .agg(journeys=('fromStation_id', 'count'), mean_duration=('duration', 'mean'))
    .reset_index()
    .assign(share_trips=lambda x: (100 * x.journeys / x.journeys.sum()))
)
```

What we did above is combining different `pandas` functions:

- `.dropna(subset=['toStation_id', 'fromStation_id'])`: Removes rows where either the start or end station ID is missing.
- `.query("fromStation_id in @bike_stations['stationID'] and toStation_id in @bike_stations['stationID']")`: Further filters the data to include only those journeys where both the start and end stations are listed in `bike_stations`.
- `.query("duration > 0 and duration <= 180*60")`: Keeps only those journeys with a duration greater than 0 and less than or equal to 180 minutes.
- `.groupby(['fromStation_id', 'toStation_id'])`: Groups the data by the combination of start and end station IDs.
- `.agg(journeys=('fromStation_id', 'size'), mean_duration=('duration', 'mean'))`: Aggregates each group to calculate the total number (`count`) and average duration (`mean`) of journeys.
- `.reset_index()`: Resets the `DataFrame`'s index, turning the grouped columns back into regular columns.
- `.assign(share_trips=lambda x: 100 * x.journeys / x.journeys.sum())`: Creates a new column `share_trips`, calculating each route's share of total journeys as a percentage.

The `.query()` method in `pandas` allows you to filter a `DataFrame` using a query string. It's a more concise and readable way to perform filtering compared to traditional Boolean indexing. It

takes the following syntax: `.query('expression')` where ‘expression’ is a string that describes the condition to filter by. Inside the expression, you can refer to DataFrame columns directly by their names and use logical operators (like `==`, `!=`, `>`, `<`, `in`, `and`, or `or`) to define the conditions. **Example:** `.query('age > 30 and city == "New York"')` filters the DataFrame to include only rows where the ‘age’ column is greater than 30 and the ‘city’ column equals “New York”

No need to remember this!

```
journeys_agg[['journeys', 'mean_duration', 'share_trips']].describe()
```

	journeys	mean_duration	share_trips
count	168459.000000	168459.000000	168459.000000
mean	4.886714	1297.513994	0.000594
std	9.681218	927.798752	0.001176
min	1.000000	60.000000	0.000121
25%	1.000000	780.000000	0.000121
50%	2.000000	1121.818182	0.000243
75%	5.000000	1530.000000	0.000607
max	679.000000	10800.000000	0.082482

Most origin/destination pairs have an average of 4.80 trips during the period. The average duration is around 21 min (1297/60). We can then filter our journeys to the top 2 percentiles in terms of counts between two stations. Most pairs do not have any trips (none goes from the furthest station in Hackney down to Oval station).

```
import numpy as np
from shapely.geometry import LineString

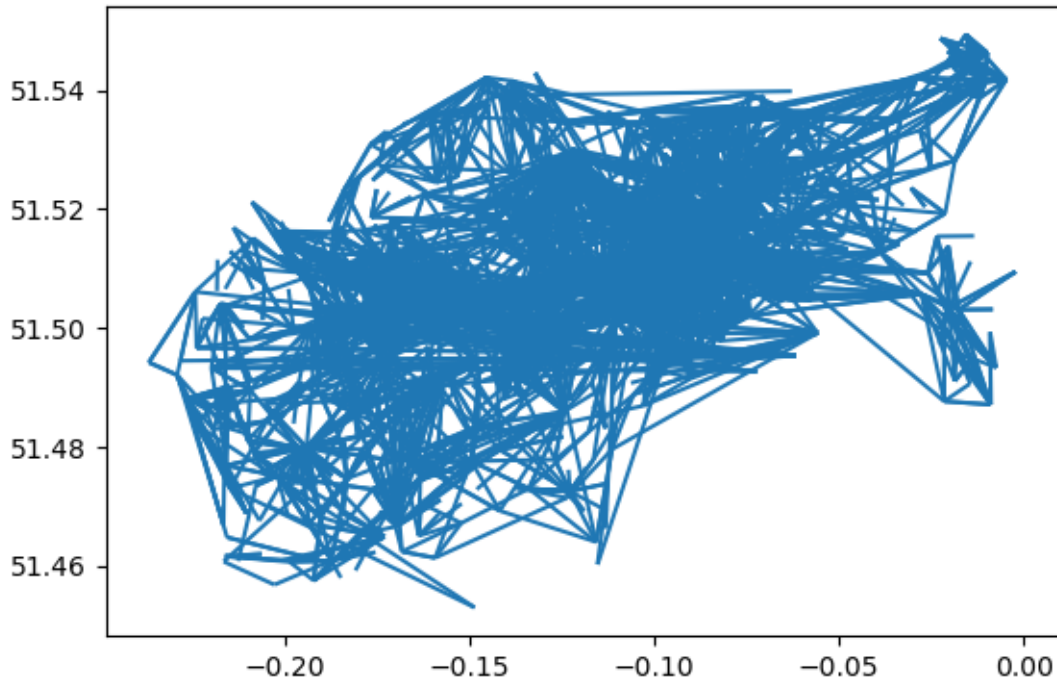
# Calculate the 98th percentile value for the number of journeys
percentile_value = np.percentile(journeys_agg['journeys'], 98)

# Filter out the top 2% of journeys based on the number of journeys
top_journeys = journeys_agg[journeys_agg['journeys'] >= percentile_value]

# Continuing with the merging and creation of desire lines as before
top_journeys = top_journeys.merge(bike_stations, left_on='fromStation_id', right_on='stationID')
top_journeys = top_journeys.merge(bike_stations, left_on='toStation_id', right_on='stationID')

geometry = [LineString([(row['lon_from'], row['lat_from']), (row['lon_to'], row['lat_to'])])]
desire_lines = gpd.GeoDataFrame(top_journeys, geometry=geometry, crs = 'EPSG:4326')

desire_lines.plot()
```



Plotting all lines is quite messy but we can use `folium` again :) Unfortunately, we can't use `Choropleth` for `LineString` geometries. We have to carry out some manual work.

```
import json
import jenkspy
import matplotlib
import matplotlib.pyplot as plt
# Convert the desire_lines GeoDataFrame to GeoJSON
desire_lines_geojson = desire_lines.to_json()

# breaks with fisher-jenks
breaks = jenkspy.jenks_breaks(desire_lines['journeys'], n_classes=5)

# Use a Matplotlib colormap
cmap = plt.get_cmap('Oranges', len(breaks)) # 'viridis' can be replaced with any Matplotlib

# Generate colors for each break
colors = [matplotlib.colors.rgb2hex(cmap(i)) for i in range(cmap.N)]
cmap
```



Let's define again a styling function for our lines.

```
# Define the style function using the generated colors
def assign_color(value):
    for i, break_point in enumerate(breaks[:-1]):
        if value < breaks[i+1]:
            return colors[i]
    return colors[-1]

style_function = lambda feature: {
    'color': assign_color(feature['properties']['journeys']),
    'weight': feature['properties']['share_trips']*150,
    'opacity': max(feature['properties']['share_trips']*25, 0.05)
}

# Create and display the Folium map
map = folium.Map(location=[51.5074, -0.1278], zoom_start=12, tiles="CartoDB.DarkMatterNoLabels")
folium.GeoJson(desire_lines_geojson, style_function=style_function).add_to(map)
map
```

<folium.folium.Map at 0x22aa33e25d0>

Exercise: Look at the documentation for one of the APIs that we discussed last week. - Think of 2 ideas of web maps you could construct from your chosen API/data. - Think about the basemap and the data you would plot on top.

Folium is just one of many packages that provide an easy way to create interactive maps using data stored in (geo-)pandas **DataFrames**. Other interesting libraries include:

- [GeoViews](#).
- [Bokeh](#).
- [kepler.gl](#).
- [pydeck](#).

Nevertheless, Folium allows for a huge degree of personalisation and it's worth exploring further its [documentation](#).

5 Web Architectures and APIs

The **Lecture slides** can be found [here](#).

This **lab's** notebook can be downloaded from [here](#).

5.1 What do APIs actually do?

In this lab, we will unpack how Application Programming Interfaces (“APIs”) work and we cover the basics of accessing an API using Python. Instead of downloading a data set, APIs allow programmers, statisticians (or students) to request data directly from a server to a local machine. When you work with web APIs, two different computers — a client and server — will interact with each other to request and provide data, respectively.

5.1.1 RESTful Web APIs are all around you.

Web APIs

- Allow you query a remote database over the internet
- Take on a variety of formats
- Adhere to a particular style known as Representation State Transfer or REST (in most cases)
- RESTful APIs are convenient because we use them to query database using URLs

Consider a simple Google search:

Ever wonder what all that extra stuff in the address bar was all about? In this case, the full address is Google’s way of sending a query to its databases requesting information related to the search term *liverpool top attractions*.

In fact, it looks like Google makes its query by taking the search terms, separating each of them with a +, and appending them to the link <https://www.google.com/#q=>. Therefore, we should be able to actually change our Google search by adding some terms to the URL:

Learning how to use RESTful APIs is all about learning how to format these URLs so that you can get the response you want.

5.1.2 Group activity

Get into groups of 5 or 6 students. Using your friend the internet, look up answers to the following questions. Each group will be assigned one question and asked to present their findings in 5 min to discuss with the entire class.

1. What is a URL and how can it help us query data? What is a response status and what are the possible categories?
2. What is a GET request? How does a GET request work?
3. What are API keys and how do you obtain them? What kinds of restrictions to they impose on users? Find an example of an API key, what does it look like?
4. (For 2 groups) More and more APIs pop up every day. Do a bit of quick research and find 2 different examples of APIs that you would be interested in using. 2 groups, 2 or 3 APIs each.

There are two ways to collect data through APIs in Python:

- **Plug-n-play packages.** Many common APIs are available through user-written Python (or R) Packages. These packages offer functions that conveniently “wrap” API queries and format the response. These packages are usually much more convenient than writing our own query, so it is worth searching for a package that works with the API we need.
- **Writing our own API request.** If no wrapper function is available, we have to write our own API request and format the response ourselves using Python. This is tricky, but definitely doable.

5.2 API Python libraries

Exercise: census pair activity:

Some Python packages “wrap” API queries and format the response. Lucky us! In pairs, let’s have a look at [census](#), a wrapper for the United States Census Bureau’s API. You can also have a look at the different APIs available from the [United States Census Bureau](#).

```
import pandas as pd
from census import Census
```

To get started working, set your Census API key. A key can be obtained from http://api.census.gov/data/key_signup.html.

```
census_api_key = "" # Replace 'YOUR_CENSUS_API_KEY_HERE' with your actual Census API key.
```

```
# Set API key
c = Census(census_api_key)
```

- Variables in tidycensus are identified by their Census ID, e.g. B19013_001.
- Entire tables of variables can be requested with the table argument, e.g. table = 'B19001'.
- Users can request multiple variables at a time, and set custom names with a named vector.

In Python we can use the library [census](#) to access the American Community Survey (ACS) 5-Year Data (2016-2020)

Exercise:

In pairs explore some of the different variables available in the 5-Year ACS (2016-2020). Make a note of 3 variables you would be interested in exploring. The [ACS2 variables](#) might help.

Let's explore income data for example.

```
# Retrieve income data by state using the ACS table B19001
# Note: The variable 'B19001_001E' represents "Estimate!!Total" in table B19001
income_data = c.acs5.state(('NAME', 'B19001_001E'), Census.ALL, year=2020)
income_data[:10] # just first ten "rows"
```

```
[{'NAME': 'Pennsylvania', 'B19001_001E': 5106601.0, 'state': '42'},
 {'NAME': 'California', 'B19001_001E': 13103114.0, 'state': '06'},
 {'NAME': 'West Virginia', 'B19001_001E': 734235.0, 'state': '54'},
 {'NAME': 'Utah', 'B19001_001E': 1003345.0, 'state': '49'},
 {'NAME': 'New York', 'B19001_001E': 7417224.0, 'state': '36'},
 {'NAME': 'District of Columbia', 'B19001_001E': 288307.0, 'state': '11'},
 {'NAME': 'Alaska', 'B19001_001E': 255173.0, 'state': '02'},
 {'NAME': 'Florida', 'B19001_001E': 7931313.0, 'state': '12'},
 {'NAME': 'South Carolina', 'B19001_001E': 1961481.0, 'state': '45'},
 {'NAME': 'North Dakota', 'B19001_001E': 320873.0, 'state': '38'}]
```

```
income_df = pd.DataFrame(income_data)
income_df.head()
```

	NAME	B19001_001E	state
0	Pennsylvania	5106601.0	42
1	California	13103114.0	06
2	West Virginia	734235.0	54

	NAME	B19001_001E	state
3	Utah	1003345.0	49
4	New York	7417224.0	36

Exercise:

Discuss the format of the data obtained with your partner and then use the function `acs5.state` to explore the 3 variables you discussed in the previous exercise.

You can also get more variables by passing the names in a **tuple**. The code below, for example, fetches all the columns from the B19001 table, which include various income brackets. The tuple in the request generates a list of column names (like 'B19001_001E', 'B19001_002E', ..., 'B19001_017E').

```
variables = tuple(f'B19001_{str(i).zfill(3)}E' for i in range(1, 18))
```

5.2.0.1 This is what the line above does:

- `range(1, 18)`: This part generates a sequence of numbers from 1 to 17. In Python, `range(start, stop)` generates numbers from start up to but not including stop.
- `for i in range(1, 18)`: This is a loop within a comprehension that iterates over each number in the range from 1 to 17.
- `str(i).zfill(3)`: For each number `i`, this converts `i` to a string. Then, `.zfill(3)` pads the string with zeros to make it 3 characters long. For example, if `i` is 2, `str(i).zfill(3)` becomes '002'.
- `f'B19001_{str(i).zfill(3)}E'`: This is an **f-string**, a way to format strings in Python. It inserts the zero-padded string into a larger string. So, for `i = 2`, you would get 'B19001_002E'.
- Finally, the comprehension is wrapped in `tuple(...)`, which converts the entire series of strings into a tuple.

```
wide_data = c.acs5.state(('NAME',) + variables, Census.ALL, year=2020)
# Convert to a Pandas DataFrame
wide_df = pd.DataFrame(wide_data)
wide_df.head()
```

	NAME	B19001_001E	B19001_002E	B19001_003E	B19001_004E	B19001_005E	B19001_006E
0	Pennsylvania	5106601.0	296733.0	206216.0	223380.0	227639.0	226678.0
1	California	13103114.0	614887.0	507398.0	435382.0	474093.0	454373.0
2	West Virginia	734235.0	62341.0	43003.0	45613.0	44635.0	40805.0

	NAME	B19001_001E	B19001_002E	B19001_003E	B19001_004E	B19001_005E	B19001_006E
3	Utah	1003345.0	36211.0	27395.0	28460.0	32497.0	36116.0
4	New York	7417224.0	471680.0	340614.0	303901.0	298025.0	276764.0

Let's make our query a bit more precise. We are going to query data on median household income and median age by county in the state of Louisiana from the 2016-2020 ACS. We can use the library `us` to get the code of each of the state by passing their name.

```
import us
# Get the state object for Louisiana
state_obj = us.states.lookup('Louisiana')
# Get the FIPS code (state code)
code = state_obj.fips

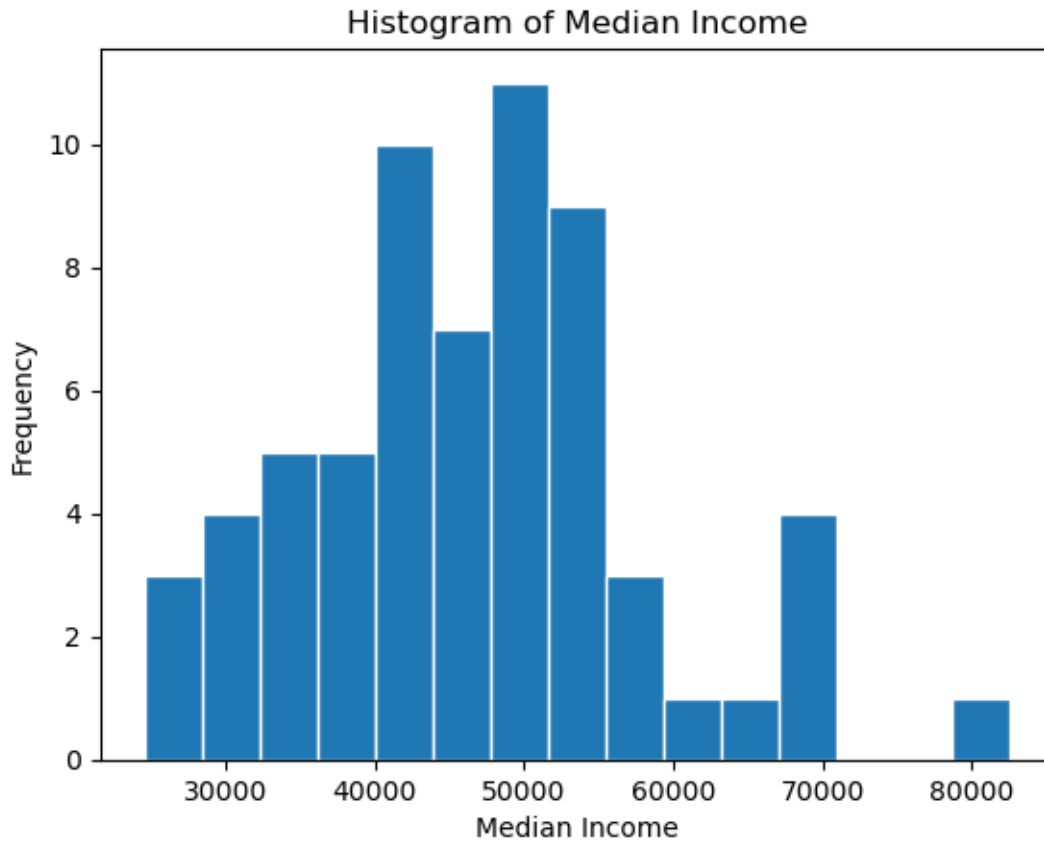
# Retrieve wide format data for median income and median age by county in Louisiana for the years 2016-2020
louisiana_data = c.acs5.state_county(('B19013_001E', 'B01002_001E'), code, Census.ALL, year=2016-2020)
# Convert to a Pandas DataFrame
louisiana_df = pd.DataFrame(louisiana_data)
# Renaming the columns for clarity
louisiana_df.rename(columns={'B19013_001E': 'median_income', 'B01002_001E': 'median_age'}, inplace=True)
louisiana_df.head() # Display the first few rows
```

	median_income	median_age	state	county
0	82594.0	36.0	22	005
1	49256.0	37.5	22	011
2	42003.0	37.8	22	017
3	56902.0	44.9	22	023
4	36294.0	37.4	22	029

Let's plot one of our variables.

```
import matplotlib.pyplot as plt

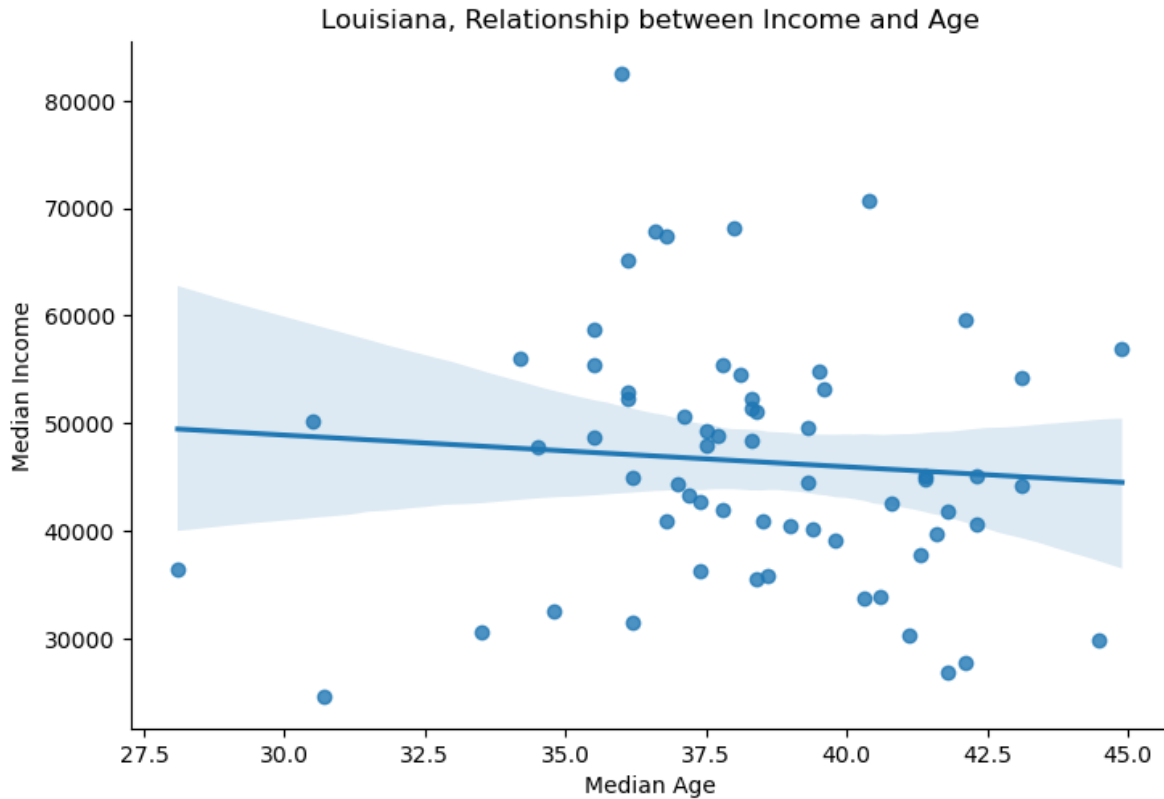
# Plotting histogram for median income with 15 bins
plt.hist(louisiana_df['median_income'], bins=15, edgecolor='white')
plt.xlabel('Median Income')
plt.ylabel('Frequency')
plt.title('Histogram of Median Income')
plt.show()
```



We can also explore correlations between variables. Let's use `seaborn`.

```
import seaborn as sns

# Creating a scatter plot with a linear regression line
sns.lmplot(x='median_age', y='median_income', data=louisiana_df, aspect=1.5)
plt.xlabel('Median Age')
plt.ylabel('Median Income')
plt.title('Louisiana, Relationship between Income and Age')
plt.show()
```



Exercise:

In pairs, modify the state, variables and year parameters following the approach adopted above and produce some other simple scatter plots (cloud of points) that suggest correlations between your variables of interest.

5.3 Your own API request demo

The Python libraries commonly used for API requests are `requests` and `json`.

JSON stands for JavaScript Object Notation. Despite its association with JavaScript, JSON is widely used due to its readability and ease of use by computers, making it the primary format for data transmission through APIs. Most APIs return their responses in JSON format. The `json` library in Python allows you to parse and convert these JSON responses into Python data structures. JSON structures are composed of key-value pairs, similar to Python dictionaries.

In Python, to make an API request and handle the response, you would typically use the `requests` library. A standard API request involves sending a `GET` request to the server's URL, which specifies the data you wish to retrieve. For instance, to request the locations of

all the hire bike stations in London from the **Transport for London API**, you would use the `requests.get()` method. This method requires a URL that directs the request to the appropriate server.

```
import requests
import json

response = requests.get("https://api.tfl.gov.uk/BikePoint/")
response
```

<Response [200]>

Good! The response code 200 indicates a successful request

Most GET request URLs for API querying have three or four components:

1. Authentication Key/Token: A user-specific character string appended to a base URL telling the server who is making the query; allows servers to efficiently manage database access.
2. Base URL: A link stub that will be at the beginning of all calls to a given API; points the server to the location of an entire database.
3. Search Parameters: A character string appended to a base URL that tells the server what to extract from the database; basically, a series of filters used to point to specific parts of a database.
4. Response Format: A character string indicating how the response should be formatted; usually one of .csv, .json, or .xml.

```
# Making the API request
response = requests.get("https://api.tfl.gov.uk/BikePoint/")

# Parsing the response content as JSON and converting it to a DataFrame
bike_stations = pd.DataFrame(response.json())

# Printing the column names
print(bike_stations.columns)
```

```
Index(['$type', 'id', 'url', 'commonName', 'placeType', 'additionalProperties',
      'children', 'childrenUrls', 'lat', 'lon'],
      dtype='str')
```



```
# Creating a new column 'Station ID' by extracting the numeric part from the 'id' column
bike_stations['Station ID'] = bike_stations['id'].str.extract(r'BikePoints_(\d+)', expand=False)
bike_stations.head()
```

	\$type	id	url	commonName
0	Tfl.Api.Presentation.Entities.Place, Tfl.Api.P...	BikePoints_1	/Place/BikePoints_1	River Street , Cle
1	Tfl.Api.Presentation.Entities.Place, Tfl.Api.P...	BikePoints_2	/Place/BikePoints_2	Phillimore Garden
2	Tfl.Api.Presentation.Entities.Place, Tfl.Api.P...	BikePoints_3	/Place/BikePoints_3	Christopher Stree
3	Tfl.Api.Presentation.Entities.Place, Tfl.Api.P...	BikePoints_4	/Place/BikePoints_4	St. Chad's Street
4	Tfl.Api.Presentation.Entities.Place, Tfl.Api.P...	BikePoints_5	/Place/BikePoints_5	Sedding Street, S

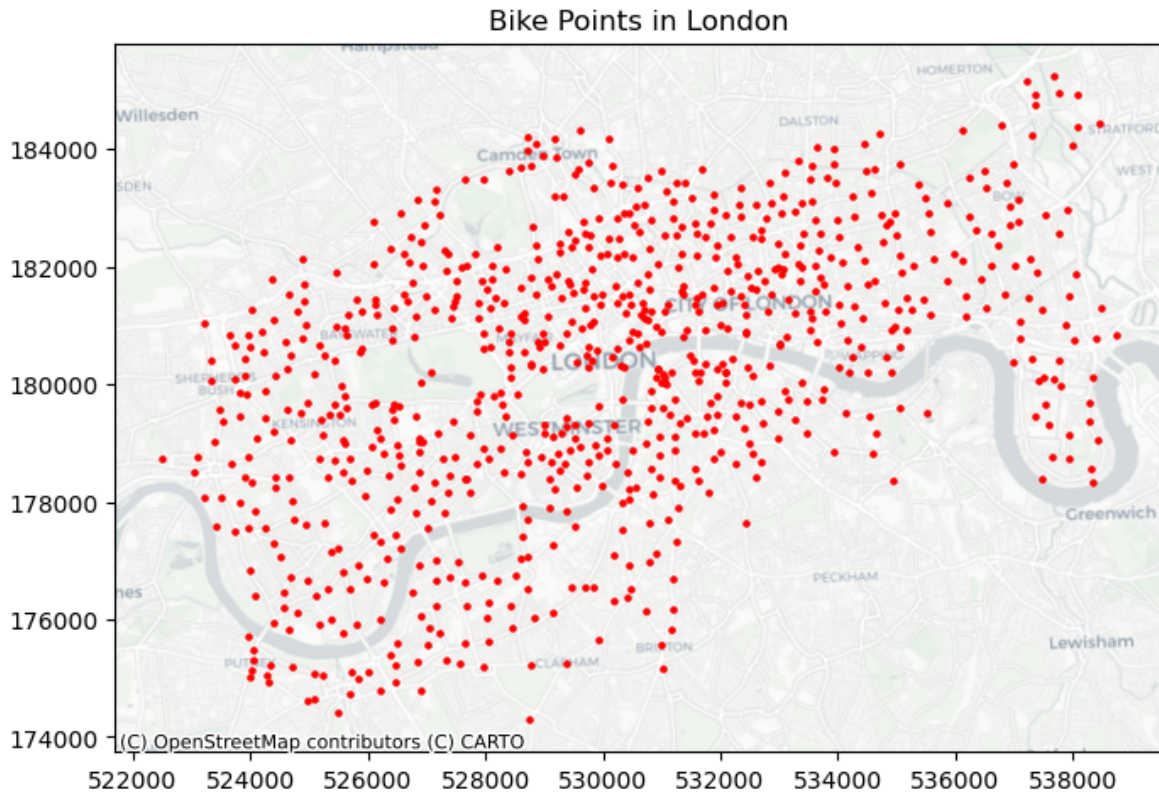
By now you should be able to visualise the data that you have obtained through this API.

```
%matplotlib inline
import geopandas as gpd
import contextily as ctx

# Convert DataFrame to GeoDataFrame
gdf = gpd.GeoDataFrame(bike_stations, geometry=gpd.points_from_xy(bike_stations.lon, bike_sta
# Setting the Coordinate Reference System (CRS) to WGS84 (EPSG:4326)
gdf.set_crs(epsg=4326, inplace=True)
# project to British Grid
gdf.to_crs(epsg= 27700, inplace = True)

# Plotting bikepoints on a map
fig, ax = plt.subplots(1, 1, figsize=(8, 6))
gdf.plot(ax = ax, color='red', markersize=5)
source = ctx.providers.CartoDB.Positron
ctx.add_basemap(ax, crs= gdf.crs.to_string(), source= source)
ax.set_title('Bike Points in London')
```

```
Text(0.5, 1.0, 'Bike Points in London')
```



5.4 Geocoding API

Below is a short exploration of the `geocoder` provided by the library `geopy`. The `geocoder` relies on various external geocoding services. When using `Nominatim`, it accesses data from OpenStreetMap. OpenStreetMap's data includes a vast array of geographical information sourced from contributors globally. This includes street addresses, points of interest, and other location-based data. When you provide an address to the `Nominatim` geocoder, it queries OpenStreetMap's databases to find the corresponding geographical coordinates.

In your own time, try to use it to automatically embed coordinates between addresses.

```
from geopy.geocoders import Nominatim
from geopy.extra.rate_limiter import RateLimiter

# Create a DataFrame with addresses
some_addresses = pd.DataFrame({
    'name': ["South Campus Teaching Hub", "Sefton Park", "Stanley Street"],
    'addr': ["140 Chatham St, Liverpool L7 7BA", "Sefton Park, Liverpool L17 1AP", "4 Stanley"]
})
```

```

}))

geolocator = Nominatim(user_agent="wma-geovis")
geocode = RateLimiter(geolocator.geocode, min_delay_seconds=1, return_value_on_exception=None)

lat longs = some_addresses["addr"].apply(lambda a: geocode(a, timeout=10)).apply(
    lambda loc: pd.Series([loc.latitude, loc.longitude], index=["latitude", "longitude"])
    if loc else pd.Series([None, None], index=["latitude", "longitude"]) # handle None
)

some_addresses = some_addresses.join(lat longs)
some_addresses

```

	name	addr	latitude	longitude
0	South Campus Teaching Hub	140 Chatham St, Liverpool L7 7BA	53.400091	-2.964452
1	Sefton Park	Sefton Park, Liverpool L17 1AP	53.386547	-2.932565
2	Stanley Street	4 Stanley St, Liverpool L1 6AA	53.407644	-2.987573

5.4.0.1 Reverse Geocoding

You can also reverse geo-code the data, open the output and see what the result is.

```

# Reverse geocoding (optional)
def reverse_geocode(lat, lon):
    location = geolocator.reverse((lat, lon))
    return location.address

# Apply reverse geocoding
some_addresses[['latitude', 'longitude']].apply(lambda x: reverse_geocode(x['latitude'], x['longitude']), axis=1)

```

```

0    152, Chatham Street, Canning / Georgian Quarte...
1    Sefton Park Cricket Club, Croxteth Drive, Wave...
2    Workforce, 14-16, Stanley Street, Pride Quarte...
dtype: str

```

5.4.0.2 Geographic Data through APIs and the web

APIs can help us generate and create spatial data. In this lab, we have both:

- Written **our own API request**.
- Used **Plug-n-play packages**.

There are many APIs where we can get data these days. A few other examples include

- [The London DataStore API](#)
- [Thames Water API](#)
- [London Air API](#)
- [Crime data](#)

5.5 *Group activity answers*

1. Uniform Resource Location (URL) is a string of characters that, when interpreted via the Hypertext Transfer Protocol (HTTP). URLs point to a data resource, notably files written in Hypertext Markup Language (HTML) or a subset of a database.
 - 1xx informational response - the request was received, continuing process
 - 2xx successful - the request was successfully received, understood, and accepted
 - 3xx redirection - further action needs to be taken in order to complete the request
 - 4xx client error - the request contains bad syntax or cannot be fulfilled
 - 5xx server error - the server failed to fulfil an apparently valid request

2. GET requests a representation of a data resource corresponding to a particular URL. The process of executing the GET method is often referred to as a **GET request** and is the main method used for querying RESTful databases. HEAD, POST, PUT, DELETE: other common methods, though mostly never used for database querying.

Surfing the web is basically equivalent to sending a bunch of GET requests to different servers and asking for different files written in HTML. Suppose, for instance, I wanted to look something up on Wikipedia. Your first step would be to open your web browser and type in `http://www.wikipedia.org`. Once you hit return, you would see the page below. Several different processes occurred, however, between you hitting “return” and the page finally being rendered:

1. The web browser took the entered character string, used the command-line tool “Curl” to write a properly formatted HTTP GET request, and submitted it to the server that hosts the Wikipedia homepage.
2. After receiving this request, the server sent an HTTP response, from which Curl extracted the HTML code for the page (partially shown below).
3. The raw HTML code was parsed and then executed by the web browser, rendering the page as seen in the window.

4. Most APIs requires a key or other user credentials before you can query their database. Getting credentials for an API requires that you register with the organization. Once you have successfully registered, you will be assigned one or more keys, tokens, or other credentials that must be supplied to the server as part of any API call you make. To make sure users are not abusing their data access privileges (e.g., by making many rapid queries), each set of keys will be given rate limits governing the total number of calls that can be made over certain intervals of time.
3. Most APIs requires a key before you can query their database. This usually requires you to register with the organization. Most APIs are set up for developers, so you will likely be asked to register an “application.” All this really entails is coming up with a name for your app/bot/project and providing your real name, organization, and email. Note that some more popular APIs (e.g., Twitter, Facebook) will require additional information, such as a web address or mobile number. Once you have registered, you will be assigned one or more keys, tokens, or other credentials that must be supplied to the server as part of any API call you make. Most API keys limits the total number of calls that can be made over certain intervals of time. This is so users do not abusing their data access privileges.

5.6 References

- [Brief History of the Internet](#), by the Internet Society, is a handy (and free!) introduction to how it all came to be.
- Haklay, M., Singleton, A., Parker, C. 2008. “[Web Mapping 2.0: The Neogeography of the GeoWeb](#)”. *Geography Compass*, 2(6):2011–2039
- [A blog post from JoeMorrison](#) commenting on the recent change of licensing for some of the core software from Mapbox
- Terman, R., 2020. [Computational Tools for Social Science](#)